E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

A Capstone Design Project
Presented to the Faculty of the
Information and Communications Technology Program
STI College of Caloocan

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Engineering

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June 2023

ENDORSEMENT FOR FINAL DEFENSE

TITLE OF RESEARCH: **E-bag: An Electronic Backpack with GPS Locator and IOT Technology**

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ACKNOWLEDGEMENTS

We would like to express our heartfelt gratitude to the individuals who have played a significant role in the success of our project.

To our thesis coordinator, Engr. Bernard A. Mendiola, MSEE-CpE, and adviser, Engr. Jimmy A. Jovero, have provided unwavering support, enthusiasm, and expertise throughout our study and research. Their valuable insights and feedback have greatly improved the quality of our work.

We also want to acknowledge the review panelists for their constructive feedback and suggestions, which have helped us enhance the output of our device.

To our parents and guardians for their continuous support, encouragement, and financial assistance that enabled us to complete the project within the given timeframe.

To our classmates and friends for their ideas, encouragement, and moral support, making our journey more enjoyable and rewarding.

To our client, Engr. Mark Lapuz, for providing valuable insights and feedback that also contributed to the success of our project.

We extend our gratitude to the respondents of our study for their responsiveness and valuable time in participating.

Lastly, we thank God for blessing us with the knowledge, courage, and strength to overcome challenges and successfully complete this project.

To all those mentioned and everyone who has supported us in various ways, we sincerely appreciate your contributions and are forever grateful for your presence in our capstone project.

Thank you.

ABSTRACT

Title of research: **E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY**

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Degree: **Bachelor of Science in Computer Engineering**

Date of Completion: **May 2023**

The E-Bag is an electronic backpack incorporating GPS locator and IoT technology for enhanced security and theft prevention. It offers a range of features to prevent potential robbery scenarios. The bag includes a built-in camera and system that can detect when someone opens the backpack without the user's permission. The anti-theft alarm is activated to alert the user and those nearby, and the image captured by the system is immediately displayed on the mobile application. The user will also be notified using SMS and mobile application. The study aims to address safety concerns and instill confidence in backpack users by significantly enhancing security and preventing theft. Using an applied research design and purposive convenient sampling, the study gathers valuable insights from a targeted participant group. Practical recommendations are developed to improve backpack security. The findings contribute to advancing backpack security technology, offering an innovative solution to prevent theft and provide peace of mind to users. People who rely on backpacks can gain from a dependable and effective system that reduces the risk of theft by taking advantage of these findings. The project is useful for the development of backpack security technology. It tackles the pressing issue of theft and its impact on individuals' safety and well-being. The study's outcomes have the potential to revolutionize the backpack user experience, making it more secure and worry-free.

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CHAPTER 1

INTRODUCTION

Backpacks have long been an essential item for many people, whether for school, work, or travel. However, with the rise of technology and increasing security concerns, the traditional backpack has evolved to include advanced security features to protect the belongings of its users.

The researchers of the electronic backpack with a GPS locator and IoT technology, known as E-bag, developed this product to provide users with a heightened sense of security and peace of mind. The system comes equipped with an armed and disarmed mechanism, which enables the user to control the access to their bag. When the backpack is armed, and someone tries to open the bag, the backpack will capture an image of the person's face using IoT technology, allowing the owner to identify the person who opened the bag. The captured images will be stored in a database for identification purposes. In addition, an anti-theft alarm will be activated, and the backpack's weight sensor will detect any changes in the weight of the bag, alerting the owner if any items have been removed. The tracking system included in the E-bag is another valuable feature. It allows users to track their bag in real-time using Web page and even SMS notification, making it an ideal solution for individuals who often misplace their baggage.

A. Mounika and Anitha Chepuru (2019) highlighted the growing concern among modern individuals regarding the safety and security of their electronic devices. One potential solution to address this concern is using GPS technology, which has already been employed in public transportation and vehicles as a deterrent against theft. GPS technology uses satellite signals to pinpoint the exact location of a lost or stolen item, providing latitude and longitude coordinates for easy tracking. Using a GSM modem now simplifies and reduces the cost of communication either way. With this improved communication technology, individuals can easily locate their devices and protect them against potential theft or loss.

This idea was designed to safeguard the user's backpack. The researchers implement an electronic backpack with a GPS locator and IoT to safeguard the owner's possessions. It provides users with a range of advanced features and capabilities to ensure their security, convenience, and peace of mind.

Background of the problem

Despite the convenience and popularity of backpacks, users remain concerned about potential security issues, especially when they are unaware that their backpacks have been opened by thieves who take advantage of the unsuspecting individuals who use backpacks. Surveillance cameras have captured a number of instances in which individuals using backpacks have been the victims of theft. The thief's strategy is to manipulate the bag since it is placed on their back and may go unnoticed, thereby taking advantage of others who carry backpacks. Ms. Nicole Cruz, a resident of #340 Maginoo St., Tondo, Manila City, had an unfortunate experience that is all too common among backpack users. As someone who often carries her most valuable possessions in her backpack, she encountered a sudden realization that items were missing from it, without any apparent explanation of how it happened. This incident highlights the importance of staying vigilant and taking necessary precautions to protect our belongings, especially when carrying them in public places.

In one instance, the client may have put her belongings inside her backpack only to find them missing later, without any signs of forced entry or tampering. Another common scenario that Ms. Nicole Cruz face is finding the zipper of his backpack open while she is on her way home and discovering that her wallet is missing. This experience can be particularly stressful, as it may be difficult to determine how the zipper was opened and how the wallet was taken without the client's knowledge.

To address these security concerns, the proposed study presents a solution: E-Bag, an electronic backpack equipped with GPS locator and IoT technology that offers a range of features to prevent potential robbery scenarios. The bag includes a built-in camera and system that can detect when someone opens the backpack without the user's permission. The anti-theft alarm is activated to alert the user and those nearby, and the image captured by the system is immediately displayed on the mobile application. The user will also be notified through the use of SMS and mobile application.

The primary objective of this study is to provide backpack users with a sense of comfort regarding the safety of their backpacks and valuables. The E-Bag system is expected to enhance the security of backpacks and prevent potential theft, giving users peace of mind when carrying their belongings. By addressing this problem, the proposed system can make a significant contribution to the safety and security of individuals in the country who use backpacks to carry their possessions.

Overview of the current state of the technology

Backpacks have undergone significant changes since their inception as simple cloth bags with straps. Today's backpacks are designed with a range of features and materials that make them more comfortable, durable, and practical. One of the most important advancements in backpack technology is the use of ergonomic design principles. Backpacks now distribute weight evenly across the wearer's back and shoulders, reducing strain and fatigue. This has led to the creation of backpacks with adjustable straps, padded back panels, and other features that enhance comfort and support.

Another area of innovation in backpack design is the use of high-tech materials. With technology constantly evolving, modern backpacks now incorporate safety technology to enhance security. These technologies include mobile apps, wearables, machine sensors, cloud-based software, predictive analytics, 3D printing, robotics, real-time employee monitoring and tracking, and PPE tracking with embedded sensors.

Concerning safety technologies, the researchers have also presented a study that focuses on the development of a safety system that safeguards the wearer's possessions and protects them from potential theft. The system includes several user-beneficial technologies, such as an integrated camera, weight sensor, GPS tracking system, and anti-theft alarm. The backpack's safety features are further enhanced by an SMS system that can alert the owner when their bag is opened, even if they are not at home. The device can be controlled using a mobile app, making it easy to monitor and manage from a distance. The safety system ensures that users' possessions are safe and secure, providing peace of mind and protecting them from potential theft.

Objectives of the Study

General Objective of the Study

To create and develop E-Bag: An Electronic Backpack with GPS Locator and IOT Technology to help bag owners secure their belongings while inside the bag giving them that sense of peace of mind and comfort.

Specific Objective of the Study

1. **To develop an anti-theft backpack with alarm features using IOT technology.**
    - The E-bag is an anti-theft backpack with alarm features and GPS locator, designed to provide travelers with peace of mind and enhanced security for their belongings. It incorporates IoT technology to detect unauthorized access and trigger an alarm, alerting the owner through a mobile app or SMS. The GPS locator allows users to track the backpack's location, while the safety camera acts as a deterrent for potential thieves. These features, accessible through a mobile app, offer travelers the ability to monitor their backpack and utilize innovative capabilities to enhance their travel experience.

2. **To develop a system that can track location and monitor your backpack in the event that it is misplaced, stolen, or lost.**
    - With the use of GPS navigation technology and SMS system enabled by the Internet of Things, it is now possible to track and monitor a backpack using a mobile phone. This feature can come in handy in situations such as misplacement, loss, or theft of the backpack. In such cases, the backpack's location can be determined using the GPS device, thereby providing an ideal solution for recovering the lost or stolen backpack.

3. **To develop an electronic backpack equipped with a camera that can be easily managed and monitored using a mobile app.**
    - The aim is to develop an electronic backpack equipped with a camera that can be easily managed and monitored using a mobile app, Oftentimes, when backpacks are left unattended, it is difficult to determine who has accessed them. The inclusion of a hidden camera inside the backpack can provide an added layer of security, allowing users to monitor their belongings and quickly identify anyone who has handled the backpack.

Scope and Limitations of the Study

Scope of the study

1. **Log In**

    - In this module, the system administrator is granted elevated privileges that enable them to modify and manage access for other users. The system administrator can perform various tasks, including clearing data, adjusting the alarm, updating the time and date, utilizing GPS features, and modifying usernames and passwords. This elevated access ensures that only authorized users can make changes to the system and prevents unauthorized access to critical information.

2. **History**

    - In this section, if the bag is set to "armed," the user can view a history of images that were taken of the individual who tried to open the bag.

3. **User Manual**

    - This section provides users with access to detailed manuals and step-by-step instructions for using both the application and backpack. By offering comprehensive guidelines, users can easily understand how to use the system's features, ensuring optimal functionality and security.

4. **Configuration**

    - Within this segment of the application, users have the ability to modify their login credentials and password, as well as erase any stored data. Additionally, they can utilize this section to verify the functionality of the GPS system, SMS notifications, and anti-theft alarm. Furthermore, users can power off the software of the device via this area.

5. **System Status**

- System status displays the system's current state. Users can configure the device's "armed" and "disarmed" functionalities in this section. This part also displays the backpack's current weight and the "Lat/Long"(latitude and longitude) coordinates of the device's location. This section is simultaneously responding.

6. **Web Page**

- The web page will feature a map that enables users to track the E-bag using an internet connection. This map will provide real-time updates on the E-bag's location, allowing users to monitor its movements and whereabouts.

7. **SMS Notification**

- SMS notifications are designed to assist users in locating their bag by providing them with a message containing the bag's location. Additionally, these notifications will alert users if their bag has been opened by someone.

Limitation

1. The app does not support simultaneous usage by multiple individuals.
2. Anything that is extremely light cannot be detected by the weight sensor. (Pens, a piece of paper, etc.).
3. Google Maps prevented third-party GPS from accessing its maps directly. It is only accessible via a link.
4. E-Bag relies on GPS locators and IoT technology for features like tracking and alerts, which can be affected in areas with limited connectivity. In locations without a signal, the bag may not be able to perform certain functions such as sending SMS notifications or updating its location in real-time on the web page.
5. The SMS cannot send a link directly to the map due to the NTC Guidelines
6. The E-Bag system will not be covered in fabric to avoid potential malfunctions that may arise from the materials used in the prototype.
7. The E-Bag is vulnerable to physical damage as its electronic components, like safety cameras and GPS locators, can be easily harmed when the bag is roughly handled, exposed to impacts, or subjected to extreme environmental conditions.
8. Security features of the E-Bag, like the weight sensor and alarm systems, might mistakenly activate alarms or provide incorrect information. This could happen due to factors such as the bag moving during transportation, external interferences, or the sensitivity of the sensors.

CHAPTER II

LITERATURE REVIEW

Review of related literature, studies or systems

This section aims to establish the significance of the study and to identify gaps in the literature that the research intends to address. It serves as the foundation of the study to develop E-Bag: An Electronic Backpack with GPS Locator and IOT Technology.

Foreign Literature and Studies

As Rosdi et al. (2016) stated, snatch stealing has become a severe challenge in society, and immediate action is essential to address it. One possible way to end it is to create a Smart Handbag system that uses a radio frequency signal and a location tracker as the primary component. Their project was implemented using the Arduino microcontroller, a Radio Frequency (RF) transmitter and receiver module, a 9dB buzzer, and a SIM900 Global System for Mobile Communications (GSM) and Global Positioning System (GPS) module. Moreover, when a snatch crime occurs, this Smart Handbag makes a loud noise to call the attention of people surrounding the scene. The technology can also use GPS to track the location of the handbag. The victim will be notified of the location tracked by the GPS module through Short Messaging Service (SMS). As a result, this successfully designed Smart Handbag project is likely to assist people in preventing the spread of this crime.

Abhang et al. (2018) proposed a novel technology to address security issues faced by travelers with their bags. The solution utilizes Raspberry Pi, GSM, GPS, a camera, an electromagnetic sensor, and a battery to provide a one-touch system for ensuring the protection of the backpack. The system uses GSM and GPS to track the bag in real-time, and when a bag is taken and crossed the geofencing area, the owner will receive an SMS text and photos of the thief will be collected and sent to a database for identification. Additionally, the system can record the opening and closing of the bag as the switch is pressed. The technology is an android-based application and can trace the location of the bag using data from the GPS. Abhang et al. (2018) concluded that their solution takes advantage of

smartphones' many functions and implemented Geofencing and Emergency Messaging to improve the system.On the other hand, Sheth (2019) pointed out a design flaw of backpacks, which is that wearing them for a long-time results in a sweat patch around the back due to frequent contact with the bag and a lack of sufficient ventilation.

As mentioned by Gajbhiye et al. (2020), bag security is a sensitive issue. Standard bags do not have smart features, which is a problem if the owner forgets them. Therefore, to solve the problem, a Multipurpose Smart Bag was designed that can be useful for everyone. A Multipurpose Smart bag was built with one or more extra features, such as USB charging connections and the ability to send its whereabouts via cellular and app. Moreover, a smart bag with an advanced GPS monitoring system is required to keep the bag secure while traveling. The system offers information on the location of the bags.

Furthermore, the Smart Bag is a design tailored to a particular application and can benefit practically every person in society. Arduino ATmega16, the proposed system's brain, can control all distinguishing aspects. A solar panel is connected to the front half of the bag that will charge electronic devices such as smartphones, power banks, and MP3 players. There is also an Arduino board and a GPS module used to keep track of everything in the bag in case of loss.

Smith (2020) conducted a study on 500 high school students to determine the effectiveness of electronic backpacks with security features in reducing theft. The study randomly assigned students to either an experimental group with electronic backpacks or a control group with traditional backpacks. Results showed that the electronic backpacks with security features were highly effective in reducing theft and did not significantly affect students' daily routines. Students using the electronic backpacks also reported feeling safer and more confident in the school's ability to prevent theft. While electronic backpacks were more expensive, the increased security outweighed the cost. However, the study had some limitations, such as its short-term nature and the use of self-report measures to assess theft. Future research should address these limitations to provide more comprehensive insights into the effectiveness of electronic backpacks in reducing theft in schools.

In order to better understand attitudes and perceptions concerning the deployment of electronic backpacks with security features in schools, Williams and Davis (2017) performed a study. A sample of 600 students, parents, and teachers from five different high schools in the United States participated

in the study. According to the study's findings, 78% of participants had favorable opinions on the use of electronic backpacks with security features and thought that such technology may improve school safety and security. However, other participants voiced worries about the technology's possible expense and potential privacy infringement. According to the report, teachers and parents are more inclined than pupils to favor the usage of electronic backpacks with security features. The study emphasizes the need for additional research to examine the advantages and disadvantages of electronic backpacks with security features and to address any potential concerns held by the users. The study's overall conclusions point to the possibility of enhancing school safety and security through the use of electronic backpacks with security measures, but it's crucial to take into account the opinions of all potential users before putting this technology into practice.

Local Literature and Studies

Duat, Bebelone, and Gallego (2017) addressed the problems that travelers often encounter such as misplaced luggage, battery depletion, theft, and stress, and designed a four-in-one luggage bag to provide a solution. The prototype emphasized features such as biometric security, GPS tracking, an emergency power bank, and a kick scooter for easy transportation. The biometric lock ensured that only the owner could access the luggage bag, while GPS tracking allowed for easy monitoring of the bag's location. In addition, the built-in power bank provided a convenient charging station for electronic devices, and the kick scooter made transportation easier. The 4-in-1 luggage bag was successfully designed and implemented for 50 passengers at the Agora Bus Terminal, who reported satisfaction with the bag's functionality and usefulness. This project highlights the importance of innovation in creating luggage bags that can deliver multiple benefits and meet the needs of modern-day travelers. The success of the prototype suggests the need for further research to explore the potential of luggage bags with additional security features and improved functionality to enhance the travel experience.

Berna, Penis, and Olila (2019) have mentioned that the Philippines was ranked 14th globally regarding the robbery rate. That is why their goal was to build an android-controlled anti-theft bag using an Arduino Uno Rev3 and an android device with enhanced security and other features to protect the valuable and essential belongings of the owner. Also, to see if the anti-theft bag's system

is accurate enough and will work and function properly.

Their project is called GARRISON, a GPS-Installed Robotic Anti-Theft Bag. The device was created by attaching electrical components to the bag's hardware layer, such as the Arduino Uno Rev3, breadboard, jumper wires, servo motors, GPS, modules, and a battery bank as a power source. It was programmed through Arduino IDE and turned into a portable application installed on an Android handset.0

Moreover, it shows that GARRISON correctly replied to the software application's designed command. It also demonstrates that linking GARRISON and the android application will not cause any issues for the owner of the robotic anti-theft bag. Servo motors were utilized to lock and unlock the GARRISON, indicating no faults in the process. The program also had no errors when accessing and closing the device, and the user could change the password at any time. In conclusion, they prove that a simple bag can be improved using Arduino technologies and helps reduce crime rates. If the bag is taken, the critical belongings and GPS will be safeguarded with a password.

Velasco et al. (2020) conduct a study that focuses on the development and implementation of an Internet of Things (IoT)-based inventory monitoring refrigerator. The system was successfully developed and verified as functional and efficient for use in industry. The sensor network system inside the refrigerator was properly installed and has been functioning properly. The wireless inventory monitoring system was a success and was tested in various locations using Android applications developed using Unity and Box it. However, there are areas for improvement in the system. Firstly, for a clearer image of the refrigerator's body, the camera that will be installed must have a high zoom-out rate so that it can cover the entire body of the fridge. Secondly, distance-measuring sensors that can withstand the temperature of the refrigerator can be utilized for the egg tray compartment to reduce the number of wires required for the study. This literature review contributes to the growing body of knowledge on the Internet of Things and its potential applications for monitoring and tracking devices. The insights gained from this study can aid researchers in developing more efficient and effective IoT-based systems for a variety of purposes.

Tayag's (2019) study focuses on the development of a Smart Motorcycle Helmet to address safety concerns among the increasing number of motorcycle riders in the Philippines. The helmet integrates various sensors and an internet connection through 3G, providing features such as an alcohol level

detector, crash/impact sensor, SMS, and cloud computing infrastructure. The device was tested through quantitative and descriptive research during the alpha and beta testing process, resulting in a viable tool for motorcycle riders during emergencies. The study suggests future enhancements, such as the integration of a kill-switch to turn off the engine when the motorcycle is stolen and an automated kill-switch for riders with high alcohol content. The anti-theft and SMS notification features of the Smart Motorcycle Helmet are significant for improving rider security. The anti-theft tracking module integrated into the helmet provides an added layer of security against motorcycle theft, while the SMS notification feature can alert family members or first-responders in case of an accident. These features can reduce motorcycle theft and provide quick response in case of emergencies, making the Smart Helmet a valuable tool for motorcycle riders. The study highlights the importance of anti-theft and SMS notification features in ensuring the safety and security of motorcycle riders. The study provides valuable insights into the potential benefits of integrating such features in wearable devices like smart helmets. As such, the current researcher plans to leverage the data gathered from the study to develop an electronic backpack that incorporates similar security measures. By utilizing the anti-theft tracking module and SMS notification feature, the proposed electronic backpack will provide an added layer of security for users.

Borromeo (2017) reported on the development of a GPS-based vehicle tracking system that enables the secure tracking and monitoring of a vehicle's whereabouts. The study found that the system was able to remotely monitor the engine start and stop of the vehicle, locate the whereabouts of the vehicle, and secure the vehicle from thieves and intruders with the use of security modules of the system. The software project was designed according to the planned design and specifications, and every module was designed according to its functionality. The system was tested several times to check the accuracy, and the results gathered were consistent. The system was evaluated to be "Very Good" with a grand mean of 3.83, signifying that the system attained its desired functionality and reliability. The system was successfully tested for install ability, accuracy, speed, reliability, and stability, and it was recommended that additional security measures be added, such as encryption of the backup map for restore purposes, hidden camera, and audio device that can be accessed within the system. The study concluded that the system could be implemented without issues and could be a generic tool for vehicle location tracking in the fields of business, medical, military, and rescue operations.

Synthesis

The literature and studies presented above highlights the growing interest in developing smart bags and backpacks with advanced security features. E-bag: An Electronic Backpack with GPS Locator and IOT Technology has been conducted to develop a safety system that safeguards the wearer's possessions and protects them from potential theft. The backpack's safety features include an integrated camera, weight sensor, GPS tracking system, and anti-theft alarm. The backpack's safety features are further enhanced by an SMS system that can alert the owner when their bag is opened, even if they are not at home. By incorporating the latest technologies and features, the backpack stands out as a practical and innovative solution that provides peace of mind to users. The SMS notification that indicates the detailed location address of the bag alerts the owner when their bag is opened without their knowledge enhances the backpack's safety features, ensuring that users' possessions are safe and secure. The literature and studies above can be used as a reference point to develop the E-bag: An Electronic Backpack with GPS Locator and IOT Technology using its findings that serve as a starting point to address the limitations and concerns raised by potential users. The study provides practical and innovative solutions to improve security and convenience for users.

CHAPTER III

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

Overview of the project

Theft is a common issue in society, and to address this concern, researchers conducted a survey among individuals who use backpacks to carry their valuable possessions in public places. In response to these security concerns, the proposed study introduces the E-Bag: an electronic backpack with GPS locator and IoT technology. The E-Bag features a built-in camera and a system that detects unauthorized opening of the backpack, triggering an anti-theft alarm and capturing an image displayed on the mobile application. Users are also notified via SMS and the app. The primary objective is to provide users with a sense of comfort and security for their belongings, contributing to the overall safety of individuals who rely on backpacks.

The researchers opted for an applied research design, aiming to address practical problems by applying research findings in real-world contexts. This design involves the use of mixed methods, combining quantitative and qualitative data collection and analysis techniques. Various sources, including surveys, interviews, consultations, and existing data, will be utilized to gather information for the development of the E-Bag and achieve the study's objectives.

For participant selection, purposive convenient sampling will be employed. This approach combines convenience sampling with specific criteria relevant to the study's objectives. By combining convenience and purposeful selection, the researchers can efficiently gather targeted data from participants who possess the desired characteristics or experiences that is relevant to the study.

The study aims to develop an innovative solution to address theft concerns through the E-Bag system. The applied research design and purposive convenient sampling will enable researchers to

gather valuable insights and develop practical recommendations to enhance backpack security, ensuring users can confidently carry their belongings.

H**ardware/Software**

In this section, a comprehensive discussion will be presented on the hardware and software components that will be used in the development of the E-Bag: an electronic backpack with GPS locator and IoT technology.

Hardware:

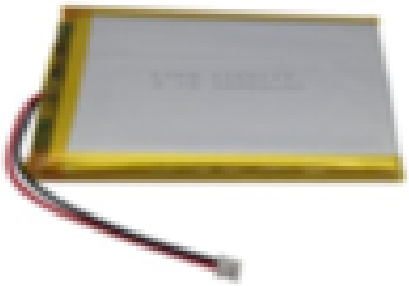


Figure 1. Power Bank 10000mAh

LiPo battery is the power bank of the prototype E-Bag: an electronic backpack with GPS locator and IoT technology. It offers advantages such as high energy density, lightweight construction, and stable power output. The compact and portable design of LiPo batteries makes them convenient for various applications. Their high energy density enables prolonged operation without frequent recharging, and their high discharge rates efficiently supply the prototype's power needs. Integrating proper charging and protection circuitry enhances safety and optimizes the battery's lifespan by preventing overcharging, over-discharging, and excessive current draw.

Figure 2. Mobile Phone

Mobile phone as part of the E-Bag's functionality enhances its maneuverability through a mobile app. The mobile app provides an interface that allows users to control and navigate the device with ease. Additionally, the mobile phone will accept notifications, enabling users to receive real-time updates and track the device's status remotely.

Figure 3. On/off Switch

The purpose of the on/off switch in E-Bag is to control its power supply. It allows users to easily turn the device on or off as needed, providing a means to start or stop its operation. It helps in conserving power by disconnecting the device from its power source when not in use. This helps prolong battery life or reduce energy consumption, promoting efficient and sustainable operation.

Figure 4. Pocket Wi-Fi

In E-Bag, the pocket Wi-Fi serves the purpose of establishing a portable and convenient internet connection, enabling seamless communication between the mobile app and the E-Bag. This connectivity allows users to track and monitor the E-Bag's status remotely, providing real-time updates and enhancing security. Additionally, the pocket Wi-Fi enables the mobile phone to maneuver the E-Bag, offering greater control and accessibility even in indoor environments where GPS signals may be limited.

Figure 5. Raspberry Pi 3b

In the E-Bag system, the Raspberry Pi 3B acts as the central processing unit, responsible for executing and managing the various functions of the system. It acts as the brain of the system, handling data processing, algorithm execution, and overall system control. Additionally, the Raspberry Pi 3B acts as the storage unit for the gathered data, providing a secure and reliable storage solution for the information collected by the E-Bag.

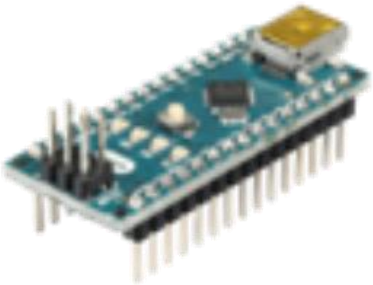


Figure 6. Arduino Nano

Arduino Nano serves as sensor reader in E-Bag System, it collects data from the various sensors integrated into the bag. It acts as the interface between the sensors and the Raspberry Pi 3b, transmitting the collected data to the main processing unit for further analysis and decision-making. Through feedback loop, the Arduino Nano provides real-time information to the Raspberry Pi 3b, enabling the system to respond and adapt based on the sensor readings, enhancing the overall functionality and responsiveness of the E-Bag.

Figure 7. GSM module

GSM module plays a vital role in sending SMS notifications to the user, providing valuable information about the location of the E-Bag. By integrating the GSM module, the E-Bag can communicate with cellular networks, enabling it to send location updates to the user's mobile phone via SMS. This feature enhances the security and convenience of the E-Bag, allowing users to track and monitor the bag's whereabouts in real-time.

Figure 8. HX711 Weight Sensor Amplifier

The HX711 Weight Sensor integrated into the E-Bag system serves a crucial purpose by determining the weight of the items inside the bag when it is in armed mode. This feature helps users detect any unnoticed or unauthorized additions or removals of items from the bag. By monitoring the weight, the system can provide users with immediate feedback or notifications, ensuring they are aware of any potential changes to the bag's contents, enhancing security and minimizing the risk of theft or tampering.

Figure 9. GPS module

The E-Bag system's integrated GPS module gives precise, real-time information on the location of the bag. The GPS module allow users to track the E-Bag's movements via satellite positioning, guaranteeing they always have visibility and awareness of its location. This feature improves security, enables immediate retrieval in the event of loss or theft, and gives users peace of mind by making it simple to pinpoint the exact position of their E-Bag at any time.

Figure 10. 16GB Micro SD Card

The SD card in the E-Bag system serves as a removable and expandable storage solution, enabling secure storage of data collected by the bag's sensors. It provides ample capacity to store various types of data, including GPS coordinates, sensor readings, and images. The SD card's flexibility allows for easy data transfer, analysis, and backup, enhancing the functionality and usability of the E-Bag system.

Figure 11. Limit Switch

In E-Bag, Limit Switch serves as a trigger point that starts the activation of other sensors. When the Limit Switch is triggered, it sends a signal to the system, indicating a specific condition or event. This signal acts as a cue for other sensors to activate, enabling them to perform their designated functions and contributing to the overall functionality and responsiveness of the E-Bag.

Figure 12. Piezoelectric Buzzer

The Piezoelectric Buzzer acts as an anti-theft alarm and warns the user audibly, is a significant component of the E-Bag system. The Piezoelectric Buzzer generates a loud and attention-grabbing sound when certain conditions, including unauthorized access or tampering, it immediately alerts the user and anybody nearby to potential theft or suspicious activity. This feature makes the E-Bag more secure by discouraging potential theft and giving users an effective way to protect their belongings.

Software:

Figure 13. MIT App Inventor

MIT App Inventor is a platform that was used to develop a mobile application that enables users to control and maneuver the E-Bag. Through its user interface and drag-and-drop

functionality, MIT App Inventor allows for the creation of a user-friendly mobile app that can interact with the various features and functionalities of the E-Bag system.

Figure 14. Geany IDE

The HTML, PHP, CSS, and JavaScript elements of the E-Bag User Interface are programmed using the versatile software platform Geany IDE. Geany IDE offers a reliable development environment for developing and executing the visual components and functionality of the E-Bag User Interface due to its user-friendly interface and substantial code editing features.

Figure 15. Python 2.7 IDE

The software environment used for writing, compiling, and uploading the code that integrates with the E-Bag system is Python 2.7 IDE. Python 2.7 IDE offers a reliable development environment for developing the E-Bag system's functionality.

Figure 16. Hermit

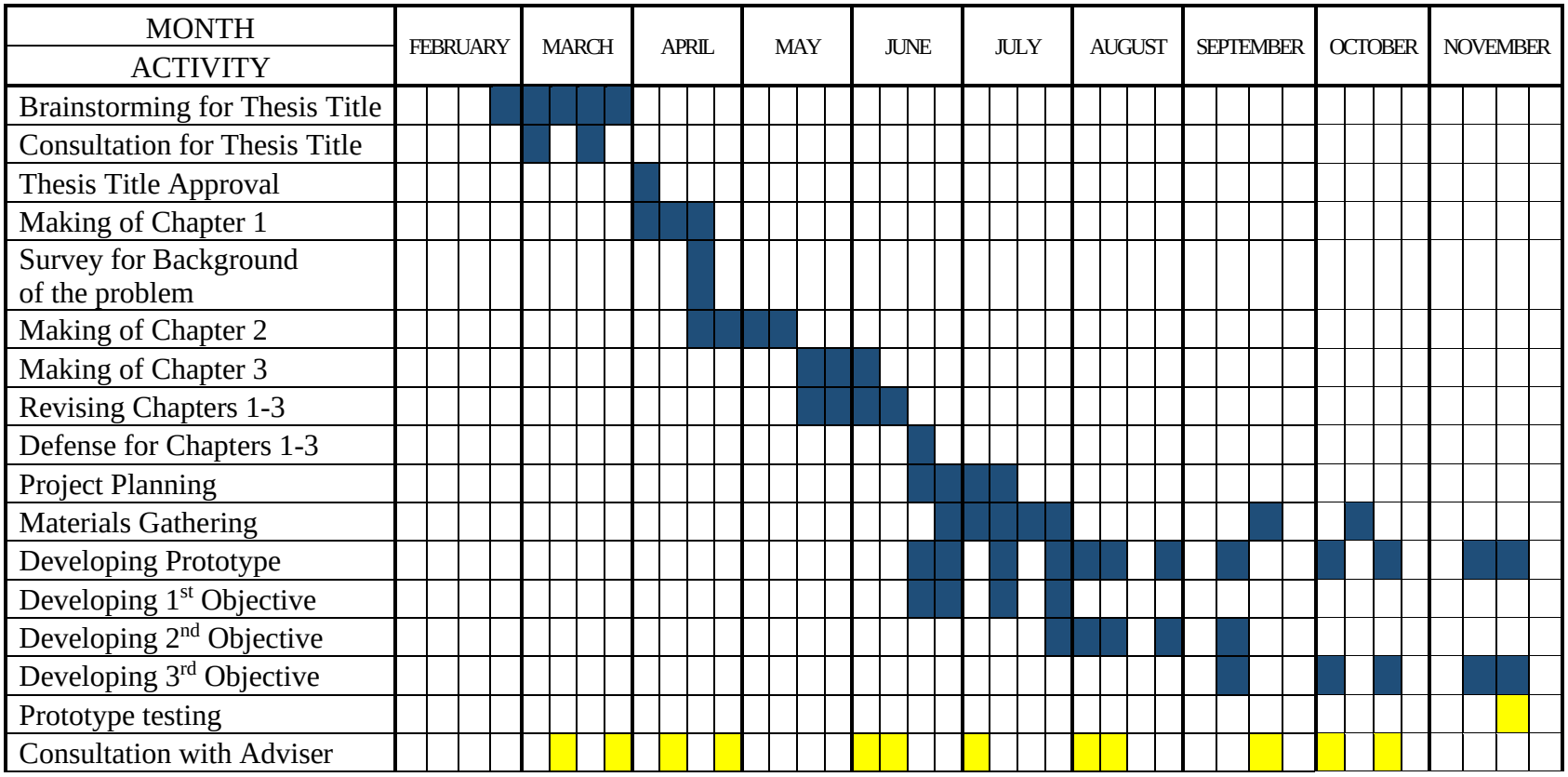
Hermit, an open-source and cross-platform tool, is utilized by the proponents to execute the mobile application developed for the E-Bag system. This versatile tool enables seamless integration and smooth functionality across various mobile platforms, ensuring compatibility and accessibility for users.

Figure 17. Web Browser (Google Chrome)

Web browser is essential for viewing and maneuvering the system's program. It allows users to access and interact with the E-Bag system through a user-friendly interface accessible from various devices. A reliable web browser ensures seamless navigation, control, and monitoring of the system, providing users with convenient access to the E-Bag's features and functionalities.

Calendar of Activities

Gantt chart of Activities



MONTH / ACTIVITY	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER
Finalizing Document for Capstone 1										
Defense for Capstone 1										
Consultation with Adviser										
Refinement of Prototype										
Prototype Testing										
Making of Chapter 4										
Evaluation of Prototype										
Making of Chapter 5										
Finalizing Document for Capstone 2										
Defense for Capstone 2										

Table 1. Gantt Cart of Activities from February to May

Budgetary Estimate

Quantity	Specifics	Approximate Cost
1pc	Arduino Nano	₱ 450
1pc	Raspberry Pi 3 Model B	₱ 4,000
1pc	Power bank with 10000mAh	₱ 1,100
1pc	Japanese Randoseru Bag	₱ 1,200
1pc	2MP USB Camera	₱ 500
1pc	HX711 Weight Sensor	₱ 1,500
1pc	Piezo Electric Buzzer	₱ 250
1pc	GSM Module	₱400
1pc	GPS Module	₱300
1pc	On/Off Push Button	₱29
4pc	Whole Sintra Board	₱400
1pc	Pocket Wi-Fi	₱750
1pc	Limit Switch	₱80

Table 2. Budgetary Estimate of E-bag: An Electronic Backpack with GPS Locator and IOT Technology

The researchers develop a budget plan to ensure the proper allocation of funds throughout the entirety of the project. The researchers list the resources required for the project from start to finish and calculate the costs of necessary modules and components through online stores. The researchers evaluate various online retailers and compare prices to identify the most cost-effective options. Furthermore, they assess the functionality of the items purchased and evaluate prior customer ratings to ensure satisfactory quality.

CHAPTER IV

METHODOLOGY, RESULTS, AND ANALYSIS

This chapter provides a detailed account of how the study was conducted and what results were obtained, as well as an analysis of the data collected regarding the development of E-bag: An Electronic Backpack with GPS Locator and IOT Technology. It includes the methods used in the study's development, such as software development, project prototyping, and customer needs assessments. A storyboard is provided to further clarify the basis for the understanding between the E-bag users and the software developers concerning the E-bag system features. The data collected during the study will present using tables, graphs, and other visual aids. Through analysis, the researcher interprets the results of the study and provides an explanation for the findings.

Requirement analysis

The E-bag: An Electronic Backpack with GPS Locator and IOT Technology is an innovative project and useful solution to a common problem faced by backpack users. It is designed to help users regarding possible security issues in their backpacks such as theft. The project's focus on addressing security issues in backpacks is particularly relevant in the current context, where people are increasingly relying on backpacks to carry their valuables, such as laptops, smartphones, and other electronic devices. There are a number of instances in which individuals using backpacks have been the victims of theft. One of the thief's strategies is to manipulate the bag since it is placed on their back and may go unnoticed, thereby taking advantage of others who carry backpacks. The researchers initiate the project at #340 Maginoo St., Tondo, Manila City where the client of the project was live. E-bag: An Electronic Backpack with GPS Locator and IOT Technology offers a variety of capabilities that can prevent these potential robbery scenarios. The E-bag's system works as a solution that is both practical and effective in preventing theft and provides users with a sense of comfort regarding the safety of backpacks and valuables.

Research Method

Prototype Model

The researchers' use of the prototype model methodology allowed them to develop a functional project by following a structured process. The seven phases of the prototype model they used, starting with requirements gathering and ending with engineering the final product, ensured that they obtained all the necessary functions and features of the project. By including customer evaluation and refinement in their process, the researchers were able to gather feedback from the target client and make necessary improvements to its software, resulting in a better-quality product.

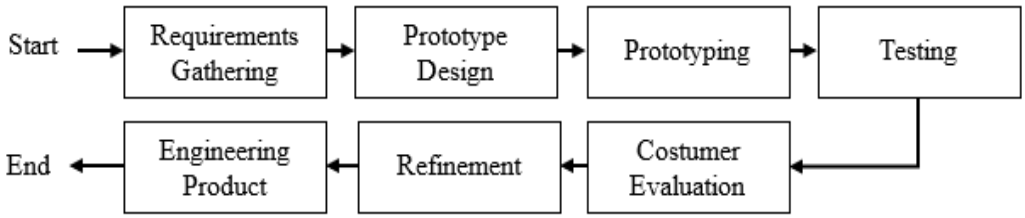


Figure 18. Prototype Model

The researchers employed the Prototyping Model in order to create E- Bags: An Electronic Backpack with GPS Locator and IOT Technology. The prototype model provided a framework that facilitated efficient and effective software development. Through multiple revisions to adapt to the changing needs of the developers. During this process, modifications may be made to either the hardware or software components, including adding or removing certain elements and adjusting software functions. The device's alterations will primarily result from the researchers' experimentation and identification of errors, with the aim of improving its effectiveness. The findings and results obtained from testing the device will guide the researchers' decision-making throughout the development cycle.

Requirements Gathering

In the E-bag project's requirements gathering, the researchers identify and document the software's functional and non-functional requirements such as determining the objectives, collecting input from the potential users of the project, analyzing existing software and systems that are related to the researcher's project, and conducting market research to understand user needs and preferences. The requirements gathered during this phase serve as the foundation for the subsequent phases of the prototype model, and any changes or modifications to the requirements are documented throughout the development process. Effective requirements gathering is essential for the successful development of a functional and user-friendly product.

Prototype Design

In the prototype design phase of the E-bag project, the researchers create a preliminary design that involves identifying the key features and functionality of the E-bag project and its system. The researchers aim to develop a functional prototype that can be tested and refined through multiple iterations to ensure that it meets the desired specifications and user requirements. The prototype design phase is critical for ensuring that the final product meets the needs of users, and it is both functional and user-friendly.

Prototyping

In the Prototyping phase of the E-bag project, the researchers plan the materials needed and then test them with the model of the system creating a preliminary version of the final product. The prototype is used to test and refine the prototype design to identify any issues or areas for improvement. The researchers aim to create a functional prototype that closely resembles the final product and can be used to validate the design.

Testing

In the Testing phase, the E-Bag prototype model was used to test its functionality and usability. This may involve conducting user testing to gather feedback and

insights from potential users, as well as performing technical tests to ensure that the product works as intended. It helps the project to reduce costs and minimize the risk of failure.

Customer Evaluation

In the Customer evaluation phase of the E-bag project, the researcher's finished prototype is tested and evaluated by the users to identify any issues or areas for improvement. Feedback is gathered from users through surveys, interviews, and observation, and is used to refine and optimize the design. This iterative process helps to ensure that the E-bag project meets the needs of the users and is user-friendly and effective.

Refinement

The refinement phase refers to improving and optimizing the E-bag project prototype through multiple iterations. The researchers gather feedback from users and used it to refine the design, addressing any issues or problems and optimizing the functionality and user-friendliness of the E-Bag project. The refinement process includes making adjustments to the design, adding new features or functionality, and optimizing existing features. By continuously refining the design, researchers can optimize the functionality, user-friendliness, and overall effectiveness of the product resulting in a better-end product.

Engineering Product

The Engineering Product phase is crucial for the success of the project, as it ensures that the final product is optimized for performance, user-friendliness, and efficiency. During the Engineering Product phase of the E-Bag project, the researchers undertake a final evaluation of the product to ensure that it meets the desired specifications and requirements. This phase involves rigorous testing and updates to identify and address any weaknesses or failures in the design, ensuring that the final product is a better-end product and functional. This will be done if the project meets the desired specifications and requirements.

Summary

In addition to outlining the objectives, software programs, materials, and hardware that will be used in the development of E-bag: An Electronic Backpack with GPS Locator and IOT Technology, the methodology prototype model provides a detailed plan for how the project will be executed. This includes a step-by-step approach to the development process, outlining the different phases and milestones that will be achieved. The prototype model also allows for testing and refinement of the product, ensuring that it meets the needs and expectations of potential users. Additionally, the flow of the plan is discussed in detail, including how different components of the project will be integrated and how the final product will be delivered. This chapter provides a comprehensive overview of the project and its development process, to provide a clear understanding of the scope and objectives of the E-bag project.

Requirement Documentation

Orthographic Projection of the Prototype

Figure 19. The Outer view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

Figure 19. shows the external appearance of the prototype model E-bag: An Electronic Backpack with GPS Locator and IOT Technology. It is the outermost layer or surface of the E-bag's system. It represents what the E-bag's system will look like from the outside when in use or being worn.

Figure 20. The Inner view and Top View of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology

Figure 20. provides an internal view of the prototype model and the top view of the system of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology. It shows the contents and organization of the backpack when it is opened by the user.

Figure 21. The Front view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System.

Figure 21. represents the front view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System. Figure 21 provides information on the construction of the front-facing components of the E-bag's system. This includes details

on how the Piezoelectric Buzzer, USB Camera, and IR sensor were built and integrated into the front of the E-bag's system.

Figure 22. The Right-side view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System.

Figure 22. shows the right-side view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System. It provides details on how the GPS module was integrated into the right side of the E-bag's system and how the right side of the E-bag's system was built.

Figure 23. The Left-side view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System.

Figure 23. provides the left-side view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System. It provides details on how the left side of the E-bag's system was built and used for cabling management.

Figure 24. The Back view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System.

Figure 24. shows the back view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System. It provides information about the construction and integration of several components into the back side of the E-bag's system. This includes details on how the indicator of the power bank, the backside of the USB camera, the Infrared sensor, GSM module, Limit switch and On/Off switch were built and integrated into the back of the E-bag's system. It provides information on the structure of the back side of the E-bag's system, which also includes a designated area for the user to place a pocket Wi-Fi.

Figure 25. The Bottom view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System.

Figure 25. provides the bottom view of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology System. It provides information about the construction and integration of several components into the bottom side of the E-bag's system. This includes details on how the power bank, Raspberry Pi 3b, Arduino Nano, and HX711 Weight sensor Amplifier were built and integrated into the bottom of the E-bag's system.

Wire Frame

Figure 26. The Login page of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The Login page of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology is an essential gateway to accessing the full suite of features available on this electronic backpack. As shown in Figure 26, this page provides a simple and user-friendly interface that enables the user to input their username and password securely. The default username and password are provided, allowing the user to quickly access their account. Once logged in, the user can change their username and password at any time, providing an additional layer of security and privacy. The Login page is just the beginning of a seamless and intuitive user journey that will enable the user to take full advantage of all that E-bag has to offer.

Figure 27. The Download page of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The Download page of E-bag: An Electronic Backpack with GPS Locator and IOT Technology, as depicted in Figure 27, features two prominent buttons that allow the user to either download the app or log out of the page. The first button, labeled "Download App Here," enables the user to access the E-bag app quickly and easily, which is available for download on both iOS and Android platforms. The second button, labeled "Logout," allows the user to log out of the page securely, ensuring that their information is protected and secure.

Figure 28. The Menu page of the Mobile App of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The Menu page of the E-bag mobile app, displayed in Figure 28, is the starting point for users to explore the various features of this advanced electronic backpack. It consists of five buttons, each providing a different function. The "History" button allows users to view details of their previous backpacking trips, including location, date and time, and even images captured inside the backpack. The "System Status" button offers real-time updates on the performance of GPS locator and IoT technology. The "Configuration" button allows the user to customize settings such as username and password, input mobile number of the owner for notification, and other features. The "User Manual" button offers a comprehensive guide to using the E-bag mobile app and E-bag. Finally, the "Logout" button enables users to securely log out of the app. The Menu page is designed to provide a user-friendly

and intuitive interface that enables the user to easily navigate and explore the full range of features and functions available on the E-bag electronic backpack.

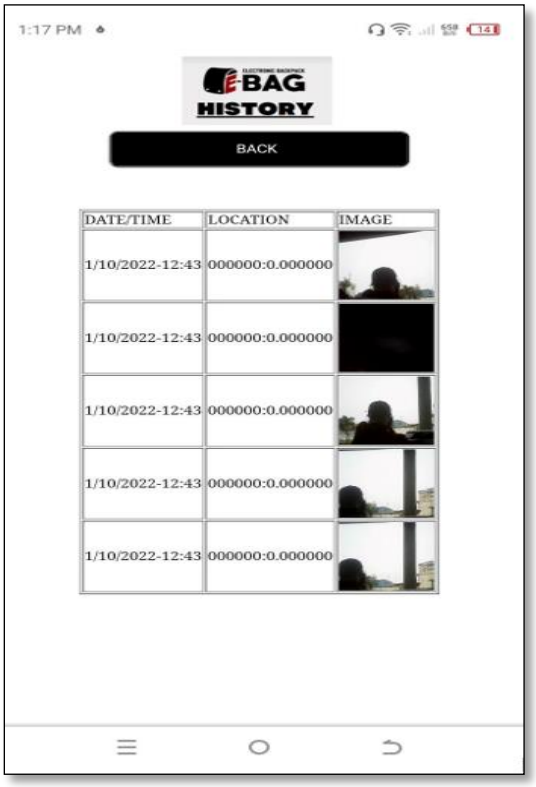


Figure 29. The History page of the Mobile App of E-bag: An Electronic with GPS Locator and IOT Technology.

The History page of the E-bag mobile app, displayed in Figure 29, provides valuable information to users about their previous trips. The stored data in the app includesthe date and time of each trip, the location of the bag using longitude and latitude, and an image captured when the bag is in "armed mode." With this information readily available on the History page, users can easily identify any unusual activity and take appropriate action, such as notifying authorities or tracking down lost or stolen personal belongings.

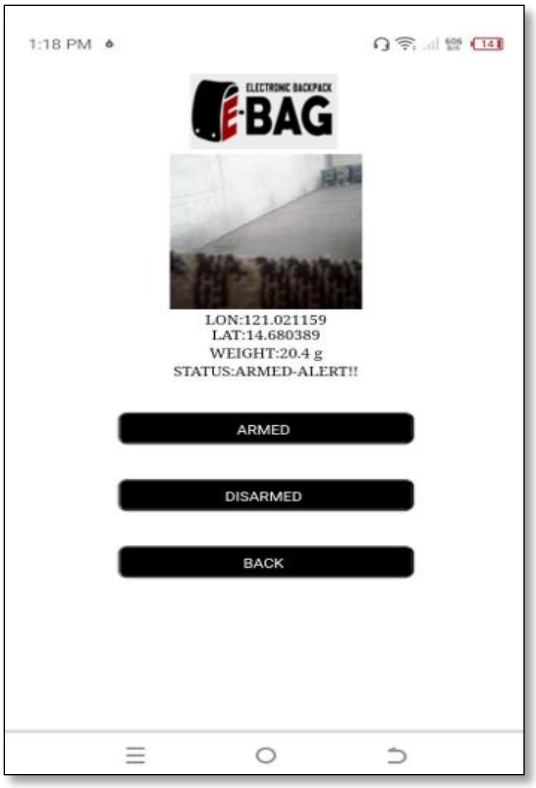


Figure 30. The System Status page of the Mobile App of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The system status page of the E-bag in figure 30 serves as a centralized location for users to monitor the functionality and security of their bag. With clear indicators for the camera function, users can quickly and easily confirm if the device is working correctly. Additionally, the page provides valuable information on the bag's location, including longitude and latitude data, helping users stay informed on the whereabouts of their bag. By displaying the weight of the bag and its status, whether it is armed or disarmed, users can keep track of their bag's condition and make informed decisions about when to use thedevice. The inclusion of an easily accessible button to activate and deactivate the device further simplifies the user experience and enhances the bag's overall security features.

Figure 31. The Configuration page of the Mobile App of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The Configuration page, as shown in Figure 31, offers users a range of options to modify their E-bag and tailor it to their specific needs. The page includes buttons that allowusers to change the default username and password, ensuring that their device is secure andprotected from unauthorized access. Additionally, users can set up a mobile number that will be notified if the E-bag is handled while in the "armed" mode, providing an added layer of security. The "Send SMS GPS" button allows users to test if the device can send notifications to their phone, ensuring that they can receive updates on the bag's location and status. The Configuration page also includes a function to test the anti-theft alarm andstop it when triggered, helping users to protect their bag and prevent theft. With a "shutdown" button to stop any process of the mobile app and a "reset data" button to clearthe history of the app, users can easily manage their device and ensure that it is functioning correctly.

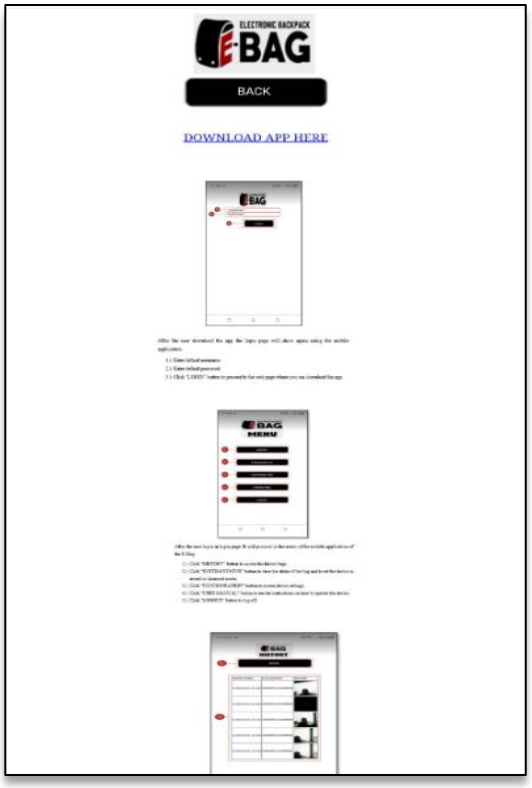


Figure 32. The User Manual page of the Mobile App of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The User Manual page of the E-bag mobile app, as shown in figure 32, provides users with a comprehensive guide to understanding and using the E-bag's various featuresand functionalities. With step-by-step instructions on how to set up the mobile app and connect it to the E-bag, even new users can easily navigate the setup process. The page also includes instructions on how to test the connection between the E-bag and the mobileapp, enabling users to confirm that the device is working correctly and that they can accessreal-time data.

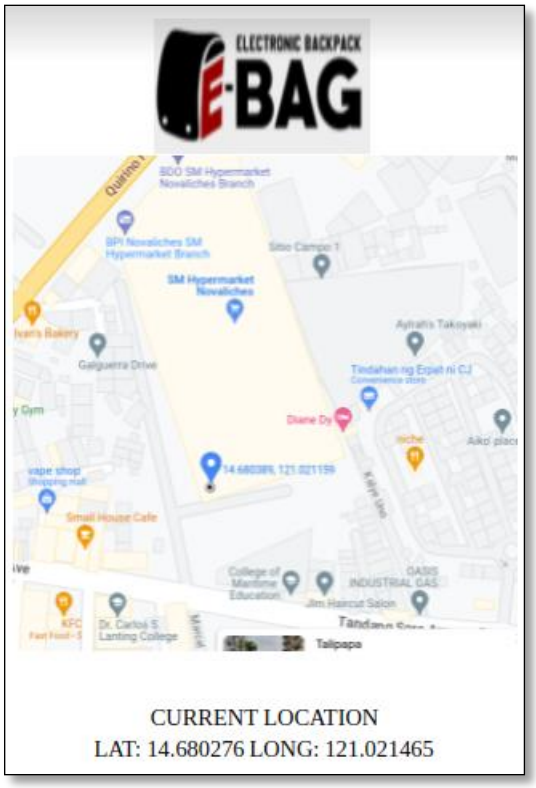


Figure 33. Map page of the Web Page of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The Map Page on the web page of the E-bag is feature for the users who needs to keep track of their backpack's location in real-time using map. With the use of GPS locator and IOT technology, the E-Bag's current location can be viewed accurately on the map page. Furthermore, the longitude and latitude information provided on the map page makes it easier for users to pinpoint the exact location of their E-bag.

User's Diagram

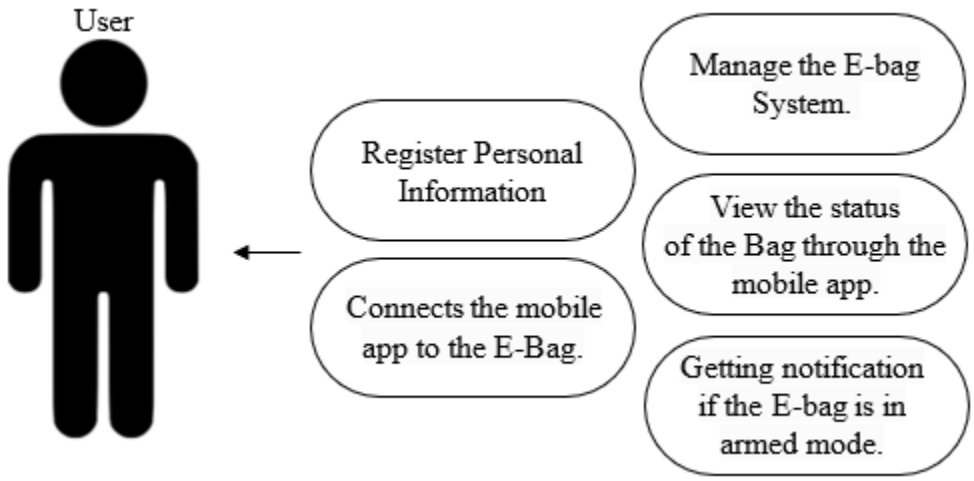


Figure 34. The User's Diagram of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

As shown in figure 34, The user diagram of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology is an essential tool for understanding how the system works and how users can interact with it. It focuses on the users of the E-bag. They are the ones who register their personal information and connect their mobile app to the E- bag using pocket Wi-Fi. This connection enables the user to view the status of the bag, including all stored data captured by the E-bag's system. The user is also responsible for managing the E-bag's system. They have the ability to activate the E-bag's system and set it up through the use of the mobile app. This provides users with complete control over the system and ensures that they can modify it to their individual needs.

Description of the prototype

Context Diagram

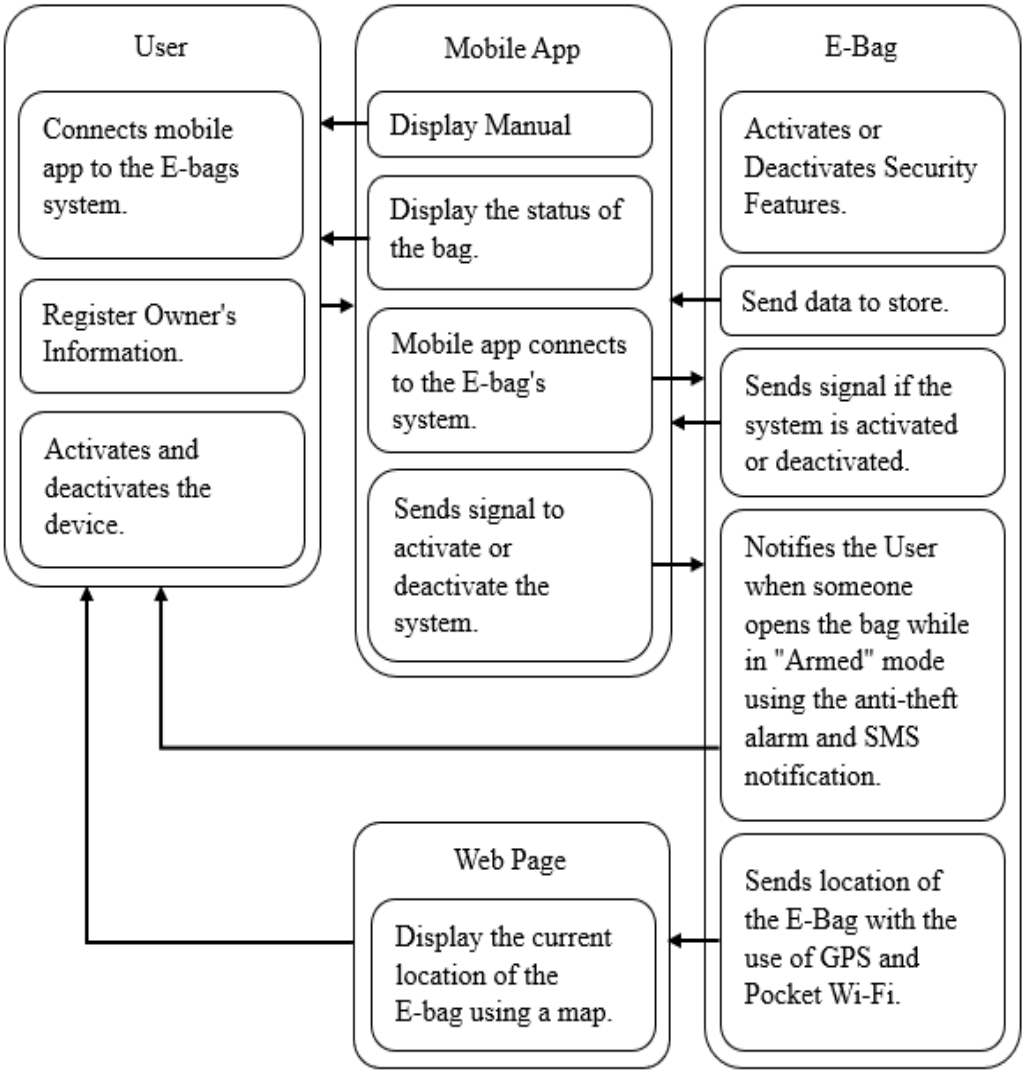


Figure 35. Context Flow Diagram of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The figure 35 shows the context flow diagram of the study which interprets how the data flows as a whole system. The system is composed of (3) three main components: the user, the mobile app, and the E-bag: An Electronic Backpack with GPS Locator and IOT Technology which is the E-bag's system. The user interacts

with the E-bag's system through the mobile app, which sends requests to the E-bag's system. The E-bag's system controls the backpack's armed and disarmed state with the use of a mobile app that is created for the E-bag's system. It receives signals from the weight sensor, and camera, and sends these signals to the mobile app to store in its database. The E-bag's System notifies the user when the bag is opened while it is in armed mode using an anti-theft alarm and SMS notification with the help of GPS. The SMS notification indicates the detailed location address.

Data Flow diagram

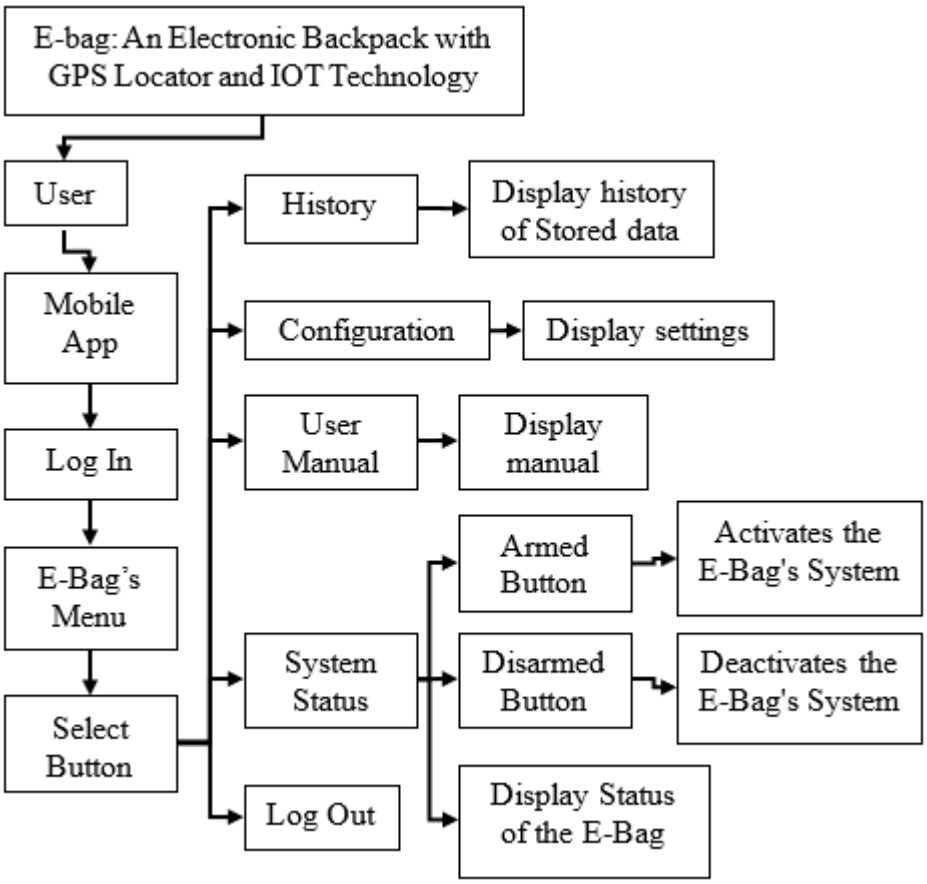


Figure 36. Data Flow Diagram of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The figure 36 illustrates the data flow diagram of E-bag: An Electronic Backpack with GPS Locator and IOT Technology, which demonstrates how data is transferred between the user, mobile app, and the E-bag's system. To use the mobile app, the user needs to register first. After successful registration, the user can log in to the mobile app. The mobile app verifies the user's login credentials and proceeds to display the manual of the app, which provides a step-by-step guide on how to use the app effectively. Once the user is familiar with the mobile app, they can proceed to the E-bag's Menu Section, which comprises five buttons. These buttons include the History button, where the user can view the captured images, weight, and status of the bag at different times. The Configuration button allows the user to set up the mobile app and check if it is working. The User Manual button provides additional information to the user about how the mobile app works. The System Status button allows the user to control the E-bag's system by selecting the "armed" or "disarmed" mode to activate or deactivate the E-bag's System. When the E-bag's system is activated, the security features are triggered. If the backpack is opened, the system captures images of the person who opened the bag using the integrated camera and sends them to the database through the pocket Wi-Fi. The weight sensor detects any changes in the backpack's weight and sends this information to the database as well. The GPS locator tracks the location of the backpack using the Internet of Things technology. The mobile app retrieves this information from the database and displays it to the user. If the system is triggered, the anti-theft alarm is activated, and the E-bag's system sends an SMS notification to the user indicating the location of the bag. Lastly, the user can log out of the mobile app by clicking the Logout button, which takes them back to the login form. In summary, the E-bag's system utilizes advanced technology to ensure the safety and security of the user's belongings. The mobile app provides a convenient and user-friendly interface for the user to monitor and control the E-bag's system.

Block Diagram

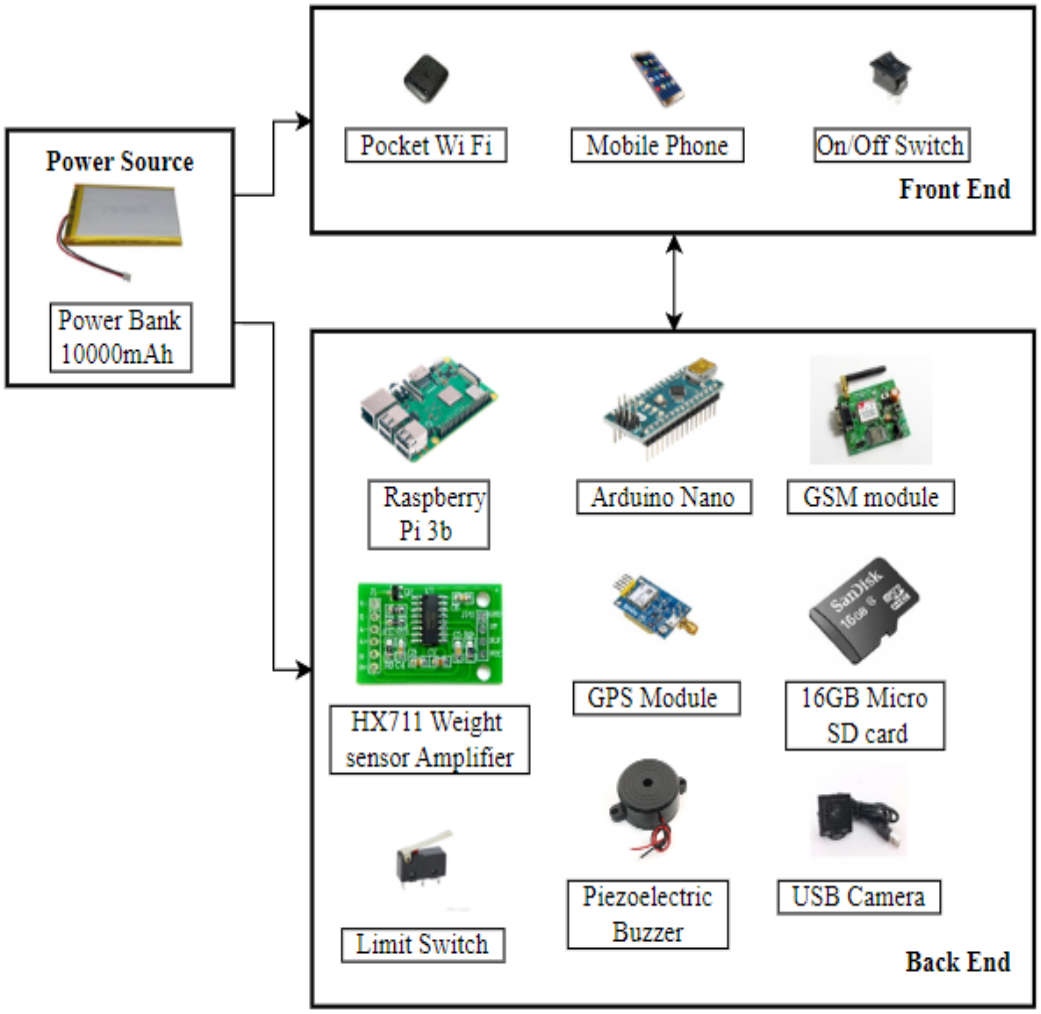


Figure 37.Block Diagram of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

The figure 37 depicts the Block Diagram of the E-bag: An Electronic Backpack with GPS Locator and IOT Technology. The power source of the device is a power bank with a capacity of 10000mAh that provides power to all the components mentioned in the diagram. The system comprises a Raspberry Pi 3B and an Arduino Nano, which are interconnected to various essential components such as the USB camera, GSM module, GPS module, HX711 weight sensor amplifier, 16GB Micro SD card, and Piezoelectric buzzer. The components are integrated to function as a cohesive unit,

enabling the backpack to perform its designated tasks. The E-Bag backpack is also equipped with an on/off switch to power on and off the device, and pocket Wi-Fi, which facilitates remote access and control of the prototype via a mobile phone. This feature allows the user to monitor the backpack's status, location, and other essential details from a distance, making it an efficient and convenient travel companion.

Hierarchical Diagram (User)

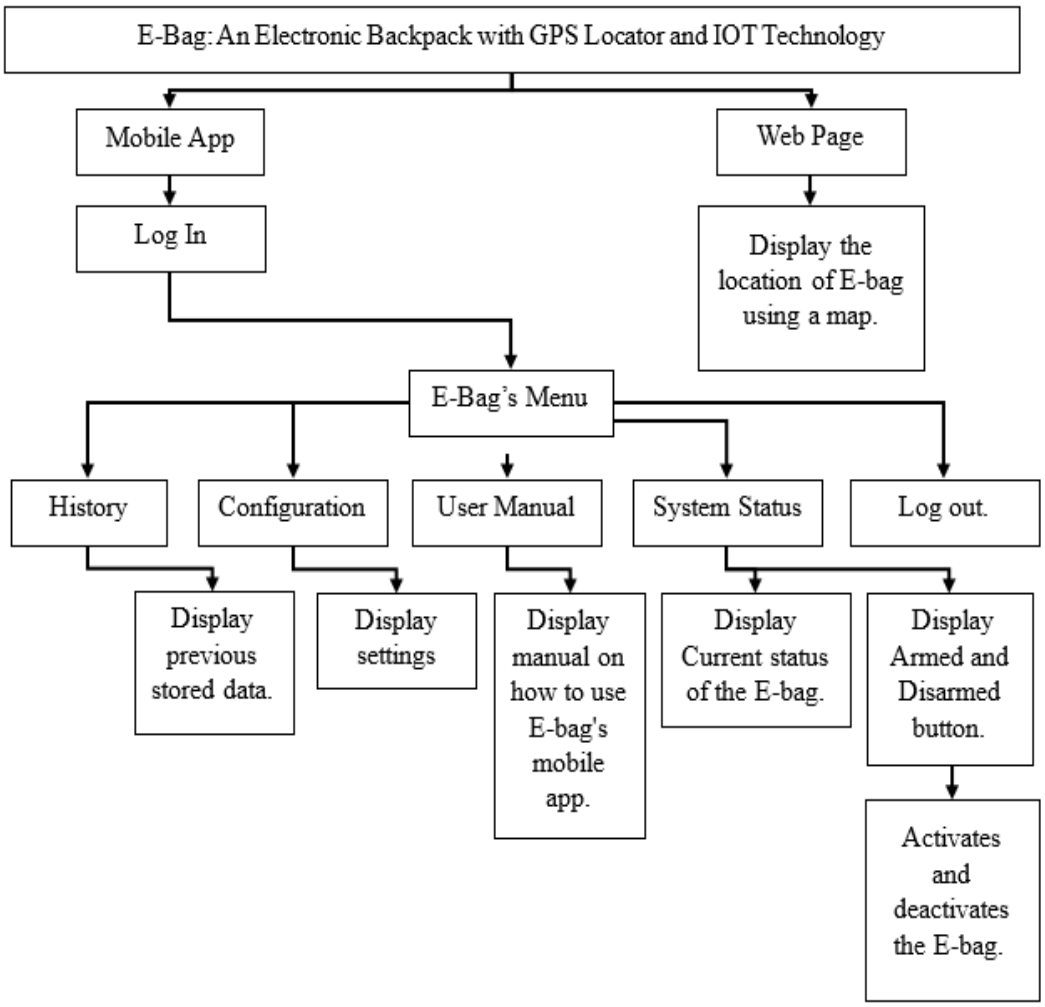


Figure 38. Hierarchical Diagram of the User of E-bag: An Electronic Backpack with GPS Locator and IOT Technology.

In Figure 38 shows the Hierarchical Diagram of the User of E-bag: An Electronic Backpack with GPS Locator and IOT Technology. The E-bag system provides a

range of security features to ensure the safety of the user's belongings. The mobile app allows the user to access the E-bag's various buttons such as History, Configuration, User Manual, and System Status buttons. When the E-bag's system is armed, it captures images of anyone who opens the bag using the integrated camera and sends them to the database through the pocket Wi-Fi. The weight sensor detects changes in the backpack's weight and sends this information to the database. The GPS locator tracks the location of the backpack using the Internet of Things technology. The mobile app retrieves this information and displays it to the user. If the system is triggered, the anti-theft alarm is activated, and the E-bag's system sends an SMS notification to the user indicating the location of the bag.

Data Dictionary

Table Name: <u>Table of main controller function</u>

No	Entity	Description	Data type	Size	Null	Auto Inc	Character	Example
1	**inyGPSPlus gps;**	GPS **declaration**	**string**	**55**	**no**	**no**	a-z **A-Z 0-9**	**ewr23r**
2	**#define calibration_ factor - 7050.0**	This value is obtained using the SparkFun_ H X711_Cali bration sketch	**string**	**55**	**no**	**no**	**a-z A-Z 0-9**	**234ewr**
3	**#define LOADCEL L_DOUT_ PIN 2**	**definition of pin 2 as sda**	**string**	**55**	**no**	**no**	**a-z A-Z 0-9**	**2**
4	**#define LOADCEL L_SCK_PI N 3**	**definition of pin 2 as sck**	**string**	**55**	**no**	**no**	**a-z A-Z 0-9**	**3**
5	String	a String to hold	string	55	yes	no	a-z A-Z	24

	inputString = "";	incoming data					0-9	
6	**int startcapture=0;**	variable for start execution	int	11	no	no	0-9	1
7	String cellnumber ="+639261 **234567"**	variable for mobile number	string	55	no	no	a-z A-Z 0-9	92423 4 52324
8	**int warning=0;**	variable for warning	int	11	no	no	0-9	1
9	**String warning1=" ";**	variable for string store	string	55	no	no	a-z A-Z 0-9	e
10	**String switch1="E ";**	variable for switch	string	55	no	no	a-z A-Z 0-9	r
11	**String switch2="G ";**	variable for switch	string	55	no	no	a-z A-Z 0-9	e
12	int delayloop= 0;String **switch2="G ";**	variable for switch	int	11	no	no	0-9	f
13	int armed=0;String switch2=" G ";	variable forswitch	int	11	no	no	0-9	g
14	SoftwareSerial gpsSerial(4 ,5);	rx, tx	string	55	no	no	a-z A-Z0-9	2
15	SoftwareSerial gsmSerial(6 ,7);	rx, tx	string	55	no	no	a-z A-Z0-9	3
16	HX711 scale;	declaration of amplifier	string	55	no	no	a-z A-Z0-9	2
17	const int buttonPin = 8;	the number of the	int	11	no	no	0-9	2

		pushbutton pin						
18	int buttonStat e = 0;	variable forreading the pushbutton status	int	11	no	no	0-9	5
19	const int buttonPin 1 = 9;	the number of the pushbutton pin	int	11	no	no	0-9	12
20	int buttonState 1 = 0;	variable forreading the pushbutton status	int	11	no	no	0-9	2

Table 3. Table of the main controller function

Development and Testing

The researchers planned to do the study based on the experiences of backpack users. By gathering data and insights from backpack users, the researchers were able to gain a thorough understanding of the challenges and issues associated with traditional backpacks. This information was further supplemented by online readings, meetings, and consultations with the adviser and experts, who provided valuable insights and suggestions for the development of the proposed system. As a result of this collaborative effort, the researchers were able to design a prototype which is E-Bag: An Electronic Backpack with GPS Locator and IOT Technology that provides users with a more secure and convenient way to carry their belongings while on-the-go.

Survey Questionnaire

To evaluate the effectiveness of the prototype, the researchers intend to conduct a survey among 50 participants. The survey will contain 10 questions related to the prototype's features and objectives, and the respondents will be requested to rate their level of agreement or disagreement with each statement based on their experience using the prototype. The researchers will use the Likert's scale to

collect the participants' responses, which will enable them to measure the feedback in a standardized and structured manner. The data gathered from the survey will be used to identify areas of the prototype that require improvement and to further develop the technology to better cater to the users' needs.

Interview

The researchers conducted an interview with a backpack user. The primary objective of the interview was to obtain the client's feedback regarding their experience using the E-bag prototype. The researchers will seek the client's permission to use their feedback for the betterment of the E-bag project.

The development process of the E-Bag: An Electronic Backpack with GPS Locator and IOT Technology presented a number of challenges to the researchers. Creating the prototype proved to be the biggest obstacle, as it required a specific set of skills and knowledge to successfully complete. The researchers had to rely on various resources, such as online readings and consultations with advisors and experts, to overcome this hurdle. However, identifying trustworthy sources proved to be a challenge as well, as the researchers needed to carefully select and cite reliable resources.

Apart from this, the researchers also faced difficulties in budgeting for the necessary equipment required for the project, as well as in interpreting technical manuals for executing the necessary steps on the device. Despite these challenges, the researchers persevered and explored various resources to overcome these hurdles and ultimately develop the E-Bag: An Electronic Backpack with GPS Locator and IOT Technology.

User Testing

The developers decided to test the system and prototype before conducting it to the potential users. User testing is an essential step in ensuring that the E-Bag: An Electronic Backpack with GPS Locator and IOT Technology meets the needs of their intended audience and is user-friendly and effective.

Backpack User

The researchers enlisted (2) two backpack users to conduct interviews and test the prototype. This allowed the researchers to evaluate whether the system and prototype effectively achieved the study's objectives, which were based on three key goals.

Design of Software, system, Product, and/or Processes

In order to conceive the idea and design of the E-Bag's system, the researchers utilized a variety of tools, methods, and requirements available on the internet andrecommended by the project adviser. The prototype model was developed using a set of ordered phases and serves as a guide for the researchers to achieve their objectives. The wireframe was created to help the users to convey to the E-bag's system, while the contextdiagram was used to illustrate the interaction between the mobile app, the E-bag's system,and the users.

To provide a more detailed understanding of the system's processes and activities,the researchers employed a data-flow diagram, which identified the inputs and outputs of each process. The block diagram was used to provide a visual representation of the overall structure of the system and the relationship between its different components. Furthermore, a hierarchical diagram was created to better understand how the E-Bag's system works and how it can benefit the users in terms of security and convenience.

Throughout the development process, the researchers relied on trial and error, as well as guidance from a dependable project adviser, to refine the software and prototype design. By utilizing a range of tools and methods, the researchers were able to create a comprehensive and effective system that addresses the security concerns of backpack users.

Implementation Plan

The researchers plan to execute the E-bag project, which involves the creation of an electronic backpack equipped with GPS locator and IOT technology to enhance user

security and safety. The implementation procedure includes multiple consultations with the project adviser, survey questionnaires, and interviews, all with the approval of the executive dean and program head. Additionally, a video presentation will be created to demonstrate how the E-bag system works. The researchers' primary objective during the implementation of the E-bag project is to receive positive feedback from users, who should find the mobile app interface user-friendly and simple to navigate. The respondents must also acknowledge that the E-bag system is fully functional and provides enhanced security and safety features. The prototype will be deemed successful based on these factors.

Survey Questionnaire

The researchers plan to administer a survey questionnaire to 50 participants,using Likert's scale to measure their responses. The questionnaire will consist of 10 questions about the prototype's features and objectives, and the respondents will beasked to rate their agreement or disagreement with each statement based on their experience with the prototype.

Interview

The researchers of E-bag: An Electronic Backpack with GPS Locator andIOT Technology will conduct an interview with a client who is located near them.The purpose of the interview is to gather feedback from the client about their experience using the E-bag prototype. The researchers will also ask the client for approval to use their feedback for the improvement of the E-bag project.

Implementation Results and Discussion

This section focuses on presenting the findings of a survey questionnaire that used Likert's scale to measure the extent to which respondents strongly agree or disagree with certain statements. The survey questionnaire was designed specifically to gather information about E-bags: An Electronic Backpack with GPS Locator and IOT Technology.

The researchers used a numerical rating system for data analysis where participants could rate their responses on a scale from 1 to 5, with 5 being strongly agree and 1 being strongly disagree. The section provides tables that summarize the participants' responses for each question, and the researchers calculated the average score for each question based on these ratings. Below is the complete formula that the researchers used.

$$x_n = Total\ Response\ of\ Respondents\ in\ every\ scale,$$

$$n\ is\ the\ number\ of\ Scale$$

$$y = Total\ Number\ of\ Respondents$$

$$m_a = mean\ of\ the\ response,$$

$$a\ is\ the\ number\ of\ question$$

$$m_a = (x_5 * 5) + (x_4 * 4) + (x_3 * 3) + (x_2 * 2) + (x_1 * 1) / y$$

To get the overall mean of the responses,

$$Overallmean = m_1 + m_2 + m_3 + m_4 + m_5 + m_6 + m_7 + m_8 + m_9 + m_{10} / y$$

The researchers use the 5-Points Likert's Scale. Define, (reference)

Scale	Range	Verbal Interpretation
5	5.0-4.21	Strongly Agree
4	4.20-3.41	Agree
3	3.40-2.61	Neutral/Uncertain
2	2.60-1.81	Disagree
1	1.80-1.00	Strongly Disagree

Table 4. Likert's Scale

(Kees V.H., & Kees V.O., 2012)

Survey Results from All Participants

	Strongly Agree (5)	Agree (4)	Neutral (3)	Disagree (2)	Strongly Disagree (1)	Weighted Mean	Verbal Interpretation
A. User							
1. E-Bag provide protection for belongings through GPS locators, alarm systems, and safety cameras integrated into the backpacks	27	22	1	0	0	4.52	Strongly Agree
2. E-bag can be monitored its tracking and other innovative capabilities via SMS and mobile app.	23	27	0	0	0	4.46	Strongly Agree
3. IOT technology in E-bag provides peace of mind while traveling by enhancing security when leaving belongings unattended.	19	24	7	0	0	4.24	Strongly Agree
4. The E-bag's location address can be displayed via SMS if it is lost, stolen, or misplaced	23	22	5	0	0	4.36	Strongly Agree
5. E-bags can capture an image of anyone who has handled the backpack using a safety camera.	21	24	5	0	0	4.32	Strongly Agree

6. E-bags can alert the user through security features if someone else handles the bag.	18	25	7	0	0	4.22	Strongly Agree
7. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app is easy to understand and use.	19	23	10	0	0	4.34	Strongly Agree
8. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can manage the backpack security system.	23	24	3	0	0	4.4	Strongly Agree
9. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can track and monitor the backpack when it is in armed mode.	22	24	4	0	0	4.36	Strongly Agree
10. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology is user-friendly, convenient, and useful.	25	21	4	0	0	4.42	Strongly Agree

Table 5. Total of Backpack User Testing Result.

Summarized findings per survey question based on the total number of respondents.

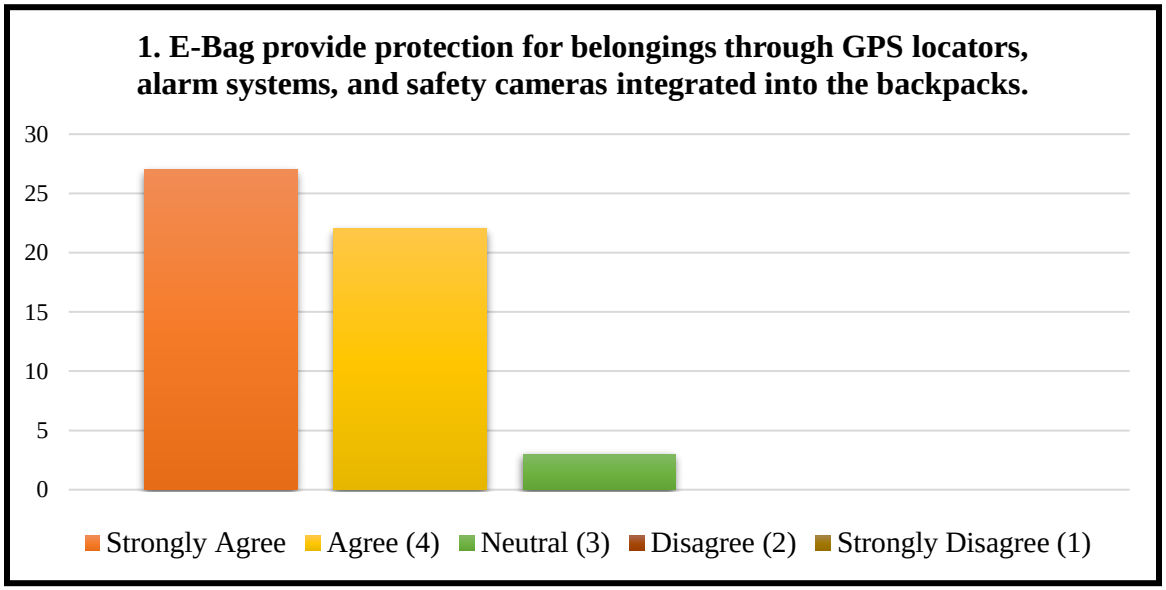


Figure 39. Findings from the 1st Question

The data presented in Figure 39 reveals that most of the participants, about 98%, either strongly agreed (54%) or agreed (44%) that the E-Bags, which have GPS locators, alarm systems, and safety cameras built into them, can effectively safeguard belongings. Only a small portion of the respondents (6%) chose the neutral option. The data indicates a weighted mean of 4.52, which corresponds to a verbal interpretation of "strongly agree".

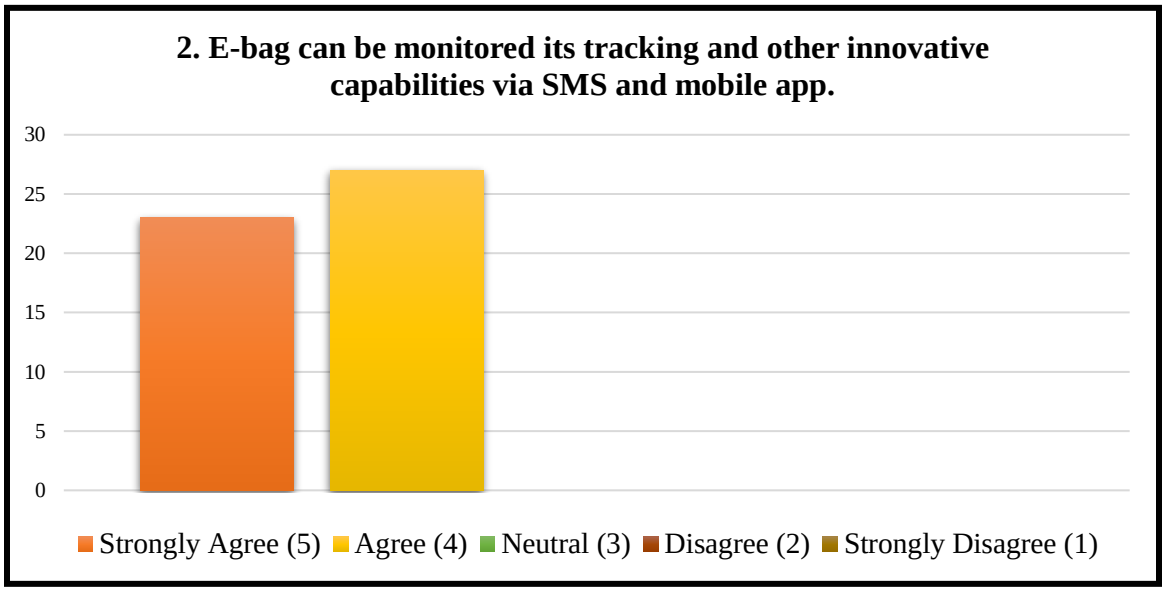


Figure 40. Findings from the 2nd Question

According to the data presented in Figure 40, all of the participants, which represents 100% of the sample, either strongly agreed (46%) or agreed (54%) that the E-Bags, which are equipped with GPS locator and IOT technology, are capable of being monitored through SMS and mobile app with additional innovative features. The weighted mean of 4.46 suggests that the participants' responses correspond to a strong agreement.

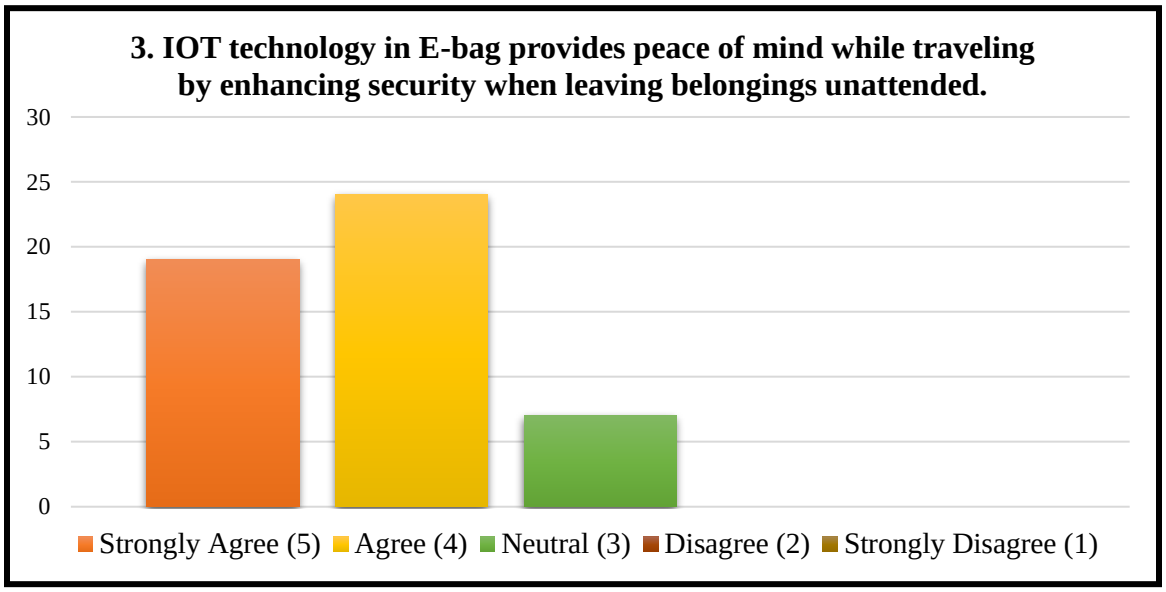


Figure 41. Findings from the 3rd Question

The information displayed in Figure 41 indicates that a large majority of the respondents, accounting for 86% of the total sample, expressed either strong agreement (38%) or agreement (48%) that the E-Bags, an electronic backpack equipped with GPS locator and IOT technology, can increase security and provide peace of mind while traveling. This is achieved through IoT technology that enhances security when leaving belongings unattended. On the other hand, the remaining respondents chose the neutral option, which represents 14% of the sample. The calculated weighted mean of 4.24 indicates that the participants' overall responses can be interpreted as a strong agreement.

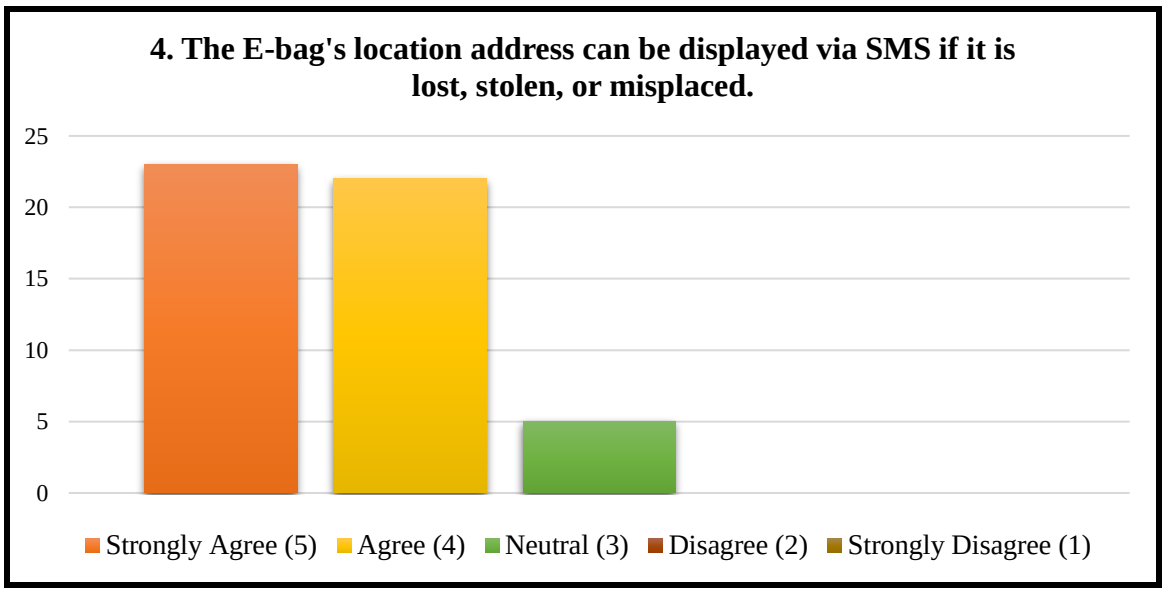


Figure 42. Findings from the 4th Question

The majority of the respondents (90%) in Figure 42 indicated strong agreement (46%) or agreement (44%) that E-Bags: An Electronic Backpack with GPS Locator and IOT Technology can display E-bag's location address via SMS if it is lost, stolen, or misplaced. While (10%) chose the neutral option. The data indicates a weighted mean of 4.36, which corresponds to a verbal interpretation of "strongly agree".

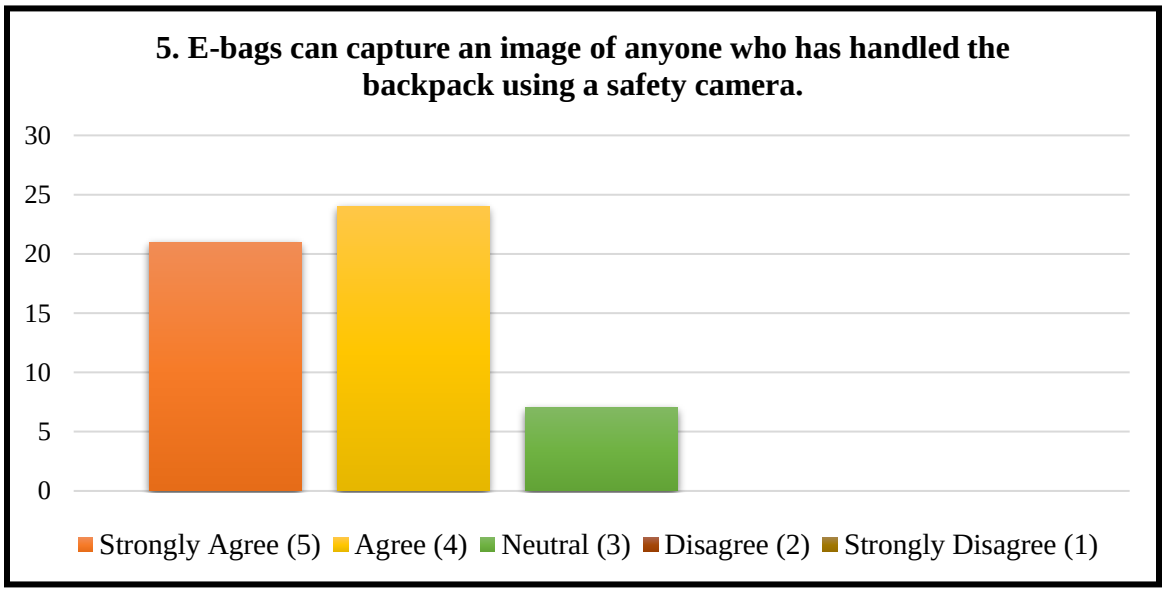


Figure 43. Findings from the 5th Question

In Figure 43 The table shows that the majority of the respondents (90%) either strongly agreed (42%) or agreed (48%) that the E-Bags: An Electronic Backpack with GPS Locator and IOT Technology can capture an image of anyone who has handled the backpack using a safety camera. Only a small percentage (10%) chose the neutral option. The data indicates a weighted mean of 4.32, which corresponds to a verbal interpretation of "strongly agree".

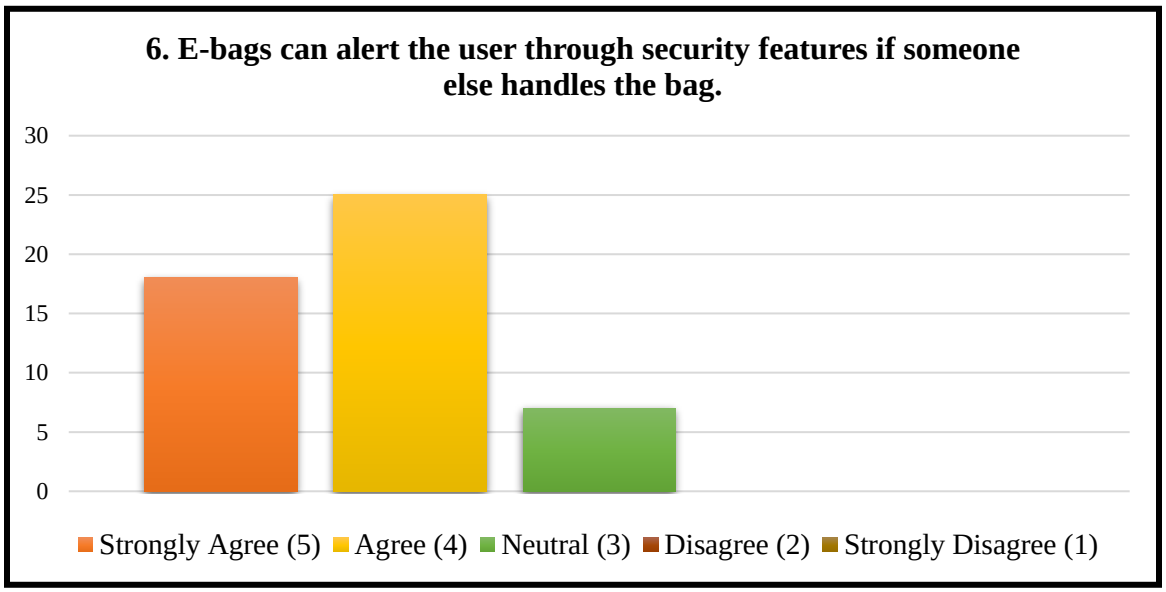


Figure 44. Findings from the 6th Question

Figure 44 illustrates that most of the respondents (86%) expressed either strong agreement (36%) or agreement (50%) towards the capability of E-Bags: An Electronic Backpack with GPS Locator and IOT Technology to alert the user through security features if someone else handles the bag. While (14%) chose the neutral option. The data indicates a weighted mean of 4.22, which corresponds to a verbal interpretation of "strongly agree".

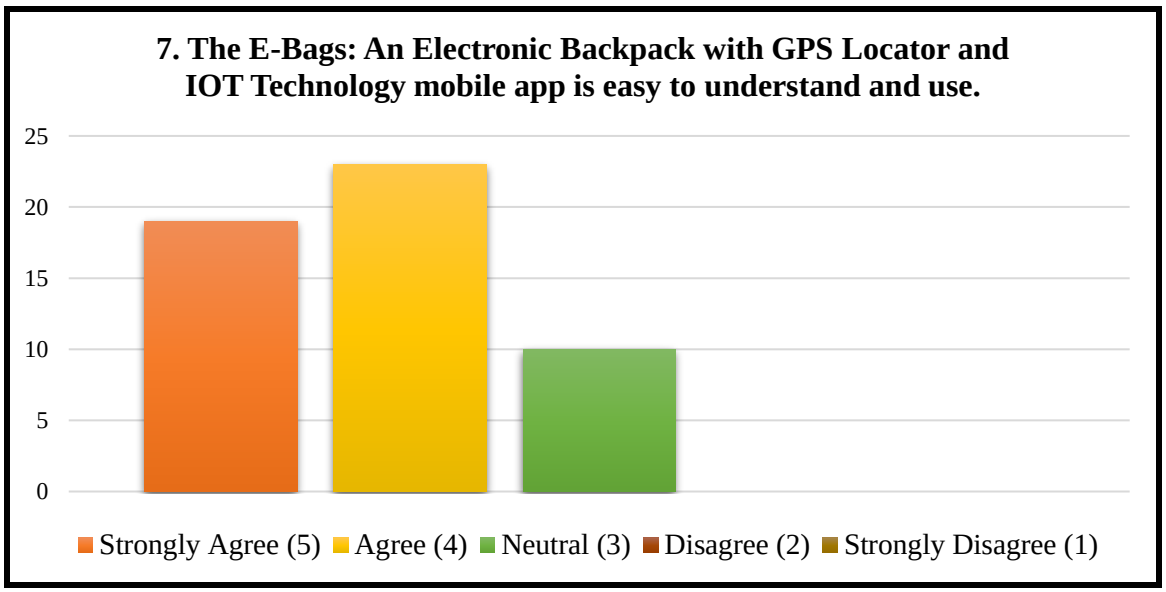


Figure 45. Findings from the 7th Question

Figure 45 depicts a table that indicates that most of the respondents (76%) expressed their strong agreement (30%) or agreement (46%) that the E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app is easy to understand and use and 20% remained neutral. The data indicates a weighted mean of 4.34, which corresponds to a verbal interpretation of "strongly agree".

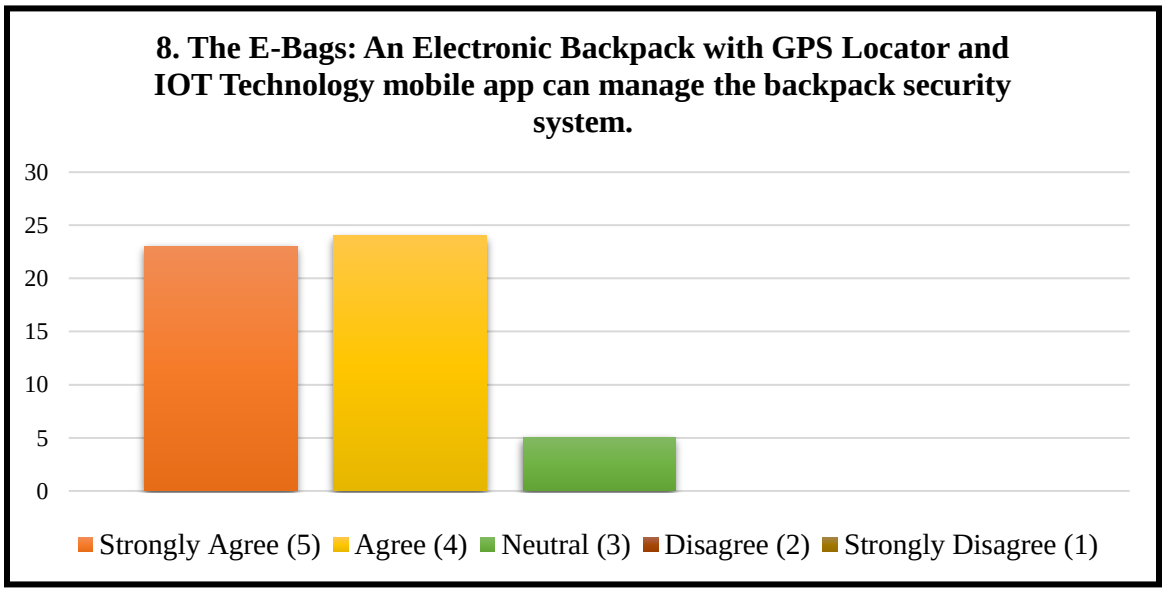


Figure 46. Findings from the 8th Question

The majority of the respondents (75%) in Figure 46 indicated strong agreement (46%) or agreement (48%) that E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can manage the backpack security system. A small percentage (5%) chose to be neutral about the statement. The data indicates a weighted mean of 4.4, which corresponds to a verbal interpretation of "strongly agree".

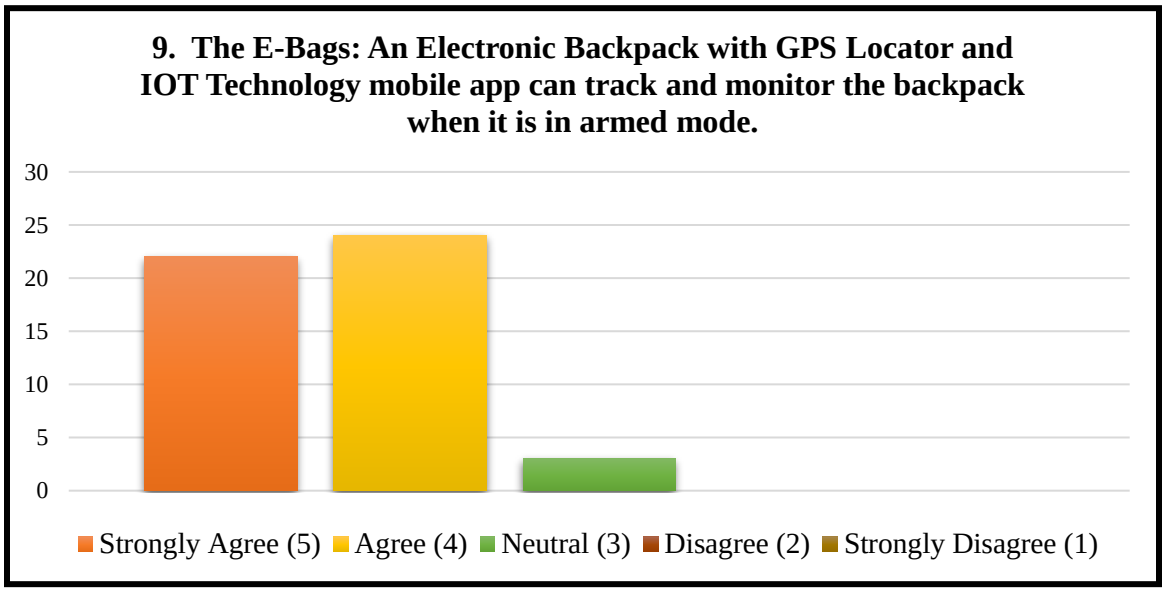


Figure 47. Findings from the 9th Question

In Figure 47 The data shows that most of the respondents (92%) either strongly agreed (44%) or agreed (48%) that the E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can track and monitor the backpack when it is in armed mode. Only a small percentage (8%) chose the neutral option. The data indicates a weighted mean of 4.36, which corresponds to a verbal interpretation of "strongly agree".

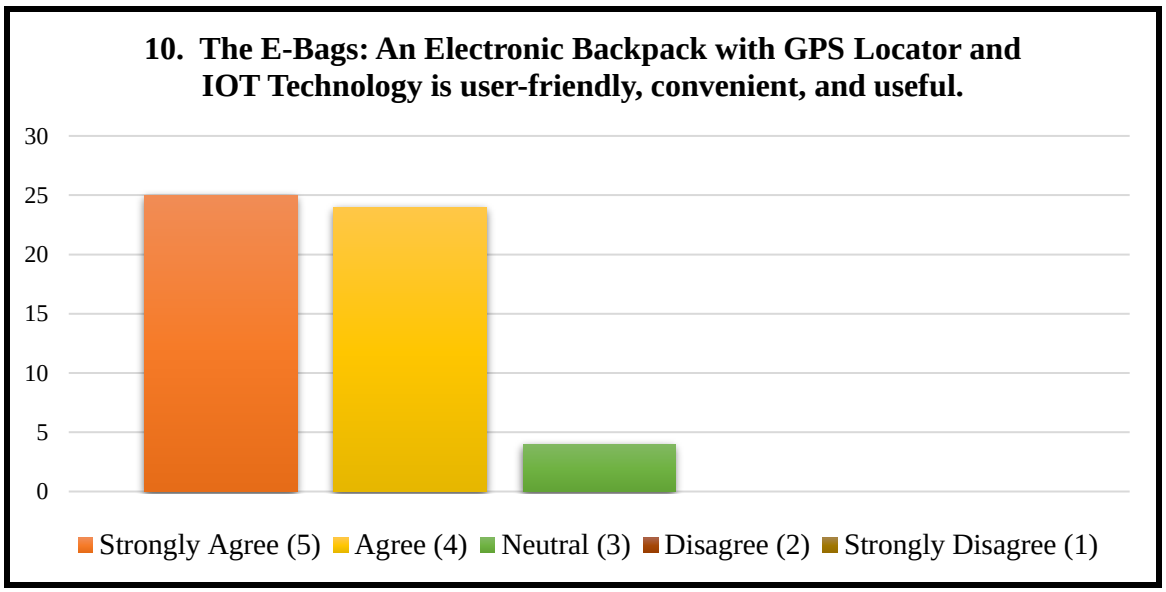


Figure 48. Findings from the 10th Question

Figure 48. illustrates that most of the respondents (92%) expressed either strong agreement (50%) or agreement (42%) towards the capability of E-Bags: An Electronic Backpack with GPS Locator and IOT Technology is user-friendly, convenient, and useful. A small proportion (8%) chose the neutral option. The data indicates a weighted mean of 4.42, which corresponds to a verbal interpretation of "strongly agree".

The survey findings demonstrate that the majority of participants have a strong agreement that E-Bag, equipped with GPS locator, alarm system, and safety camera, effectively safeguard belongings. They also agree that the E-bag can be monitored through SMS and a mobile app, providing additional innovative features. The use of IoT technology in E-Bag is perceived positively, with respondents acknowledging the increased security and peace of mind while traveling. The ability of E-Bag to display location information, capture images of handlers, and alert users in case of unauthorized access received favorable responses. The mobile app accompanying E-Bags was considered user-friendly and capable of managing the backpack security system. The survey indicates a positive perception and agreement towards the effectiveness, functionality, and convenience of the E-Bag: An Electronic Backpack with GPS Locator and IoT Technology.

CHAPTER V

CONCLUSION AND RECOMMENDATION

Conclusion

The results of the survey questionnaire and interview provided valuable insights that contributed to the development of the E-Bags: An Electronic Backpack with GPS Locator and IOT Technology. The study's findings reveal that the researchers successfully designed and developed a traveling backpack that incorporates innovative security features to enhance the safety of an ordinary bag. The backpack, equipped with IOT technology, provides travelers with peace of mind while carrying their belongings by allowing them to track and monitor the backpack in the event of loss or theft. The researchers also developed an electronic backpack equipped with a camera that can be easily managed and monitored using a mobile app to help users quickly identify anyone who has handled it. Through user testing, the researchers achieved their three objectives and ensured that the system and prototype met the required performance standards.

Recommendation

Based on the findings and suggestions provided by the client and the thesis adviser regarding the study on E-Bag, an electronic backpack with GPS Locator and IoT technology, future researchers may take into account the following recommendations:

> **Further explore user experience:** Future research should focus on exploring users' experiences and perceptions of electronic backpack with GPS locator and IOT technology. This would allow for a better understanding of how these features impact users' daily routines, as well as any potential issues that users may encounter when using these backpacks.

Assess the durability and reliability of technology: An electronic backpack with GPS locator and IOT technology rely heavily on technology, future researchers should evaluate the durability and reliability of the technology integrated into these backpacks. This includes the GPS tracking system, alarm systems, and safety cameras, among other features.

Evaluate the impact on the environment: Electronic backpacks with GPS locator and IOT technology may contain electronic components that may have an impact on the environment, so future researchers should investigate the environmental impact of these backpacks.

Investigate the impact on security: Future research could evaluate the effectiveness of electronic backpacks with GPS locator and IOT technology in enhancing security. This includes exploring how these backpacks impact theft prevention and if they can increase the likelihood of stolen items being recovered.

To summarize, future researchers should focus on user experience, the durability and reliability of technology, the environmental impact, and the impact on security to develop an Electronic Backpack.

APPENDICES

Appendix A: References

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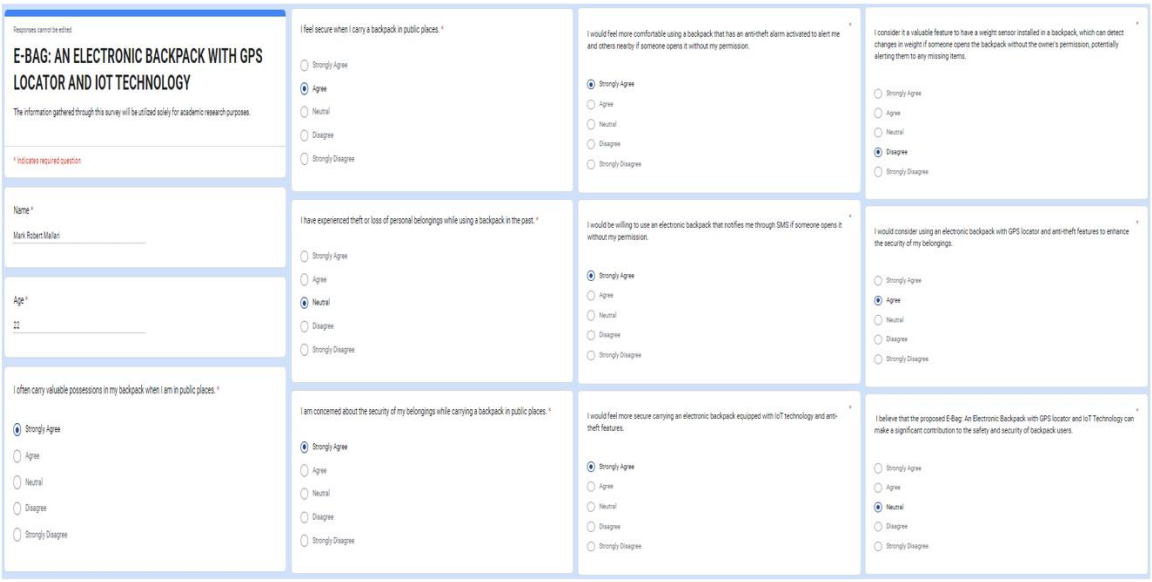
Appendix B: Transcript of Interview for Background of the Problem

Online Survey:

1.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Mark Robert Mallari

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I feel secure when I carry a backpack in public places. *
- ( ) Strongly Agree
- (•) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- ( ) Strongly Agree
- ( ) Agree
- (•) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I would be willing to use an electronic backpack that notifies me through SMS if someone opens it without my permission. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

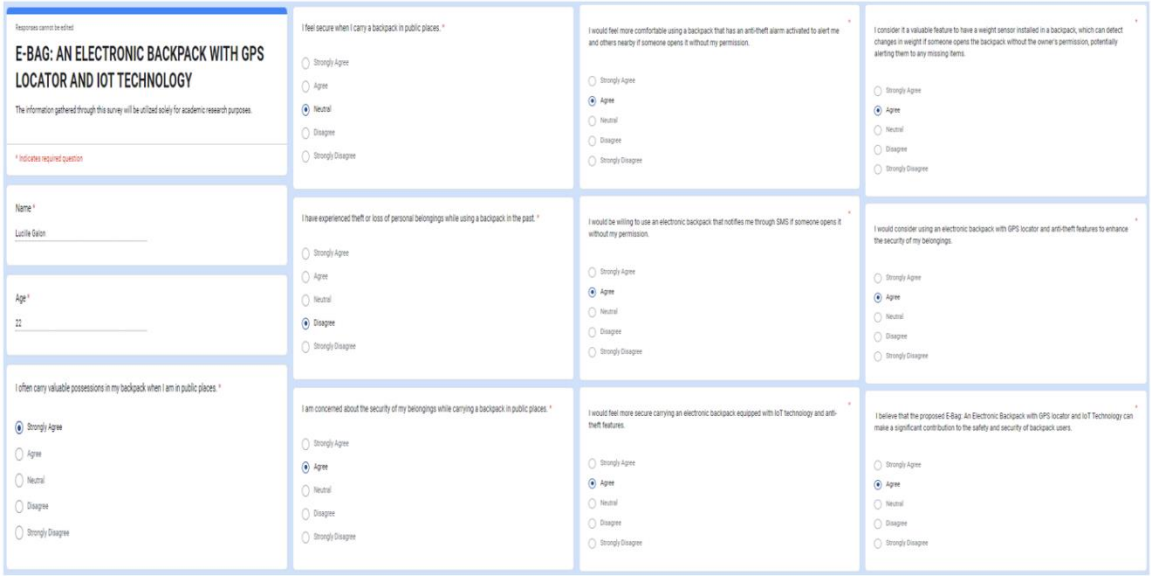
I would feel more secure carrying an electronic backpack equipped with IoT technology and anti-theft features. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I consider it a valuable feature to have a weight sensor installed in a backpack, which can detect changes in weight if someone opens the backpack without the owner's permission, potentially alerting them to any missing items. *
- ( ) Strongly Agree
- ( ) Agree
- ( ) Neutral
- (•) Disagree
- ( ) Strongly Disagree

I would consider using an electronic backpack with GPS locator and anti-theft features to enhance the security of my belongings. *
- ( ) Strongly Agree
- (•) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I believe that the proposed E-Bag: An Electronic Backpack with GPS locator and IoT Technology can make a significant contribution to the safety and security of backpack users. *
- ( ) Strongly Agree
- ( ) Agree
- (•) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

2.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Lucile Galon

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- (•) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I feel secure when I carry a backpack in public places. *
- ( ) Strongly Agree
- ( ) Agree
- (•) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Strongly Agree
- ( ) Agree
- ( ) Neutral
- (•) Disagree
- ( ) Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- ( ) Strongly Agree
- (•) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Neutral
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- ( ) Strongly Disagree

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- ( ) Disagree
- ( ) Strongly Disagree

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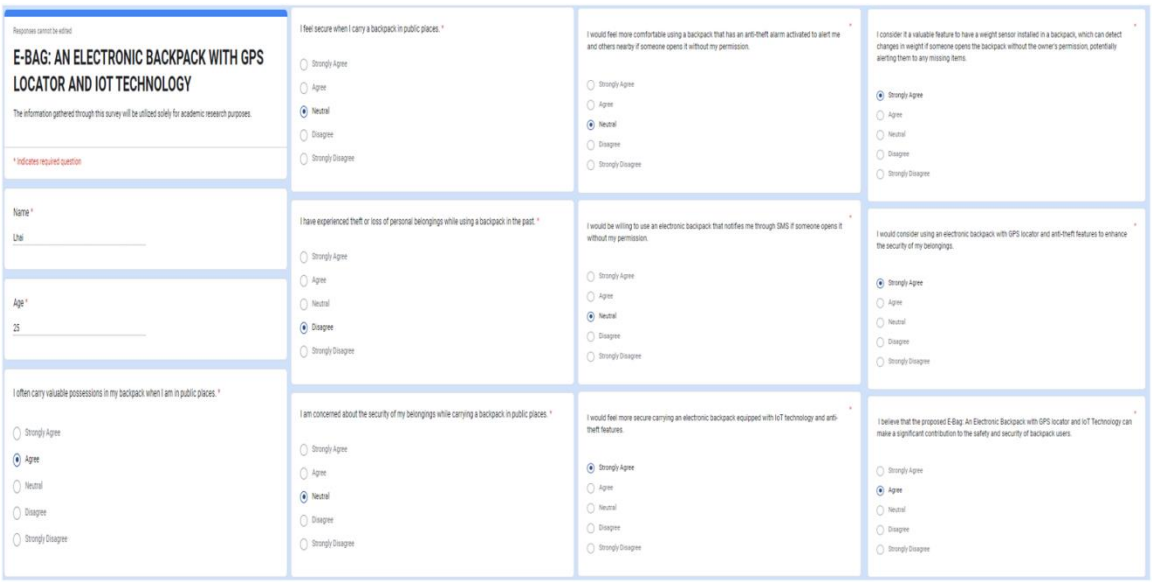
3.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Lhei

Age *
25

I often carry valuable possessions in my backpack when I am in public places. *
- Strongly Agree
- (•) Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- Strongly Agree
- Agree
- (•) Neutral
- Disagree
- Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- Strongly Agree
- Agree
- (•) Neutral
- Disagree
- Strongly Disagree

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- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- Strongly Agree
- Agree
- Neutral
- (•) Disagree
- Strongly Disagree

I would be willing to use an electronic backpack that notifies me through SMS if someone opens it without my permission. *
- Strongly Agree
- Agree
- (•) Neutral
- Disagree
- Strongly Disagree

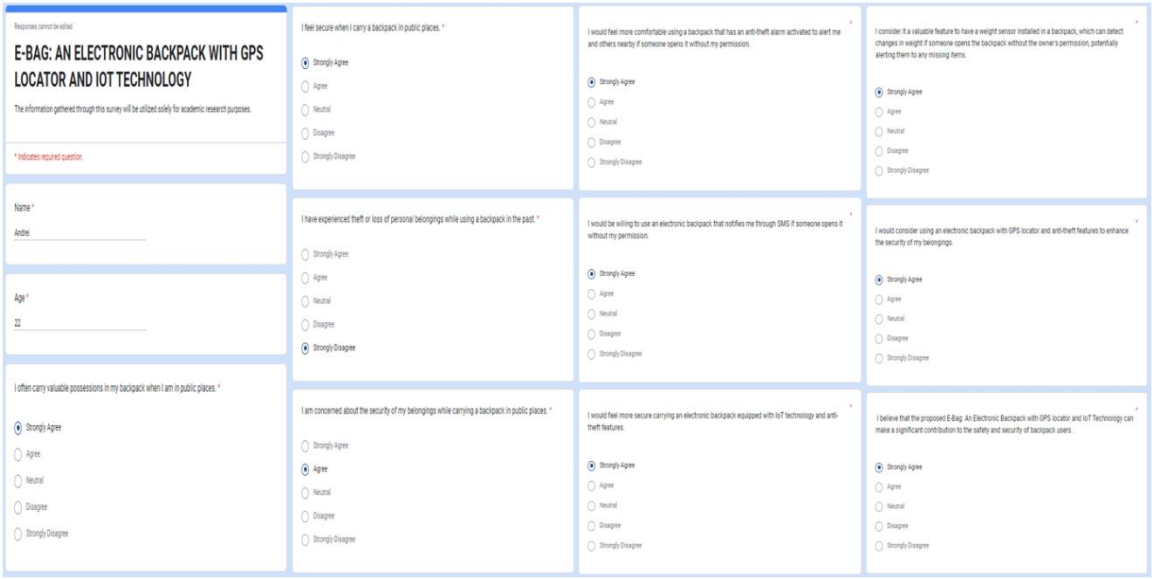
I would consider using an electronic backpack with GPS locator and anti-theft features to enhance the security of my belongings. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- Strongly Agree
- Agree
- (•) Neutral
- Disagree
- Strongly Disagree

I would feel more secure carrying an electronic backpack equipped with IoT technology and anti-theft features. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I believe that the proposed E-Bag: An Electronic Backpack with GPS locator and IoT Technology can make a significant contribution to the safety and security of backpack users. *
- Strongly Agree
- (•) Agree
- Neutral
- Disagree
- Strongly Disagree

4.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Andrei

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- (•) Strongly Disagree

I would be willing to use an electronic backpack that notifies me through SMS if someone opens it without my permission. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

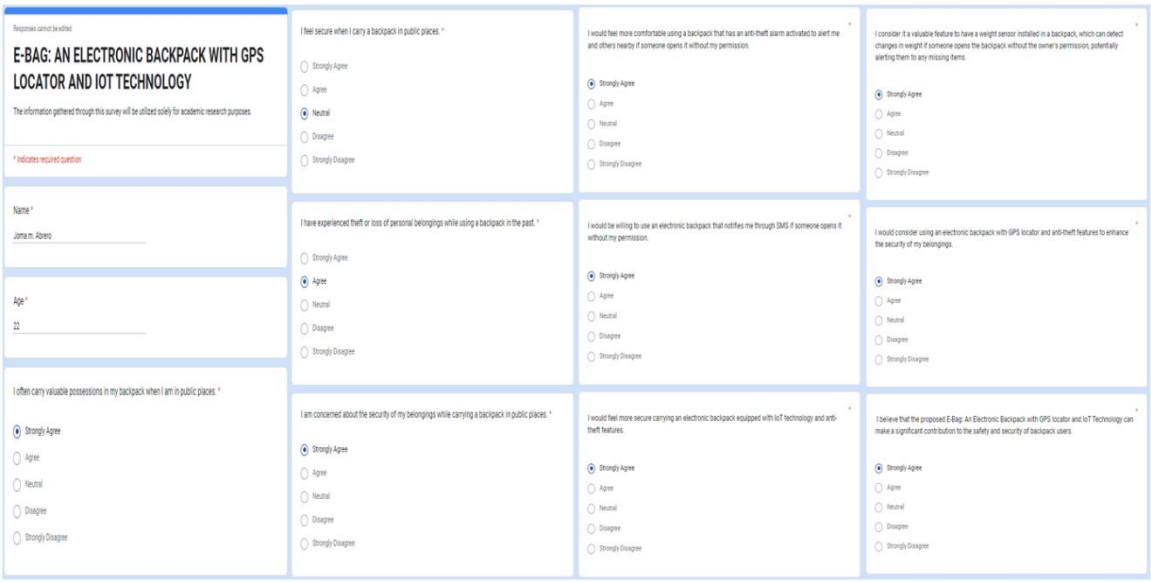
I would consider using an electronic backpack with GPS locator and anti-theft features to enhance the security of my belongings. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- Strongly Agree
- (•) Agree
- Neutral
- Disagree
- Strongly Disagree

I would feel more secure carrying an electronic backpack equipped with IoT technology and anti-theft features. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I believe that the proposed E-Bag: An Electronic Backpack with GPS locator and IoT Technology can make a significant contribution to the safety and security of backpack users. *
- (•) Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

5.



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E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

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* Indicates required question

Name *
Joma m. Abrero

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I feel secure when I carry a backpack in public places. *
- ( ) Strongly Agree
- ( ) Agree
- (●) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- ( ) Strongly Agree
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- (●) Strongly Agree
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- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
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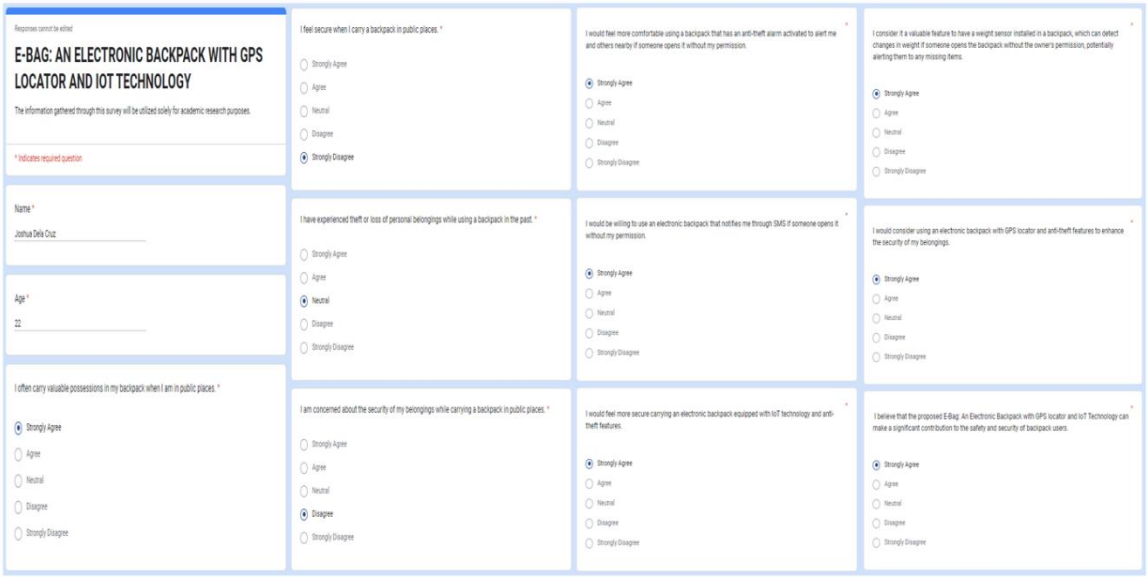
6.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Joshua Dela Cruz

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I feel secure when I carry a backpack in public places. *
- ( ) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- (●) Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

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- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- ( ) Strongly Agree
- ( ) Agree
- (●) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I would be willing to use an electronic backpack that notifies me through SMS if someone opens it without my permission. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
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- ( ) Strongly Disagree

I would consider using an electronic backpack with GPS locator and anti-theft features to enhance the security of my belongings. *
- (●) Strongly Agree
- ( ) Agree
- ( ) Neutral
- ( ) Disagree
- ( ) Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- ( ) Strongly Agree
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- (●) Disagree
- ( ) Strongly Disagree

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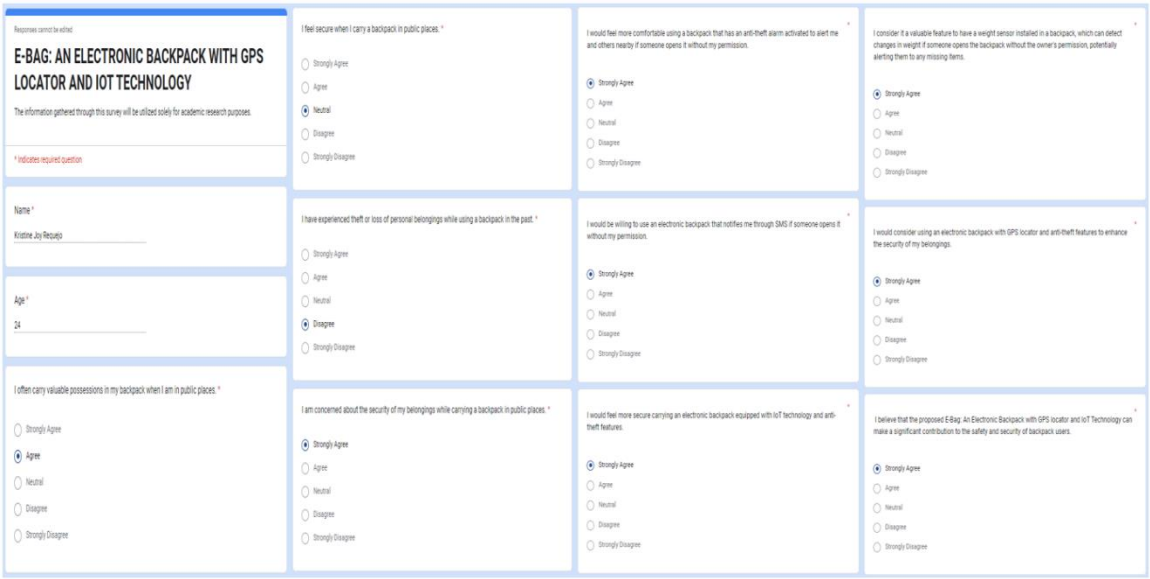

Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Kristine Joy Recuejo

Age *
24

I often carry valuable possessions in my backpack when I am in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- Strongly Agree
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- Strongly Agree
- Agree
- Neutral
- Disagree
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- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- Strongly Agree
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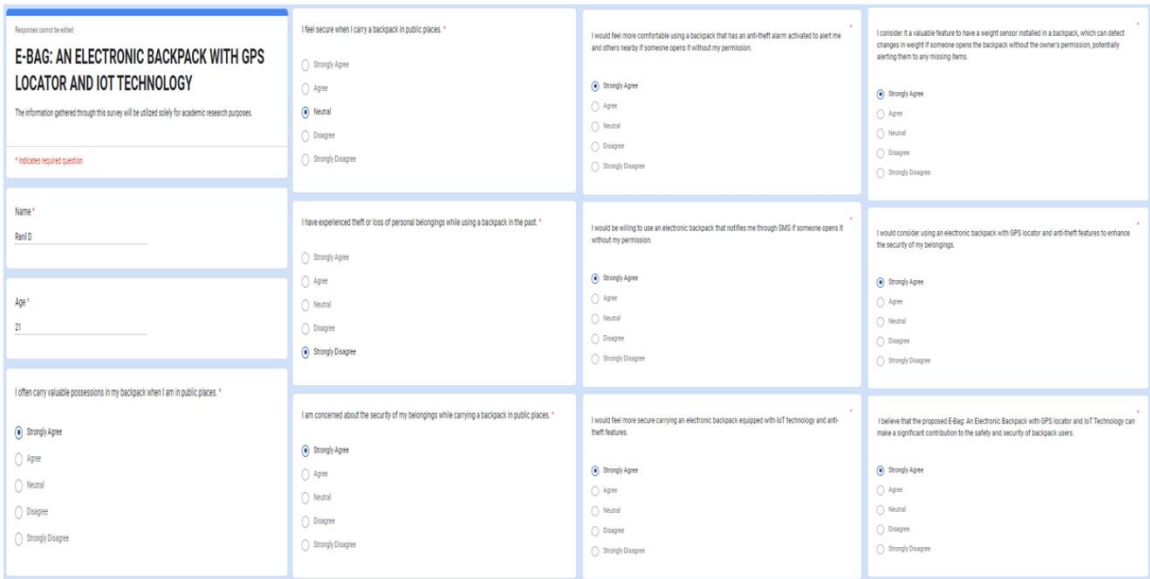
- Strongly Disagree

8.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Rexil D

Age *
21

I often carry valuable possessions in my backpack when I am in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- Strongly Agree
- Agree
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- Disagree
- Strongly Disagree

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- Strongly Agree
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- Strongly Disagree

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- Strongly Agree
- Agree
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- Agree
- Neutral
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- Strongly Agree
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- Neutral
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- Strongly Disagree

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- Strongly Agree
- Agree
- Neutral
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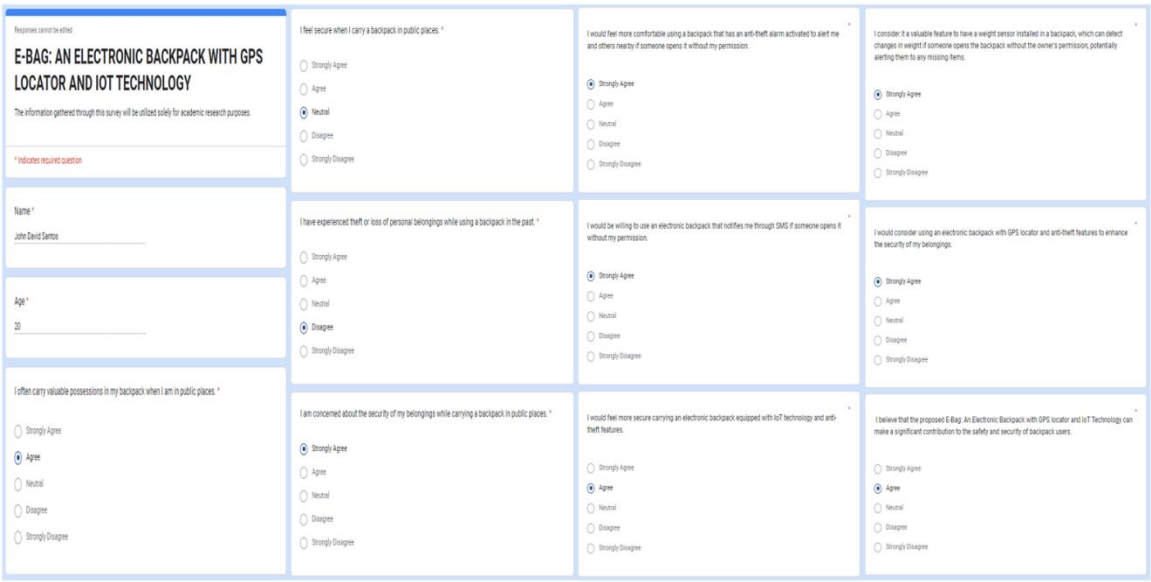
9.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
John David Santos

Age *
20

I often carry valuable possessions in my backpack when I am in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

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- Strongly Agree
- Agree
- Neutral
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- Strongly Disagree

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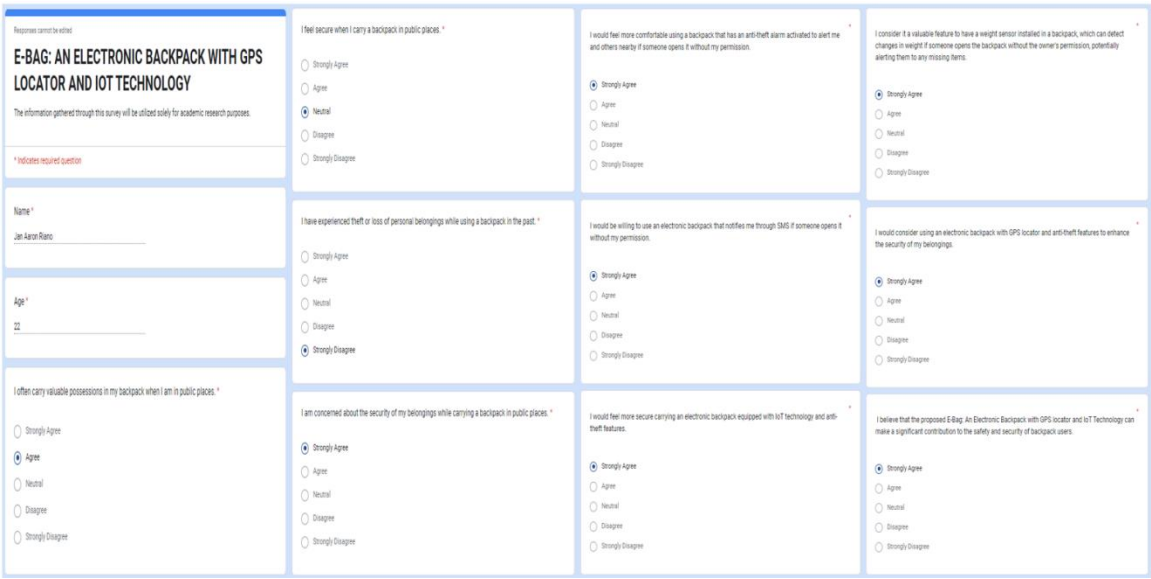
- Agree
- Neutral
- Disagree
- Strongly Disagree

10.



Responses cannot be edited

E-BAG: AN ELECTRONIC BACKPACK WITH GPS LOCATOR AND IOT TECHNOLOGY

The information gathered through this survey will be utilized solely for academic research purposes.

* Indicates required question

Name *
Jan Aaron Riano

Age *
22

I often carry valuable possessions in my backpack when I am in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I feel secure when I carry a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I have experienced theft or loss of personal belongings while using a backpack in the past. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I am concerned about the security of my belongings while carrying a backpack in public places. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I would feel more comfortable using a backpack that has an anti-theft alarm activated to alert me and others nearby if someone opens it without my permission. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I would be willing to use an electronic backpack that notifies me through SMS if someone opens it without my permission. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

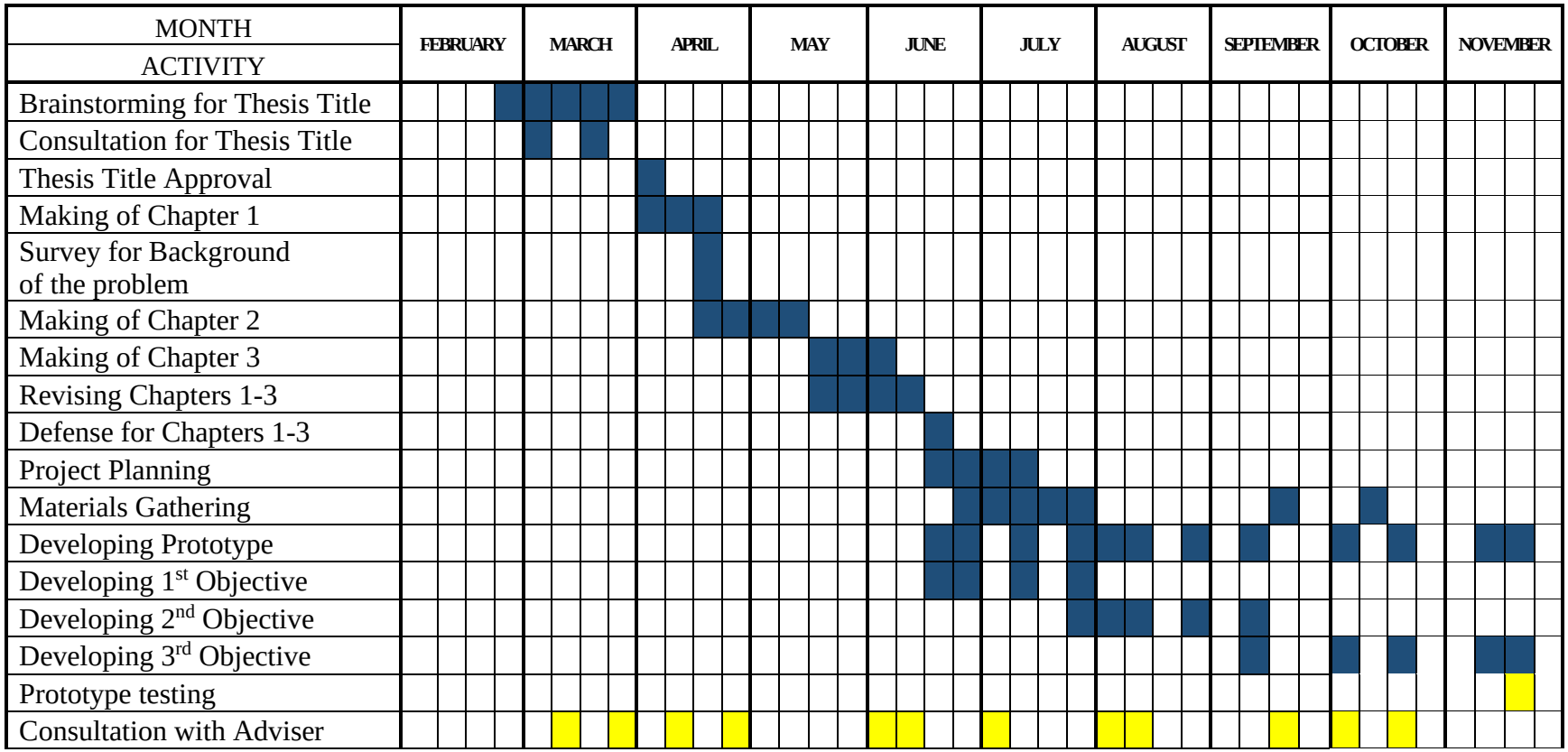
I would feel more secure carrying an electronic backpack equipped with IoT technology and anti-theft features. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I consider it a valuable feature to have a weight sensor installed in a backpack, which can detect changes in weight if someone opens the backpack without the owner's permission, potentially alerting them to any missing items. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I would consider using an electronic backpack with GPS locator and anti-theft features to enhance the security of my belongings. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

I believe that the proposed E-Bag: An Electronic Backpack with GPS locator and IoT Technology can make a significant contribution to the safety and security of backpack users. *
- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

Appendix C: Gantt chart of Activities



MONTH / ACTIVITY	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER
Finalizing Document for Capstone 1										
Defense for Capstone 1										
Consultation with Adviser										
Refinement of Prototype										
Prototype Testing										
Making of Chapter 4										
Evaluation of Prototype										
Making of Chapter 5										
Defense for Capstone 2										
Finalizing Document for Capstone 2										

Appendix D: Budgetary Estimate

Quantity	Specifics	Approximate Cost
1pc	Arduino Nano	₱ 450
1pc	Raspberry Pi 3 Model B	₱ 4,000
1pc	Power bank with 10000mAh	₱ 1,100
1pc	Japanese Randoseru Bag	₱ 1,200
1pc	2MP USB Camera	₱ 500
1pc	HX711 Weight Sensor	₱ 1,500
1pc	Piezo Electric Buzzer	₱ 250
1pc	GSM Module	₱400
1pc	GPS Module	₱300
1pc	On/Off Push Button	₱29
4pc	Whole Sintra Board	₱400
1pc	Pocket Wi-Fi	₱860
1pc	Limit Switch	₱80

Appendix E: Resource Person(s)

Interviewee's Information (Backpack User)

Ms. Nicole Cruz
Working Student
STI Recto

The focus of the inquiry is the user's experience with a regular backpack.

1. Do you often carry your valuable possessions using a backpack in public places?
   - Oo, lalo na pag babayahe ako papuntang trabaho. Madalas ako dumaan sa mga public places.
2. Do you encounter theft or loss of items inside your backpack?
   - Oo, One-time habang papauwi na ko nasa likod ko kasi yung bag ko. Noong nasa jeep na ko kukunin ko na sana cellphone ko sa bag di ko na makita. Akala ko na iwan ko binalikan ko pa sa trabaho ko wala talaga. Tapos tinawagan na ng katrabaho ko wala talaga noong una nag riring kaso mga ilang minutes lang nakapatay na dun ko narealize na nakawan ako.
3. After the occurrence what do you feel about it?
   - Noong una kinabahan ako noong di ko makita cellphone ko akala ko na misplace ko lang sa trabaho kaya bumalik ako sa workplace ko kaso wala talaga. Pinatawagan ko pa sa mga katrabaho ko nagriring daw tapos mga ilang mins lang nakapatay na daw yung cellphone ko kaya dun ko nasiguro na nawala na cellphone ko kasi di naman lowbat yun nung last time na chineck ko. Natakot na ko simula nung nangyari yun, kaya nilalagay ko na sa harap yung bag ko.
4. How can you ensure that your valuable possessions are safe when you are in public?
   - Simula nung nangyari yun lagi ko na nilalagay yung bag ko sa harap ko.
5. Do you think the proposed project can help you to secure your valuable

possessions inside your backpack?

- Yes, Kasi madami kayong dinagdag sa ordinary bag. Dun palang sa nag aalarm yung bag pag may nag bubukas na iba makakatulong na sakin para maalert ako kung may nangengeelam ba sa gamit ko pano pa kaya yung ibang security features na idadag nyo pa.

Interviewee's Information (Client)

Mr. Christian Mark G. Lapuz **President** **Innovative International Development Solutions Inc.**

The inquiry revolves around the satisfaction of the client and their recommendation regarding the electronic backpack.

1. Why will your company acquire the E-bag: an electronic backpack with GPS locator and IoT technology?
   - I aacquire namin yung bag kasi nakitaan namin ng madaming benefits din samin. So pwede pa naming ito idevelop as a new product den namin, then at the same time, kung pwede naming imerge to sainyo pwede naman. Sa nakikita naman namin may potential naman yung bag maging commercial product as of testing naman may papalitan kayong parts kasi pina stress test ko sa mga engineers ko e. So yun may papalitan kaming parts siguro during collaboration may isususuggest kami kung ano yung papalitan. So far ang nakikita palang naming problema is yung switches pero the rest gumagana yung system.
2. What are the benefits of the E-Bag to your Company?
   - Ah syempre unang una profit pag nadevelop namin to. Wala namang ibang makikinabang nito kundi mga tao ko during implementation. Para atleast pag pinapunta ko sila sa quiapo or pinaalis ko maiinform sila na ok naman, na

nawalan sila ng gamit or anything. Pero commonly nangyayari yun sa mga tao ko pag na sa mall. So, during na sa mall or nasa lugar na delikado ahh usable naman na talaga sya. Ang concern ko lang syempre yung load. Magiging empleyado din kasi yung bag siguro gagawan namin ng work around para di magkaroon ng load deduction ganun pero pwede din namn loadan pero siguro pag nas alabas yung gagamit.

3. Is it feasible to fabricate to your company?
   - Oo tulad nga ng sabi ko may potential yung product idedevelop lang ng pangcommercial yung materials na ginamit.
4. During testing what did you experience using the E-bag?
   - Ayun so during testing, since pang protype yung mga ginamit nyong parts meron mga bumigay syempre during stress test. Iniistress test kasi namin to e commonly kasi bago mo ibigay to sa client kailangan ng 3 days stress test. Kailangan susunugin mo muna yung buong system.Normal yun na ginagawa kasi pag di nyo inistress test baka pag dating sa kliyente nyan baka masunog. Pag di yan bumigay ready na yan for client's use. So normal naman na may bibigay na parts during testing since pang prototype nga ito hindi pa sya pang commercial. So magiiba pa ito commonly mas liliit tapos yung functionality
5. Are you satisfied with the E-bag? Do you have recommendation for the product to improve?
   - Oo, Ok naman yung product nyo ang ano lang ang concern ko lang since pang academic research may mga parts talaga na bibigay or masisira during testing kasi di nakadesign yun for commercial use. Kung mapapansin nyo pag commercial use its either maliliit,compat or naka embedded sa iisang board so dito kasi makikita mo pa na magkakahiwalay sila. So yun lang namna yung concern ko yung may bumibigay na parts during testing. Kasi pina istress test ko to sa mga engineer ko parang tatlo ata sila nag testing so far ok na man yung feedback namin.

E.1: Images of the Interview

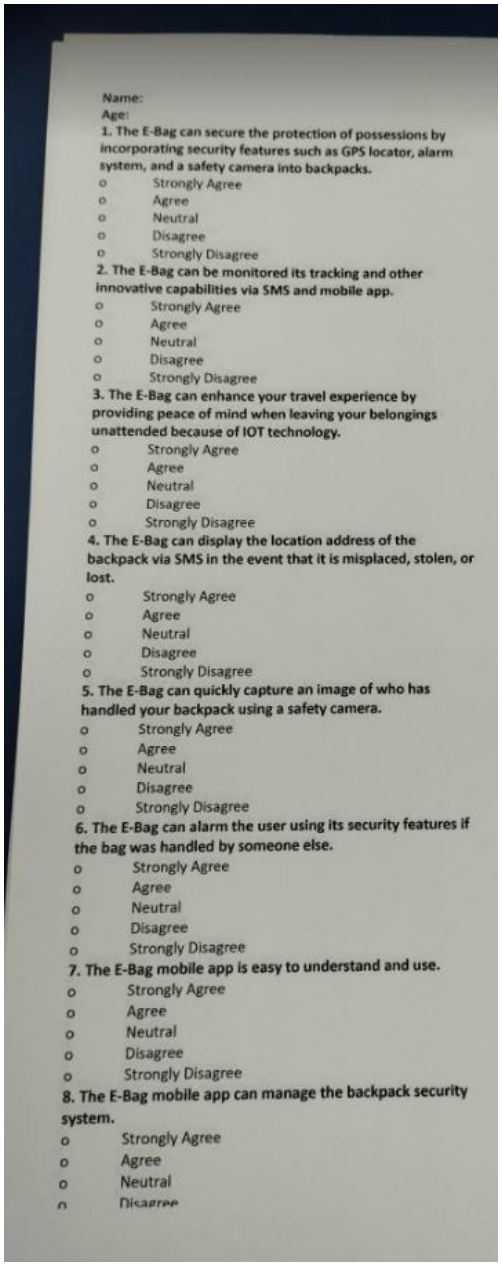
The images provided below showcase researchers conducting interviews and surveys while introducing the E-bag, an electronic backpack integrated with GPS locator and IoT technology.

E2: Images of Implementation

The images below show researchers discussing on how to use E-bag: an electronic backpack with GPS locator and IoT technology. The client will be testing the backpack's effectiveness in a stress test in three days.

Appendix F: Evaluation Tools or Test Documents

The respondents were provided with a printed version of the survey questionnaire. A sample representation is presented below:



Name:
Age:

1. The E-Bag can secure the protection of possessions by incorporating security features such as GPS locator, alarm system, and a safety camera into backpacks.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

2. The E-Bag can be monitored its tracking and other innovative capabilities via SMS and mobile app.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

3. The E-Bag can enhance your travel experience by providing peace of mind when leaving your belongings unattended because of IOT technology.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

4. The E-Bag can display the location address of the backpack via SMS in the event that it is misplaced, stolen, or lost.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

5. The E-Bag can quickly capture an image of who has handled your backpack using a safety camera.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

6. The E-Bag can alarm the user using its security features if the bag was handled by someone else.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

7. The E-Bag mobile app is easy to understand and use.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree
- o Strongly Disagree

8. The E-Bag mobile app can manage the backpack security system.

- o Strongly Agree
- o Agree
- o Neutral
- o Disagree

Figure 49. Sample of Printed Version of Survey Questionnaire

Survey Questionnaire for Backpack Users

Name:
Age:

1. The E-Bag can secure the protection of possessions by incorporating security features such as GPS locator, alarm system, and a safety camera into backpacks.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
2. The E-Bag can be monitored its tracking and other innovative capabilities via SMS and mobile app.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
3. The E-Bag can enhance your travel experience by providing peace of mind when leaving your belongings unattended because of IOT technology.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
4. The E-Bag can display the location address of the backpack via SMS in the event that it is misplaced,
stolen, or lost.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
5. The E-Bag can quickly capture an image of who has handled your backpack using a safety camera.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
6. The E-Bag can alarm the user using its security features if the bag was handled by someone else.
o Strongly Agree
o Agree

o Neutral
o Disagree
o Strongly Disagree
7. The E-Bag mobile app is easy to understand and use.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
8. The E-Bag mobile app can manage the backpack security system.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
9. The E-Bag mobile app can track and monitor the backpack when it is in armed mode.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree
10. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology is user-friendly, convenient, and useful.
o Strongly Agree
o Agree
o Neutral
o Disagree
o Strongly Disagree

Appendix G: Implementation Result Computation

	Strongly Agree (5)	**Agree (4)**	**Neutral (3)**	**Disagree (2)**	**Strongly Disagree (1)**	**Weighted Mean**	**Verbal Interpretation**
A. User							
1. E-Bag provide protection for belongings through GPS locators, alarm systems, and	27 x 5 = 135	22 x 4 = 88	1 x 3 = 3	0 x 2 = 0	0 x 1 = 0	4.52	Strongly Agree

safety cameras integrated into the backpacks							
2. E-bag can be monitored its tracking and other innovative capabilities via SMS and mobile app.	23 x 5 = 115	27 x 4 = 108	0 x 3 = 0	0 x 2 = 0	0 x 1 = 0	4.46	Strongly Agree
3. IOT technology in E-bag provides peace of mind while traveling by enhancing security when leaving belongings unattended.	19 x 5 = 95	24 x 4 = 96	7 x 3 = 21	0 x 2 = 0	0 x 1 = 0	4.24	Strongly Agree
4. The E-bag's location address can be displayed via SMS if it is lost, stolen, or misplaced	23 x 5 = 115	22 x 4 = 88	5 x 3 = 15	0 x 2 = 0	0 x 1 = 0	4.36	Strongly Agree
5. E-bags can capture an image of anyone who has handled the backpack using a safety camera.	21 x 5 = 105	24 x 4 = 96	5 x 3 = 15	0 x 2 = 0	0 x 1 = 0	4.32	Strongly Agree
6. E-bags can alert the user through security features if someone else handles the bag.	18 x 5 = 90	25 x 4 = 100	7 x 3 = 21	0 x 2 = 0	0 x 1 = 0	4.22	Strongly Agree

7. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app is easy to understand and use.	19 x 5 = 95	23 x 4 = 92	10 x 3 = 30	0 x 2 = 0	0 x 1 = 0	4.34	Strongly Agree
8. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can manage the backpack security system.	23 x 5 = 115	24 x 4 = 96	3 x 3 = 9	0 x 2 = 0	0 x 1 = 0	4.4	Strongly Agree
9. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can track and monitor the backpack when it is in armed mode.	22 x 5 = 110	24 x 4 = 96	4 x 3 = 12	0 x 2 = 0	0 x 1 = 0	4.36	Strongly Agree
10. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology is user-friendly, convenient, and useful.	25 x 5 = 125	21 x 4 = 84	4 x 3 = 12	0 x 2 = 0	0 x 1 = 0	4.42	Strongly Agree

Computation for Weighted Mean:

(27 x 5) + (22 x 4) + (1 x 3) + (0 x 2) + (0 x 1) = 226 / 50 = 4.52

(23 x 5) + (27 x 4) + (0 x 3) + (0 x 2) + (0 x 1) = 223 / 50 = 4.46

(19 x 5) + (24 x 4) + (7 x 3) + (0 x 2) + (0 x 1) = 212 / 50 = 4.24

(23 x 5) + (22 x 4) + (5 x 3) + (0 x 2) + (0 x 1) = 218 / 50 = 4.36

(21 x 5) + (24 x 4) + (5 x 3) + (0 x 2) + (0 x 1) = 216 / 50 = 4.32

(18 x 5) + (25 x 4) + (7 x 3) + (0 x 2) + (0 x 1) = 211 / 50 = 4.22

(19 x 5) + (23 x 4) + (10 x 3) + (0 x 2) + (0 x 1) = 217 / 50 = 4.34

(23 x 5) + (24 x 4) + (3 x 3) + (0 x 2) + (0 x 1) = 220 / 50 = 4.4

(22 x 5) + (24 x 4) + (4 x 3) + (0 x 2) + (0 x 1) = 218 / 50 = 4.36

(25 x 5) + (21 x 4) + (4 x 3) + (0 x 2) + (0 x 1) = 221 / 50 = 4.42

Appendix H: Relevant Source Codes

Source Code for Arduino Nano

```
// Libraries
#include "HX711.h"
#include <SoftwareSerial.h>
#include <Wire.h>
#include <TinyGPS++.h>
```

```
// Global Variables
TinyGPSPlus gps;
#define calibration_factor -7050.0
#define LOADCELL_DOUT_PIN 2
#define LOADCELL_SCK_PIN 3
String inputString = "";
int startcapture = 0;
String cellnumber = "+639261234567";
int warning = 0;
String warning1 = "";
String switch1 = "E";
String switch2 = "G";
int delayloop = 0;
int armed = 0;
float lat = 14.657866, lon = 120.976732;
SoftwareSerial gpsSerial(4, 5);
SoftwareSerial gsmSerial(6, 7);
HX711 scale;
const int buttonPin = 8;
int buttonState = 0;
const int buttonPin1 = 9;
int buttonState1 = 0;

// Function to send SMS
void sendsms() {
  delay(2000);
  gsmSerial.println("AT\r\n");
  delay(2000);
  gsmSerial.println("AT+CMGF=1\r\n");
  delay(1000);
```

```
  gsmSerial.print("AT+CMGS=\"" + cellnumber + "\"\r\n");
  delay(1000);
  gsmSerial.print("Emergency!! lat" + String(gps.location.lat(), 6) + " long" +
String(gps.location.lng(), 6));
  delay(3500);
  gsmSerial.write(0x1A);
  delay(3000);
  Serial.println("sms snt p");
}

void setup() {
  Serial.begin(9600);
  Serial.println("HX711 scale demo");
  gsmSerial.begin(9600);
  gpsSerial.begin(9600);
  pinMode(buttonPin, INPUT_PULLUP);
  inputString.reserve(200);
  scale.begin(LOADCELL_DOUT_PIN, LOADCELL_SCK_PIN);
  scale.set_scale(calibration_factor);
  scale.tare();
  Serial.println("Readings:");
}

void loop() {
  delayloop++;
  buttonState = digitalRead(buttonPin);
  if (buttonState == LOW) {
    switch1 = "E";
    warning1 = "DETECTED 3!";
    warning = 1;
```

```
}
buttonState1 = digitalRead(buttonPin1);
if (buttonState1 == LOW) {
  switch2 = "F";
  warning1 = "DETECTED 4!";
}
String latitude = String(lat, 6);
String longitude = String(lon, 6);

while (gpsSerial.available() > 0) {
  gps.encode(gpsSerial.read());
  if (gps.location.isUpdated()) {}
}

if (delayloop == 200 || delayloop == 400 || delayloop == 600 || delayloop == 800) {
  if (armed == 1) {
    float weightdata = scale.get_units();
    if (weightdata > 0.2) {
      warning1 = "DETECTED 1!";
      warning = 1;
    }
    if (weightdata < (-0.2)) {
      warning1 = "DETECTED 2!";
      warning = 1;
    }
  }
}

if (delayloop > 1000) {
  delayloop = 0;
```

```
    if (armed == 1) {
      float weightdata = scale.get_units();
      if (weightdata > 0.2) {
        warning1 = "DETECTED 1!";
        warning = 1;
  }
  if (weightdata < (-0.2)) {
    warning1 = "DETECTED 2!";
    warning = 1;
  }
}

Serial.print(gps.location.lat(), 6);
Serial.print("A");
Serial.print(gps.location.lng(), 6);
Serial.print("B");
Serial.print(scale.get_units(), 1);
Serial.print("C");
Serial.print(switch1);
Serial.print(switch2);

switch1 = "D";
switch2 = "G";

Serial.println("");

float weightdata = scale.get_units();

if (armed == 1) {
  if (warning == 1) {
```

```
    Serial.println(warning1);
    warning = 0;
    sendsms();
  }
}
}
delay(1);
}

void serialEvent() {
while (Serial.available()) {
char inChar = (char)Serial.read();
if (inChar == 'a') {
Serial.println(inputString);
startcapture = 0;
inputString = "";
scale.tare();
armed = 1;
}
if (startcapture == 1) {
inputString += inChar;
}
if (inChar == 'B') {
armed = 0;
}
if (inChar == 'C') {
sendsms();
}
if (inChar == '\n') {}
if (inChar == 'A') {
```

```
startcapture = 1;
}
}
}
```

Source Code for Data Base and User Interface

Capture.php:

```
<?php
$myfilename= "count.txt"; if(file_exists($myfilename)){$count= file_get_contents($myfilename);}
$count=$count+1;
   $ret = file_put_contents('count.txt',$count);
echo copy("1.jpg","images/".$count.".jpg");

  $url='test.php';
  echo '<META HTTP-EQUIV=REFRESH CONTENT="0.001; '.$url.'">';
?>
```

Capture1.php:

```
<?php
$myfilename= "count.txt"; if(file_exists($myfilename)){$count= file_get_contents($myfilename);}
$count=$count+1;
   $ret = file_put_contents('count.txt',$count);
echo copy("1.jpg","images/".$count.".jpg");

  $url='test.php';
  echo '<META HTTP-EQUIV=REFRESH CONTENT="0.001; '.$url.'">';
?>
```

datafrombag.php:

```
<?php

$myfilename= "cmd1.txt"; if(file_exists($myfilename)){$cmd1=
file_get_contents($myfilename);}


$myfilename= "status.txt"; if(file_exists($myfilename)){$status= file_get_contents($myfilename);}

$dataA=strpos($cmd1,"A");
$dataR=strpos($cmd1,"B");
$dataS=strpos($cmd1,"C");
```

```
$datab=strpos($cmd1,"b");


?>

                <Meta HTTP-EQUIV="Refresh" Content="2; URL=datafrombag.php">
<style >
   html, body {margin: 0; height: 100%; overflow: hidden}
   </style>

     <body style=" background-color:white;" scroll="no" style="overflow: hidden">
<table  align="center">
  <tr>
   <td align="center"><img  src="1.jpg" width="195" height="195"></img>
</td>

  </tr>

  <tr>
   <td align="center">
     LON:<?php $cmd2=substr($cmd1, $datab+2, $dataA-2); echo $cmd2;?>
     <BR>
     LAT:<?php $cmd3=substr($cmd1, $dataA+1, $dataR-$dataA-1); echo $cmd3;?>

</td>


  </tr>
  <tr>
   <td align="center">
     WEIGHT:<?php $kg=substr($cmd1, $dataR+1, $dataS-$dataR-1);echo $kg." g";?>

</td>
```

db.php:

```
     <?php
                $connection = @mysql_connect('localhost', 'root', '');

                if (!$connection)
                {
                            die("Database Connection Failed" . mysql_error());
                }

                $select_db = mysql_select_db('payroll');
```

```
		if (!$select_db)
		{
				die("Database Selection Failed" . mysql_error());
		}
?>
```

db_con.php:

```
<?php
		$connection = @mysql_connect('localhost', 'root', '');

		if (!$connection)
		{
				die("Database Connection Failed" . mysql_error());
		}

		$select_db = mysql_select_db('payroll');
		if (!$select_db)
		{
				die("Database Selection Failed" . mysql_error());
		}
?>
```

dispense.php:

```
<?php

$data1= $_GET['s'];
$data2=  $_GET['m'];

echo $data2;

 $day1=  date("D");
include('db_con.php');

		$myfilename= "date1.txt"; if(file_exists($myfilename)){$date1=
file_get_contents($myfilename);}
	$myfilename= "date2.txt"; if(file_exists($myfilename)){$date2=
file_get_contents($myfilename);}
		$myfilename= "date3.txt"; if(file_exists($myfilename)){$date3=
file_get_contents($myfilename);}
	$myfilename= "date4.txt"; if(file_exists($myfilename)){$date4=
file_get_contents($myfilename);}
		$myfilename= "date5.txt"; if(file_exists($myfilename)){$date5=
file_get_contents($myfilename);}
$myfilename= "date6.txt"; if(file_exists($myfilename)){$date6= file_get_contents($myfilename);}
```

```
$myfilename= "date7.txt"; if(file_exists($myfilename)){$date7= file_get_contents($myfilename);}

$myfilename= "sched/p1.txt"; if(file_exists($myfilename)){$sp1=
file_get_contents($myfilename);}
$myfilename= "sched/p2.txt"; if(file_exists($myfilename)){$sp2=
file_get_contents($myfilename);}
$myfilename= "sched/p3.txt"; if(file_exists($myfilename)){$sp3=
file_get_contents($myfilename);}
$myfilename= "sched/p4.txt"; if(file_exists($myfilename)){$sp4=
file_get_contents($myfilename);}
$myfilename= "sched/p5.txt"; if(file_exists($myfilename)){$sp5=
file_get_contents($myfilename);}


if ($date1>23){$date1=0;}
if ($date3>12){$date3=1;}
if ($date4>30){$date4=0;}

if ($date1<12){$date11="AM";}else{$date11="PM";}


if ($date7<12){$date12="AM";}else{$date12="PM";}



if ($date6>59){$date6=0;}

if ($date2>59){$date2=0; $date1=$date1+1;}


if ($date1>12){$date1=$date1-12;}


 if ($date1<10){$date1="0".$date1;}
 if ($date2<10){$date2="0".$date2;}
 if ($date3<10){$date3="0".$date3;}
 if ($date4<10){$date4="0".$date4;}
 if ($date5<10){$date5="0".$date5;}



   $date11= $date1.":". $date2.":00";

   $date12= $date3."/". $date4."/". $date5;
```

```
$date111=$date12." ".$date11;
$date = strtotime($date111);
$W1= date('W', $date);
$date11= $date1.":". $date2.":00 ".$date11;

echo $W1;


      $url='process1.php';

   echo '<META HTTP-EQUIV=REFRESH CONTENT="0.001; '.$url.'">';

  if ($data2==1){
  $ret = file_put_contents('cmd.txt','a');$ret = file_put_contents('cmd3.txt','1'); $ret =
file_put_contents('cmd2.txt',$data1);
  $sql = "INSERT INTO history (date1,time1,med1,med2,med3,med4,med5,
med1c,med2c,med3c,med4c,med5c,med1p,med2p,med3p,med4p,med5p,week1,year1)VALUES
('".$date12."','".$date11."','TAKEN','NONE','NONE','NONE','NONE','1','0','0','0','0','".$sp1."','".$sp2
."','".$sp3."','".$sp4."','".$sp5."','".$W1."','".$date5."')";

  if ($conn->query($sql) === TRUE) {}

      }
  if ($data2==2){$ret = file_put_contents('cmd.txt','b');$ret = file_put_contents('cmd3.txt','2'); $ret =
file_put_contents('cmd2.txt',$data1);
  $sql = "INSERT INTO history (date1,time1,med1,med2,med3,med4,med5,
med1c,med2c,med3c,med4c,med5c,med1p,med2p,med3p,med4p,med5p,week1,year1)VALUES
('".$date12."','".$date11."','NONE','TAKEN','NONE','NONE','NONE','0','1','0','0','0','".$sp1."','".$sp2
."','".$sp3."','".$sp4."','".$sp5."','".$W1."','".$date5."')";

  if ($conn->query($sql) === TRUE) {}


      }
  if ($data2==3){$ret = file_put_contents('cmd.txt','c');$ret = file_put_contents('cmd3.txt','3'); $ret =
file_put_contents('cmd2.txt',$data1);

  $sql = "INSERT INTO history (date1,time1,med1,med2,med3,med4,med5,
med1c,med2c,med3c,med4c,med5c,med1p,med2p,med3p,med4p,med5p,week1,year1)VALUES
('".$date12."','".$date11."','NONE','NONE','TAKEN','NONE','NONE','0','0','1','0','0','".$sp1."','".$sp2
."','".$sp3."','".$sp4."','".$sp5."','".$W1."','".$date5."')";

  if ($conn->query($sql) === TRUE) {}
```

```
    }
  if ($data2==4){$ret = file_put_contents('cmd.txt','d');$ret = file_put_contents('cmd3.txt','4'); $ret =
file_put_contents('cmd2.txt',$data1);
  $sql = "INSERT INTO history (date1,time1,med1,med2,med3,med4,med5,
med1c,med2c,med3c,med4c,med5c,med1p,med2p,med3p,med4p,med5p,week1,year1)VALUES
("'.$date12.'","'.$date11.'','NONE','NONE','NONE','TAKEN','NONE','0','0','0','1','0','"'.$sp1.'","'.$sp2
.'","'.$sp3.'","'.$sp4.'","'.$sp5.'","'.$W1.'","'.$date5.'")";

  if ($conn->query($sql) === TRUE) {}


    }
  if ($data2==5){$ret = file_put_contents('cmd.txt','e');$ret = file_put_contents('cmd3.txt','5'); $ret =
file_put_contents('cmd2.txt',$data1);

  $sql = "INSERT INTO history (date1,time1,med1,med2,med3,med4,med5,
med1c,med2c,med3c,med4c,med5c,med1p,med2p,med3p,med4p,med5p,week1,year1)VALUES
("'.$date12.'","'.$date11.'','NONE','NONE','NONE','NONE','TAKEN','0','0','0','0','1','"'.$sp1.'","'.$sp2
.'","'.$sp3.'","'.$sp4.'","'.$sp5.'","'.$W1.'","'.$date5.'")";

  if ($conn->query($sql) === TRUE) {}


}

      $ret = file_put_contents('sched/'.$data1.'.txt','0');



                $myfilename= "date1.txt"; if(file_exists($myfilename)){$date1=
file_get_contents($myfilename);}
      $myfilename= "date2.txt"; if(file_exists($myfilename)){$date2=
file_get_contents($myfilename);}
                $myfilename= "date3.txt"; if(file_exists($myfilename)){$date3=
file_get_contents($myfilename);}
      $myfilename= "date4.txt"; if(file_exists($myfilename)){$date4=
file_get_contents($myfilename);}
                $myfilename= "date5.txt"; if(file_exists($myfilename)){$date5=
file_get_contents($myfilename);}
$myfilename= "date6.txt"; if(file_exists($myfilename)){$date6= file_get_contents($myfilename);}
$myfilename= "date7.txt"; if(file_exists($myfilename)){$date7= file_get_contents($myfilename);}


$myfilename= "timespeed.txt"; if(file_exists($myfilename)){$timespeed=
file_get_contents($myfilename);}
```

```
if ($date1>23){$date1=0;}
if ($date3>12){$date3=1;}
if ($date4>30){$date4=0;}

if ($date1<12){$date11="AM";}else{$date11="PM";}


if ($date7<12){$date12="AM";}else{$date12="PM";}

?>
```

history.php:

```
                <Meta HTTP-EQUIV="Refresh" Content="5; URL=history.php">
<?php


$myfilename= "status.txt"; if(file_exists($myfilename)){$status= file_get_contents($myfilename);}


$myfilename= "cmd.txt"; if(file_exists($myfilename)){$cmd= file_get_contents($myfilename);}
$myfilename= "date1.txt"; if(file_exists($myfilename)){$date1a=
file_get_contents($myfilename);}
$myfilename= "date2.txt"; if(file_exists($myfilename)){$date2a=
file_get_contents($myfilename);}
$myfilename= "date3.txt"; if(file_exists($myfilename)){$date3a=
file_get_contents($myfilename);}
$myfilename= "date4.txt"; if(file_exists($myfilename)){$date4a=
file_get_contents($myfilename);}
$myfilename= "date5.txt"; if(file_exists($myfilename)){$date5a=
file_get_contents($myfilename);}



if (strpos($cmd, 'A') !== false)
{

$ret = file_put_contents('cmd.txt','0');

$myfilename= "count.txt"; if(file_exists($myfilename)){$count= file_get_contents($myfilename);}
$count=$count+1;
$ret = file_put_contents('count.txt',$count);
echo copy("1.jpg","images/".$count.".jpg");

$ret = file_put_contents("images/date".$count.".txt",$date3a."/".$date4a."/".$date5a);
$ret = file_put_contents("images/t".$count.".txt",$date1a.":".$date2a);
```

```
$ret = file_put_contents("images/a".$count.".txt",$date1a);
$ret = file_put_contents("images/w".$count.".txt",$date1a);


}


$myfilename= "images/date1.txt"; if(file_exists($myfilename)){$date1=
file_get_contents($myfilename);}
$myfilename= "images/date2.txt"; if(file_exists($myfilename)){$date2=
file_get_contents($myfilename);}
$myfilename= "images/date3.txt"; if(file_exists($myfilename)){$date3=
file_get_contents($myfilename);}
$myfilename= "images/date4.txt"; if(file_exists($myfilename)){$date4=
file_get_contents($myfilename);}
$myfilename= "images/date5.txt"; if(file_exists($myfilename)){$date5=
file_get_contents($myfilename);}


$myfilename= "images/t1.txt"; if(file_exists($myfilename)){$t1=
file_get_contents($myfilename);}
$myfilename= "images/t2.txt"; if(file_exists($myfilename)){$t2=
file_get_contents($myfilename);}
$myfilename= "images/t3.txt"; if(file_exists($myfilename)){$t3=
file_get_contents($myfilename);}
$myfilename= "images/t4.txt"; if(file_exists($myfilename)){$t4=
file_get_contents($myfilename);}
$myfilename= "images/t5.txt"; if(file_exists($myfilename)){$t5=
file_get_contents($myfilename);}

$myfilename= "images/a1.txt"; if(file_exists($myfilename)){$a1=
file_get_contents($myfilename);}
$myfilename= "images/a2.txt"; if(file_exists($myfilename)){$a2=
file_get_contents($myfilename);}
$myfilename= "images/a3.txt"; if(file_exists($myfilename)){$a3=
file_get_contents($myfilename);}
$myfilename= "images/a4.txt"; if(file_exists($myfilename)){$a4=
file_get_contents($myfilename);}
$myfilename= "images/a5.txt"; if(file_exists($myfilename)){$a5=
file_get_contents($myfilename);}

$myfilename= "images/w1.txt"; if(file_exists($myfilename)){$w1=
file_get_contents($myfilename);}
$myfilename= "images/w2.txt"; if(file_exists($myfilename)){$w2=
file_get_contents($myfilename);}
```

```
$myfilename= "images/w3.txt"; if(file_exists($myfilename)){$w3=
file_get_contents($myfilename);}
$myfilename= "images/w4.txt"; if(file_exists($myfilename)){$w4=
file_get_contents($myfilename);}
$myfilename= "images/w5.txt"; if(file_exists($myfilename)){$w5=
file_get_contents($myfilename);}


?>


<html >
                                                <style>

 .button-88  {
                    font-size: 16px;
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:65px;
    margin:2px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 2px;
    text-align: center;
    line-height: 1em;
}
</style>
                <meta charset="utf-8" />
                  <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">


                  <body   >
<table align="center" border=0>
<tr>
                <td>
```

```
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG HISTORY</h1>
</td>

</tr>
</table>

<div align="center">

<table>
                <tr>


<td>

                                <form action='main.php' method='post'>

<input type='submit'  class="button-88" style='height:50px;width:300px;'  value='BACK'
/><br><br><br>


</form>

</table>


<body >
  <table border=1 >
      <tr>
      <td>DATE</td>
      <td>TIME</td>
                <td>IMAGE</td>
      <td>WEIGHT </td>
      <td>STATUS </td>

      </tr>
                <tr>
      <td><?php echo $date1; ?></td>
      <td><?php echo $t1; ?></td>
        <td><img  src="images/1.jpg" width="95" height="95"></img></td>

      <td><?php echo $w1; ?></td>
      <td><?php echo $a1; ?></td>
```

```
      </tr>
                    <tr>
 <td><?php echo $date2; ?></td>
      <td><?php echo $t2; ?></td>
                    <td><img  src="images/2.jpg" width="95" height="95"></img></td>
        <td><?php echo $w2; ?></td>
      <td><?php echo $a2; ?></td>
      </tr>
                    <tr>
     <td><?php echo $date3; ?></td>
      <td><?php echo $t3; ?></td>
                    <td><img  src="images/3.jpg" width="95" height="95"></img></td>
         <td><?php echo $w3; ?></td>
      <td><?php echo $a3; ?></td>
      </tr>
                                                  <tr>
     <td><?php echo $date4; ?></td>
      <td><?php echo $t4; ?></td>
                    <td><img  src="images/4.jpg" width="95" height="95"></img></td>
      <td><?php echo $w4; ?></td>
      <td><?php echo $a4; ?></td>
      </tr>
                                                  <tr>
    <td><?php echo $date5; ?></td>
      <td><?php echo $t5; ?></td>
                    <td><img  src="images/5.jpg" width="95" height="95"></img></td>
       <td><?php echo $w5; ?></td>
      <td><?php echo $a5; ?></td>
      </tr>



</table>



</div>
```

index.php:

```
<?php

$un=$_POST['d1'];
$pw=$_POST['d2'];
```

```
                $myfilename= "un.txt"; if(file_exists($myfilename)){$p1=
file_get_contents($myfilename);}
                $myfilename= "pw.txt"; if(file_exists($myfilename)){$p2=
file_get_contents($myfilename);}


if (isset($_POST['d1']))
{if ($p1==$un)
   {
if ($p2==$pw)
   {
                header('Location: main.php');

                }
                else
   {$warning="INCORRECT PASSWORD";}


                }
                else
   {$warning="INCORRECT USERNAME";}


   }

?>

<html >
                                                <style>

 .button-88  {
                  font-size: 16px;
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

   width:175px;
   height:65px;
   margin:20px auto;
   border-radius:10px;
   -webkit-border-radius:10px;
   box-shadow:0 1px 2px #5e5d5b;
}
```

```
.button-88  span {
   display: block;
   padding-top: 22px;
   text-align: center;
   line-height: 1em;
}
</style>
                <meta charset="utf-8" />
                  <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                  <body  >
<table align="center" border=0>
<tr>
                <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG MENU</h1>
</td>

</tr>
</table>

<div align="center">

<table align="center" border=0>

                <form action='index.php' method='post'>
<tr>
<td>

USERNAME
</td>
<td>

<input type='text' id="d1" name="d1"  />
</td>

</tr>

<tr>
<td>

PASSWORD
```

```
</td>
<td>

<input type='password'  id="d2" name="d2"  />
</td>

</tr>

<tr>
  <td>
</td>

<td>

<input type='submit'  class="button-88" style='height:40px;width:160px;'  value='LOGIN' />

</td>
</tr>
</form>

</table>
<div align="center"><?php echo $warning;?></div>
<body >

</div>
```

index1.php :

```
<style >
   html, body {margin: 0; height: 100%; overflow: hidden}
   </style>

      <body style=" background-color:black;" scroll="no" style="overflow: hidden">
<table  >
  <tr>
   <td><iframe  width="640" height="640"  frameBorder="0"  src="time.php"
scrolling="no"></iframe></td>
  </tr>
   <tr>
 <table  >
    <tr>
                <td>
                <iframe   width="350" height="70"  frameBorder="0" src="soundwatch.php"
scrolling="no"></iframe>
                </td></tr>
```

```
        <tr>
                <td>
                <iframe   width="350" height="70"  frameBorder="0" src="sound1.php"
scrolling="no"></iframe>
                </td></tr>
   <tr>
                <td>
                <iframe   width="350" height="170"  frameBorder="0" src="processformissed.php"
scrolling="no"></iframe>
                </td></tr>

 </table>  </tr>
</table>

</body>
```

main.php:

```
<html >
                <style>

 .button-88  {
                  font-size: 16px;
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:65px;
    margin:20px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 22px;
    text-align: center;
    line-height: 1em;
}
</style>
                <meta charset="utf-8" />
```

```
                <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                <body   >
<table align="center" border=0>
<tr>
              <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG MENU</h1>
</td>

</tr>
</table>

<div align="center">

<table align="center" border=0>


              <form action='history.php' method='post'>

<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='HISTORY' />

</td>
</tr>
</form>

              <form action='status.php?s=0' method='post'>

<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='SYSTEM
STATUS' />

</td>
</tr>
</form>


</form>
```

```
                <form action='test.php' method='post'>

<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'
value='CONFIGURATION' />

</td>
</tr>
</form>


                <form action='manual.php' method='post'>


<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='USER
MANUAL' />

</td>
</tr>
</form>




                <form action='index.php' method='post'>


<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='LOGOUT' />

</td>
</tr>
</form>

</table>
<body >
```

manual.php:

```
<html >
                                                <style>

 .button-88  {
                font-size: 16px;
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:45px;
    margin:1px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 22px;
    text-align: left;
    line-height: 1em;
}
</style>
              <meta charset="utf-8" />
                <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                <body   >
<table align="center" border=0>
<tr>
              <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG MANUAL</h1>
</td>

</tr>
</table>

<div align="center">
```

```
<table align="center" border=0>
<tr>
                <form action='main.php' method='post'>

<td>

<input type='submit'  class="button-88"value='BACK' />
</td>

</form>

                <form action='manualedit.php' method='post'>

<td>

<input type='submit'  class="button-88"value='EDIT' />
</td>

</form>
</tr>

</table>

<br>
<br>
<a href="EBag.apk">DOWNLOAD APP HERE</a>

<br>
<br>
<br>

<body >



   <?php

$myfilename= "M1.txt"; if(file_exists($myfilename)){$M1= file_get_contents($myfilename);}
$myfilename= "M2.txt";if(file_exists($myfilename)){$M2= file_get_contents($myfilename);}
$myfilename= "M3.txt";if(file_exists($myfilename)){$M3= file_get_contents($myfilename);}
$myfilename= "M4.txt";if(file_exists($myfilename)){$M4= file_get_contents($myfilename);}
$myfilename= "M5.txt";if(file_exists($myfilename)){$M5= file_get_contents($myfilename);}
$myfilename= "M6.txt"; if(file_exists($myfilename)){$M6= file_get_contents($myfilename);}
$myfilename= "M7.txt";if(file_exists($myfilename)){$M7= file_get_contents($myfilename);}
```

```
$myfilename= "M8.txt";if(file_exists($myfilename)){$M8= file_get_contents($myfilename);}
$myfilename= "M9.txt";if(file_exists($myfilename)){$M9= file_get_contents($myfilename);}
$myfilename= "M10.txt";if(file_exists($myfilename)){$M10= file_get_contents($myfilename);}
$myfilename= "M11.txt"; if(file_exists($myfilename)){$M11= file_get_contents($myfilename);}
$myfilename= "M12.txt";if(file_exists($myfilename)){$M12= file_get_contents($myfilename);}
$myfilename= "M13.txt";if(file_exists($myfilename)){$M13= file_get_contents($myfilename);}
$myfilename= "M14.txt";if(file_exists($myfilename)){$M14= file_get_contents($myfilename);}
$myfilename= "M15.txt";if(file_exists($myfilename)){$M15= file_get_contents($myfilename);}
$myfilename= "M16.txt"; if(file_exists($myfilename)){$M16= file_get_contents($myfilename);}
$myfilename= "M17.txt";if(file_exists($myfilename)){$M17= file_get_contents($myfilename);}
$myfilename= "M18.txt";if(file_exists($myfilename)){$M18= file_get_contents($myfilename);}
$myfilename= "M19.txt";if(file_exists($myfilename)){$M19= file_get_contents($myfilename);}
$myfilename= "M20.txt";if(file_exists($myfilename)){$M20= file_get_contents($myfilename);}
$myfilename= "M21.txt"; if(file_exists($myfilename)){$M21= file_get_contents($myfilename);}
$myfilename= "M22.txt";if(file_exists($myfilename)){$M22= file_get_contents($myfilename);}
$myfilename= "M23.txt";if(file_exists($myfilename)){$M23= file_get_contents($myfilename);}
$myfilename= "M24.txt";if(file_exists($myfilename)){$M24= file_get_contents($myfilename);}
$myfilename= "M25.txt";if(file_exists($myfilename)){$M25= file_get_contents($myfilename);}
$myfilename= "M26.txt"; if(file_exists($myfilename)){$M26= file_get_contents($myfilename);}
$myfilename= "M27.txt";if(file_exists($myfilename)){$M27= file_get_contents($myfilename);}
$myfilename= "M28.txt";if(file_exists($myfilename)){$M28= file_get_contents($myfilename);}
$myfilename= "M29.txt";if(file_exists($myfilename)){$M29= file_get_contents($myfilename);}
$myfilename= "M30.txt";if(file_exists($myfilename)){$M30= file_get_contents($myfilename);}
$myfilename= "M31.txt"; if(file_exists($myfilename)){$M31= file_get_contents($myfilename);}
$myfilename= "M32.txt";if(file_exists($myfilename)){$M32= file_get_contents($myfilename);}
$myfilename= "M33.txt";if(file_exists($myfilename)){$M33= file_get_contents($myfilename);}
$myfilename= "M34.txt";if(file_exists($myfilename)){$M34= file_get_contents($myfilename);}
$myfilename= "M35.txt";if(file_exists($myfilename)){$M35= file_get_contents($myfilename);}
$myfilename= "M36.txt"; if(file_exists($myfilename)){$M36= file_get_contents($myfilename);}
$myfilename= "M37.txt";if(file_exists($myfilename)){$M37= file_get_contents($myfilename);}
$myfilename= "M38.txt";if(file_exists($myfilename)){$M38= file_get_contents($myfilename);}
$myfilename= "M39.txt";if(file_exists($myfilename)){$M39= file_get_contents($myfilename);}
$myfilename= "M40.txt";if(file_exists($myfilename)){$M40= file_get_contents($myfilename);}



    ?>
<table border=0>
 <tr>
 <td><?php echo $M1;?></td>
 <td width=300px;><?php echo $M2;?></td>
 </tr>
 <tr>
 <td><?php echo $M3;?></td>
 <td><?php echo $M4;?></td>
```

```
</tr>
<tr>
<td><?php echo $M5;?></td>
<td><?php echo $M6;?></td>
</tr>
<tr>
<td><?php echo $M7;?></td>
<td><?php echo $M8;?></td>
</tr>
<tr>
<td><?php echo $M9;?></td>
<td><?php echo $M10;?></td>
</tr>
<tr>
<td><?php echo $M11;?></td>
<td><?php echo $M12;?></td>
</tr>
<tr>
<td><?php echo $M13;?></td>
<td><?php echo $M14;?></td>
</tr>
<tr>
<td><?php echo $M15;?></td>
<td><?php echo $M16;?></td>
</tr>
<tr>
<td><?php echo $M17;?></td>
<td><?php echo $M18;?></td>
</tr>
<tr>
<td><?php echo $M19;?></td>
<td><?php echo $M20;?></td>
</tr>
<tr>
<td><?php echo $M21;?></td>
<td><?php echo $M22;?></td>
</tr>
<tr>
<td><?php echo $M23;?></td>
<td><?php echo $M24;?></td>
</tr>
<tr>
<td><?php echo $M25;?></td>
<td><?php echo $M26;?></td>
</tr>
```

```
 <tr>
 <td><?php echo $M27;?></td>
 <td><?php echo $M28;?></td>
 </tr>
 <tr>
 <td><?php echo $M29;?></td>
 <td><?php echo $M30;?></td>
 </tr>
 <tr>
 <td><?php echo $M31;?></td>
 <td><?php echo $M32;?></td>
 </tr>
 <tr>
 <td><?php echo $M33;?></td>
 <td><?php echo $M34;?></td>
 </tr>
 <tr>
 <td><?php echo $M35;?></td>
 <td><?php echo $M36;?></td>
 </tr>
 <tr>
 <td><?php echo $M37;?></td>
 <td><?php echo $M38;?></td>
 </tr>
 <tr>
 <td><?php echo $M39;?></td>
 <td><?php echo $M40;?></td>
 </tr>

            </table>


</div>

</div>
```

manualEdit.php:

```
<html >
                                                <style>

 .button-88  {
                font-size: 16px;
```

```
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:45px;
    margin:1px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 22px;
    text-align: left;
    line-height: 1em;
}
</style>
                <meta charset="utf-8" />
                  <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                  <body   >
<table align="center" border=0>
<tr>
                <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG EDIT MANUAL</h1>
</td>

</tr>
</table>

<div align="center">

<table align="center" border=0>
<tr>
                  <form action='main.php' method='post'>

<td>

<input type='submit'  class="button-88"value='BACK' />
```

```
</td>

</form>

                    <form action='manualedit.php' method='post'>

<td>

<input type='submit'  name="Submit1" class="button-88"value='SAVE' />
</td>
</tr>

</table>


<body >

    <?php

if(isset($_POST['Submit1']))
{

    $d1=$_POST['d1'];
    $d2=$_POST['d2'];
    $d3=$_POST['d3'];
    $d4=$_POST['d4'];
    $d5=$_POST['d5'];
    $d6=$_POST['d6'];
    $d7=$_POST['d7'];
    $d8=$_POST['d8'];
    $d9=$_POST['d9'];
    $d10=$_POST['d10'];
    $d11=$_POST['d11'];
    $d12=$_POST['d12'];
    $d13=$_POST['d13'];
    $d14=$_POST['d14'];
    $d15=$_POST['d15'];
    $d16=$_POST['d16'];
    $d17=$_POST['d17'];
    $d18=$_POST['d18'];
    $d19=$_POST['d19'];
    $d20=$_POST['d20'];
    $d21=$_POST['d21'];
    $d22=$_POST['d22'];
    $d23=$_POST['d23'];
```

```
$d24=$_POST['d24'];
$d25=$_POST['d25'];
$d26=$_POST['d26'];
$d27=$_POST['d27'];
$d28=$_POST['d28'];
$d29=$_POST['d29'];
$d30=$_POST['d30'];
$d31=$_POST['d31'];
$d32=$_POST['d32'];
$d33=$_POST['d33'];
$d34=$_POST['d34'];
$d35=$_POST['d35'];
$d36=$_POST['d36'];
$d37=$_POST['d37'];
$d38=$_POST['d38'];
$d39=$_POST['d39'];
$d40=$_POST['d40'];



$ret = file_put_contents('M1.txt',$d1);
$ret = file_put_contents('M2.txt',$d2);
$ret = file_put_contents('M3.txt',$d3);
$ret = file_put_contents('M4.txt',$d4);
$ret = file_put_contents('M5.txt',$d5);
$ret = file_put_contents('M6.txt',$d6);
$ret = file_put_contents('M7.txt',$d7);
$ret = file_put_contents('M8.txt',$d8);
$ret = file_put_contents('M9.txt',$d9);
$ret = file_put_contents('M10.txt',$d10);

$ret = file_put_contents('M11.txt',$d11);
$ret = file_put_contents('M12.txt',$d12);
$ret = file_put_contents('M13.txt',$d13);
$ret = file_put_contents('M14.txt',$d14);
$ret = file_put_contents('M15.txt',$d15);
$ret = file_put_contents('M16.txt',$d16);
$ret = file_put_contents('M17.txt',$d17);
$ret = file_put_contents('M18.txt',$d18);
$ret = file_put_contents('M19.txt',$d19);
$ret = file_put_contents('M20.txt',$d20);
```

```
$ret = file_put_contents('M21.txt',$d21);
$ret = file_put_contents('M22.txt',$d22);
$ret = file_put_contents('M23.txt',$d23);
$ret = file_put_contents('M24.txt',$d24);
$ret = file_put_contents('M25.txt',$d25);
$ret = file_put_contents('M26.txt',$d26);
$ret = file_put_contents('M27.txt',$d27);
$ret = file_put_contents('M28.txt',$d28);
$ret = file_put_contents('M29.txt',$d29);
$ret = file_put_contents('M30.txt',$d30);

$ret = file_put_contents('M31.txt',$d31);
$ret = file_put_contents('M32.txt',$d32);
$ret = file_put_contents('M33.txt',$d33);
$ret = file_put_contents('M34.txt',$d34);
$ret = file_put_contents('M35.txt',$d35);
$ret = file_put_contents('M36.txt',$d36);
$ret = file_put_contents('M37.txt',$d37);
$ret = file_put_contents('M38.txt',$d38);
$ret = file_put_contents('M39.txt',$d39);
$ret = file_put_contents('M40.txt',$d40);
```

```
$myfilename= "M1.txt"; if(file_exists($myfilename)){$M1= file_get_contents($myfilename);}
$myfilename= "M2.txt";if(file_exists($myfilename)){$M2= file_get_contents($myfilename);}
$myfilename= "M3.txt";if(file_exists($myfilename)){$M3= file_get_contents($myfilename);}
$myfilename= "M4.txt";if(file_exists($myfilename)){$M4= file_get_contents($myfilename);}
$myfilename= "M5.txt";if(file_exists($myfilename)){$M5= file_get_contents($myfilename);}
$myfilename= "M6.txt"; if(file_exists($myfilename)){$M6= file_get_contents($myfilename);}
$myfilename= "M7.txt";if(file_exists($myfilename)){$M7= file_get_contents($myfilename);}
$myfilename= "M8.txt";if(file_exists($myfilename)){$M8= file_get_contents($myfilename);}
$myfilename= "M9.txt";if(file_exists($myfilename)){$M9= file_get_contents($myfilename);}
$myfilename= "M10.txt";if(file_exists($myfilename)){$M10= file_get_contents($myfilename);}
$myfilename= "M11.txt"; if(file_exists($myfilename)){$M11= file_get_contents($myfilename);}
$myfilename= "M12.txt";if(file_exists($myfilename)){$M12= file_get_contents($myfilename);}
$myfilename= "M13.txt";if(file_exists($myfilename)){$M13= file_get_contents($myfilename);}
$myfilename= "M14.txt";if(file_exists($myfilename)){$M14= file_get_contents($myfilename);}
$myfilename= "M15.txt";if(file_exists($myfilename)){$M15= file_get_contents($myfilename);}
$myfilename= "M16.txt"; if(file_exists($myfilename)){$M16= file_get_contents($myfilename);}
$myfilename= "M17.txt";if(file_exists($myfilename)){$M17= file_get_contents($myfilename);}
$myfilename= "M18.txt";if(file_exists($myfilename)){$M18= file_get_contents($myfilename);}
$myfilename= "M19.txt";if(file_exists($myfilename)){$M19= file_get_contents($myfilename);}
$myfilename= "M20.txt";if(file_exists($myfilename)){$M20= file_get_contents($myfilename);}
```

```
$myfilename= "M21.txt"; if(file_exists($myfilename)){$M21= file_get_contents($myfilename);}
$myfilename= "M22.txt";if(file_exists($myfilename)){$M22= file_get_contents($myfilename);}
$myfilename= "M23.txt";if(file_exists($myfilename)){$M23= file_get_contents($myfilename);}
$myfilename= "M24.txt";if(file_exists($myfilename)){$M24= file_get_contents($myfilename);}
$myfilename= "M25.txt";if(file_exists($myfilename)){$M25= file_get_contents($myfilename);}
$myfilename= "M26.txt"; if(file_exists($myfilename)){$M26= file_get_contents($myfilename);}
$myfilename= "M27.txt";if(file_exists($myfilename)){$M27= file_get_contents($myfilename);}
$myfilename= "M28.txt";if(file_exists($myfilename)){$M28= file_get_contents($myfilename);}
$myfilename= "M29.txt";if(file_exists($myfilename)){$M29= file_get_contents($myfilename);}
$myfilename= "M30.txt";if(file_exists($myfilename)){$M30= file_get_contents($myfilename);}
$myfilename= "M31.txt"; if(file_exists($myfilename)){$M31= file_get_contents($myfilename);}
$myfilename= "M32.txt";if(file_exists($myfilename)){$M32= file_get_contents($myfilename);}
$myfilename= "M33.txt";if(file_exists($myfilename)){$M33= file_get_contents($myfilename);}
$myfilename= "M34.txt";if(file_exists($myfilename)){$M34= file_get_contents($myfilename);}
$myfilename= "M35.txt";if(file_exists($myfilename)){$M35= file_get_contents($myfilename);}
$myfilename= "M36.txt"; if(file_exists($myfilename)){$M36= file_get_contents($myfilename);}
$myfilename= "M37.txt";if(file_exists($myfilename)){$M37= file_get_contents($myfilename);}
$myfilename= "M38.txt";if(file_exists($myfilename)){$M38= file_get_contents($myfilename);}
$myfilename= "M39.txt";if(file_exists($myfilename)){$M39= file_get_contents($myfilename);}
$myfilename= "M40.txt";if(file_exists($myfilename)){$M40= file_get_contents($myfilename);}


    ?>

    <br>
    <br>

<table border=1>
  <tr><td><input type="text" name="d1"  size="8" title="NAME" method="post" value="<?php
echo $M1;?>"/></td>
     <td><input type="text" name="d2"  size="40"title="NAME" method="post" value="<?php
echo $M2;?>"/></td></tr>
  <tr><td><input type="text" name="d3"  size="8" title="NAME" method="post" value="<?php
echo $M3;?>"/></td>
     <td><input type="text" name="d4"  size="40"title="NAME" method="post" value="<?php
echo $M4;?>"/></td></tr>
  <tr><td><input type="text" name="d5"  size="8" title="NAME" method="post" value="<?php
echo $M5;?>"/></td>
     <td><input type="text" name="d6"  size="40"title="NAME" method="post" value="<?php
echo $M6;?>"/></td></tr>
  <tr><td><input type="text" name="d7"  size="8" title="NAME" method="post" value="<?php
echo $M7;?>"/></td>
     <td><input type="text" name="d8"  size="40"title="NAME" method="post" value="<?php
echo $M8;?>"/></td></tr>
```

```
 <tr><td><input type="text" name="d9"  size="8" title="NAME" method="post" value="<?php
echo $M9;?>"/></td>
    <td><input type="text" name="d10" size="40"title="NAME" method="post" value="<?php
echo $M10;?>"/></td></tr>
 <tr><td><input type="text" name="d11" size="8" title="NAME" method="post" value="<?php
echo $M11;?>"/></td>
    <td><input type="text" name="d12" size="40"title="NAME" method="post" value="<?php
echo $M12;?>"/></td></tr>
 <tr><td><input type="text" name="d13" size="8" title="NAME" method="post" value="<?php
echo $M13;?>"/></td>
    <td><input type="text" name="d14" size="40"title="NAME" method="post" value="<?php
echo $M14;?>"/></td></tr>
 <tr><td><input type="text" name="d15" size="8" title="NAME" method="post" value="<?php
echo $M15;?>"/></td>
    <td><input type="text" name="d16" size="40"title="NAME" method="post" value="<?php
echo $M16;?>"/></td></tr>
 <tr><td><input type="text" name="d17" size="8" title="NAME" method="post" value="<?php
echo $M17;?>"/></td>
    <td><input type="text" name="d18" size="40"title="NAME" method="post" value="<?php
echo $M18;?>"/></td></tr>
 <tr><td><input type="text" name="d19" size="8" title="NAME" method="post" value="<?php
echo $M19;?>"/></td>
    <td><input type="text" name="d20" size="40"title="NAME" method="post" value="<?php
echo $M20;?>"/></td></tr>
 <tr><td><input type="text" name="d21" size="8" title="NAME" method="post" value="<?php
echo $M21;?>"/></td>
    <td><input type="text" name="d22" size="40"title="NAME" method="post" value="<?php
echo $M22;?>"/></td></tr>
 <tr><td><input type="text" name="d23" size="8" title="NAME" method="post" value="<?php
echo $M23;?>"/></td>
    <td><input type="text" name="d24" size="40"title="NAME" method="post" value="<?php
echo $M24;?>"/></td></tr>
 <tr><td><input type="text" name="d25" size="8" title="NAME" method="post" value="<?php
echo $M25;?>"/></td>
    <td><input type="text" name="d26" size="40"title="NAME" method="post" value="<?php
echo $M26;?>"/></td></tr>
 <tr><td><input type="text" name="d27" size="8" title="NAME" method="post" value="<?php
echo $M27;?>"/></td>
    <td><input type="text" name="d28" size="40"title="NAME" method="post" value="<?php
echo $M28;?>"/></td></tr>
 <tr><td><input type="text" name="d29" size="8" title="NAME" method="post" value="<?php
echo $M29;?>"/></td>
    <td><input type="text" name="d30" size="40"title="NAME" method="post" value="<?php
echo $M30;?>"/></td></tr>
```

```
  <tr><td><input type="text" name="d31" size="8" title="NAME" method="post" value="<?php
echo $M31;?>"/></td>
     <td><input type="text" name="d32" size="40"title="NAME" method="post" value="<?php
echo $M32;?>"/></td></tr>
  <tr><td><input type="text" name="d33" size="8" title="NAME" method="post" value="<?php
echo $M33;?>"/></td>
     <td><input type="text" name="d34" size="40"title="NAME" method="post" value="<?php
echo $M34;?>"/></td></tr>
  <tr><td><input type="text" name="d35" size="8" title="NAME" method="post" value="<?php
echo $M35;?>"/></td>
     <td><input type="text" name="d36" size="40"title="NAME" method="post" value="<?php
echo $M36;?>"/></td></tr>
  <tr><td><input type="text" name="d37" size="8" title="NAME" method="post" value="<?php
echo $M37;?>"/></td>
     <td><input type="text" name="d38" size="40"title="NAME" method="post" value="<?php
echo $M38;?>"/></td></tr>
  <tr><td><input type="text" name="d39" size="8" title="NAME" method="post" value="<?php
echo $M39;?>"/></td>
     <td><input type="text" name="d40" size="40"title="NAME" method="post" value="<?php
echo $M40;?>"/></td></tr>

            </table>
</form>

</div>
```

reset.php:

```
<?php

echo copy("images/P1.jpg","images/1.jpg");
echo copy("images/P1.jpg","images/2.jpg");
echo copy("images/P1.jpg","images/3.jpg");
echo copy("images/P1.jpg","images/4.jpg");
echo copy("images/P1.jpg","images/5.jpg");
  $ret = file_put_contents('images/date1.txt','0');
  $ret = file_put_contents('images/date2.txt','0');
  $ret = file_put_contents('images/date3.txt','0');
  $ret = file_put_contents('images/date4.txt','0');
  $ret = file_put_contents('images/date5.txt','0');

  $ret = file_put_contents('images/t1.txt','0');
  $ret = file_put_contents('images/t2.txt','0');
  $ret = file_put_contents('images/t3.txt','0');
  $ret = file_put_contents('images/t4.txt','0');
```

```
 $ret = file_put_contents('images/t5.txt','0');

  $ret = file_put_contents('images/w1.txt','0');
 $ret = file_put_contents('images/w2.txt','0');
 $ret = file_put_contents('images/w3.txt','0');
 $ret = file_put_contents('images/w.txt','0');
 $ret = file_put_contents('images/w5.txt','0');


  $ret = file_put_contents('images/a1.txt','0');
 $ret = file_put_contents('images/a2.txt','0');
 $ret = file_put_contents('images/a3.txt','0');
 $ret = file_put_contents('images/a4.txt','0');
 $ret = file_put_contents('images/a5.txt','0');
 $ret = file_put_contents('count.txt','0');


  $url='test.php';
  echo '<META HTTP-EQUIV=REFRESH CONTENT=''0.001; '.$url.'">';

?>
```

status.php:

```
<?php

$data= $_GET['s'];
                $myfilename= "un.txt"; if(file_exists($myfilename)){$p6=
file_get_contents($myfilename);}
                $myfilename= "pw.txt"; if(file_exists($myfilename)){$p7=
file_get_contents($myfilename);}
                $myfilename= "cp.txt"; if(file_exists($myfilename)){$d7=
file_get_contents($myfilename);}


if ($data==1){    $ret = file_put_contents('status.txt',"ARMED");$ret =
file_put_contents('cmd.txt',"A".$d7."a");}
if ($data==2){ $ret = file_put_contents('cmd.txt',"B");   $ret =
file_put_contents('status.txt',"DISARMED");}

?>
<html >                                        <style>

 .button-88  {
                font-size: 16px;
```

```
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:65px;
    margin:20px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 22px;
    text-align: center;
    line-height: 1em;
}
</style>
                <meta charset="utf-8" />
                  <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                  <body   >
<table align="center" border=0>
<tr>
                <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG</h1>
</td>

</tr>
</table>

<div align="center">

<table align="center" border=0>

<tr>
<td align="center"></td>
<iframe   width="280" height=300"  frameBorder="0" src="datafrombag.php"
scrolling="no"></iframe>
```

```
</td>
</tr>

                <form action='status.php?s=1' method='post'>

<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='ARMED' />

</td>
</tr>
</form>
                <form action='status.php?s=2' method='post'>
<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='DISARMED' />

</td>
</tr>
</form>
                <form action='main.php' method='post'>
<tr>
<td>

<input type='submit'  class="button-88" style='height:40px;width:300px;'  value='BACK' />

</td>
</tr>
</form>

</table>
<body >
</div>
```

test.php:

```
<html >
                <style>

 .button-88  {
                  font-size: 16px;
  padding: 0.7em 1.4em 0.7em 1.1em;
  color: white;
```

```
  background: #ad5389;
  background: linear-gradient(0deg, rgba(20,167,62,1) 0%, rgba(102,247,113,1) 100%);

    width:175px;
    height:65px;
    margin:20px auto;
    border-radius:10px;
    -webkit-border-radius:10px;
    box-shadow:0 1px 2px #5e5d5b;
}

.button-88  span {
    display: block;
    padding-top: 22px;
    text-align: center;
    line-height: 1em;
}
</style>
                <meta charset="utf-8" />
                  <meta name="viewport" content="width=device-width, initial-scale=0.7,initial-
scale=0.68, maximum-scale=0.68, user-scalable=0;">
                  <body   >
<table align="center" border=0>
<tr>

                <?php


                $myfilename= "date1.txt"; if(file_exists($myfilename)){$date1=
file_get_contents($myfilename);}
      $myfilename= "date2.txt"; if(file_exists($myfilename)){$date2=
file_get_contents($myfilename);}
                $myfilename= "date3.txt"; if(file_exists($myfilename)){$date3=
file_get_contents($myfilename);}
      $myfilename= "date4.txt"; if(file_exists($myfilename)){$date4=
file_get_contents($myfilename);}
                $myfilename= "date5.txt"; if(file_exists($myfilename)){$date5=
file_get_contents($myfilename);}
                $myfilename= "un.txt"; if(file_exists($myfilename)){$p6=
file_get_contents($myfilename);}
                $myfilename= "pw.txt"; if(file_exists($myfilename)){$p7=
file_get_contents($myfilename);}
                $myfilename= "cp.txt"; if(file_exists($myfilename)){$d7=
file_get_contents($myfilename);}
```

```
$d1=$_GET['s'];

if ($d1=="z"){$d1="";

  $ret = file_put_contents('cmd.txt',"z");
  }

                                if ($d1=="q"){$d1="";
                                  $d7=$_POST['d7'];

  $ret = file_put_contents('cp.txt',$d7);
  }

if ($d1=="r"){$d1="";
                                  $p6=$_POST['d6'];
$p7=$_POST['d7'];
  $ret = file_put_contents('un.txt',$p6);
  $ret = file_put_contents('pw.txt',$p7);
  }
if ($d1=="m"){$d1="";
                                  $p1=$_POST['d1'];
$p2=$_POST['d2'];
  $ret = file_put_contents('date1.txt',$p1);
  $ret = file_put_contents('date2.txt',$p2);
  }
if ($d1=="n"){$d1="";
  $p3=$_POST['d3'];
$p4=$_POST['d4'];
$p5=$_POST['d5'];

  $ret = file_put_contents('date3.txt',$p3);
  $ret = file_put_contents('date4.txt',$p4);
  $ret = file_put_contents('date5.txt',$p5);

  }

if ($d1=="C"){$ret = file_put_contents('cmd.txt',"A".$d7."aC");$d1="";}
```

```
                ?>
                <td>
<img  src="P1.png" width="95" height="95"></img>
</td>
<td ><br>
<h1>EBAG SCHEDULE</h1>
</td>

</tr>
</table>

<div align="center">

<table align="center" border=0>

                <form action='main.php' method='post'>

<td>


<input type='submit'  class="button-88"style='height:40px;width:250px;'  value='BACK' />

</td>

</form>



                                <form action='test.php?s=z' method='post'>

<td>

<input type='submit'   class="button-88" style='height:40px;width:250px;'  value='SHUTDOWN'
/>


</form>

</td>
</tr>
<tr>
                                <form action='test.php?s=C' method='post'>

<td>
```

```
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='SEND SMS
GPS' />


</form>
</td>



<form action='reset.php' method='post'>
<td>
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='RESET DATA'
/>
</form>
</td>
</tr>



<tr>
                                <form action='test.php?s=m' method='post'>

<td>

<input type="text" name="d1" size="10"  method="post" value="<?php echo $date1;?>"/>
 : <input type="text" name="d2" size="10"  method="post" value="<?php echo $date2;?>"/>


</td>

<td>
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='UPDATE TIME'
/>
</form>
</td>
</tr>

<tr>
                                <form action='test.php?s=n' method='post'>

<td>
<input type="text" name="d3" size="5"  method="post" value="<?php echo $date3;?>"/>
/<input type="text" name="d4" size="5"  method="post" value="<?php echo $date4;?>"/>
/<input type="text" name="d5" size="5"  method="post" value="<?php echo $date5;?>"/>
```

```
</td>
<td>
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='UPDATE DATE'
/>
</form>
</td>
</tr>

<tr>
                                <form action='test.php?s=r' method='post'>

<td>

<table >
<tr>
<td>
USERNAME:</td><td><input type="text" name="d6" size="5"  method="post" value="<?php
echo $p6;?>"/>
</td>
</tr>
<tr>
  <td>
PASSSWORD:</td><td><input type="text" name="d7" size="5"  method="post" value="<?php
echo $p7;?>"/>
</td>
  </tr>
</table>


</td>
<td>
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='SAVE LOGIN'
/>
</form>
</td>
</tr>


<tr>
                                <form action='test.php?s=q' method='post'>

<td>

CELL NUMBER
 : <input type="text" name="d7" size="10"  method="post" value="<?php echo $d7;?>"/>
```

```
</td>

<td>
<input type='submit'  class="button-88" style='height:40px;width:250px;'  value='SAVE
NUMBER' />
</form>
</td>
</tr>

</table>


<?php
/*  $myfilename = "s6.txt";if(file_exists($myfilename)){$d7= file_get_contents($myfilename);}

include('db.php');
$count=0;
 $sql = "SELECT * FROM data  ORDER by dn  DESC limit ".$d7."";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
  while($row = $result->fetch_assoc()) {


  echo "<tr>

  <td align='center' >" . $row['dn']. "</td>
  <td align='center' >" . $row['dattime']. "</td>
  <td align='center' >" . $row['g1']. "</td>
  <td align='center' >" . $row['g2']. "</td>

 <td align='center' >" . $row['g3']. "</td>
<td align='center' >" . $row['g4']. "</td>
<td align='center' >" . $row['g5']. "</td>


  </tr>";

}}

*/
?>
```

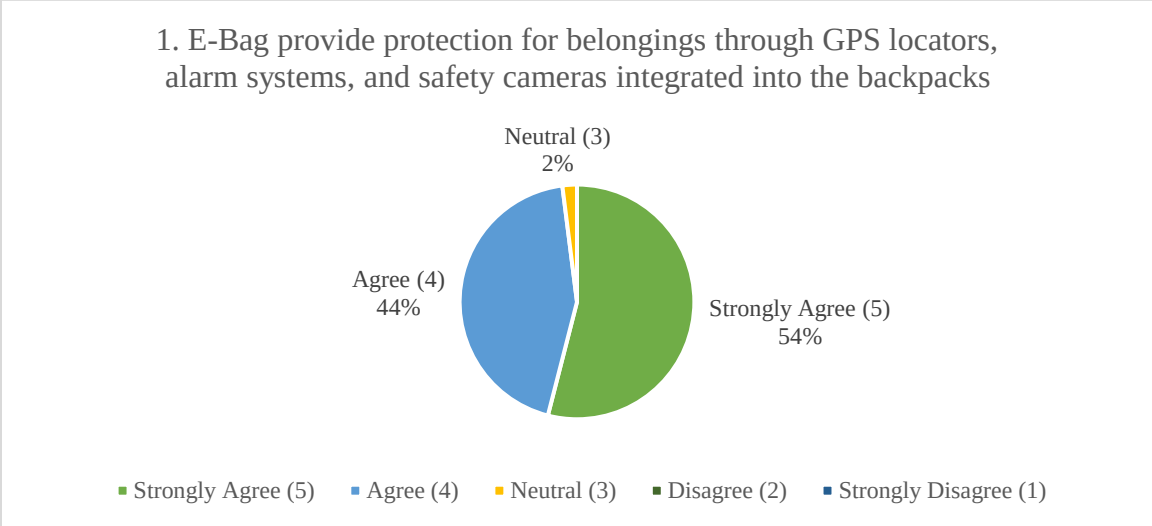
```
</div>
```

Time.php:

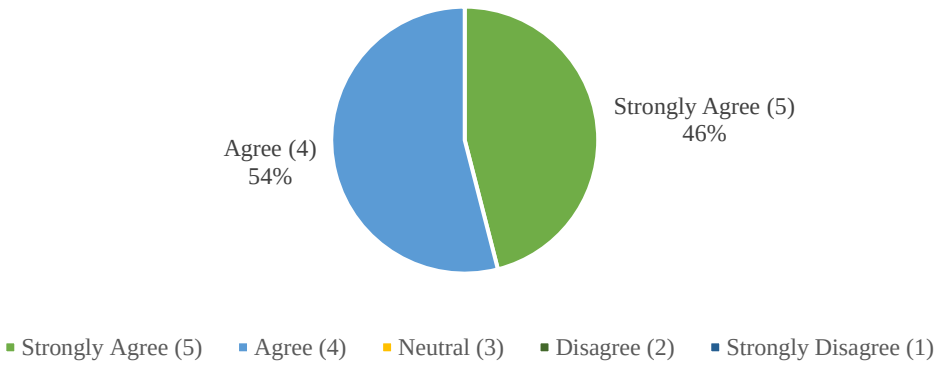
```
<Meta HTTP-EQUIV="Refresh" Content="1; URL=time.php">
<?php
$myfilename= "cmd.txt"; if(file_exists($myfilename)){$cmd= file_get_contents($myfilename);}
if ($cmd=='A')
{


 echo $alarm1;
 $url='cature1.php';
 echo '<META HTTP-EQUIV=REFRESH CONTENT="0.001; '.$url.'">';

 }
```

Appendix I. Sample Input/Output/Reports

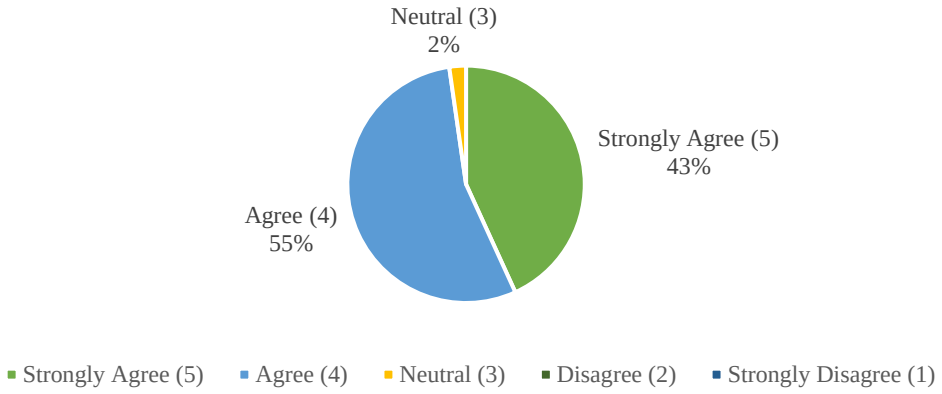
I.1: Summary Report of Survey



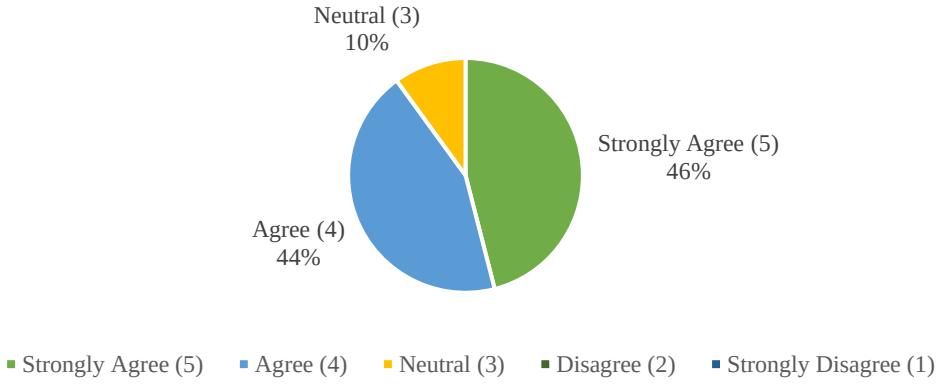
2. E-bag can be monitored its tracking and other innovative capabilities via SMS and mobile app.



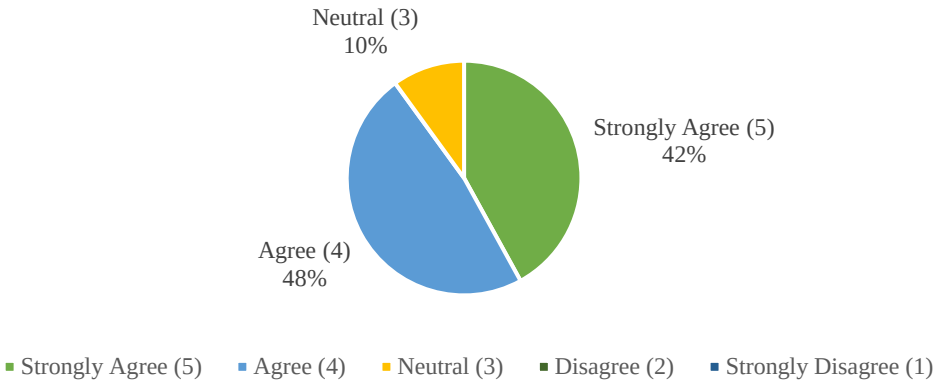
3. IOT technology in E-bag provides peace of mind while traveling by enhancing security when leaving belongings unattended.



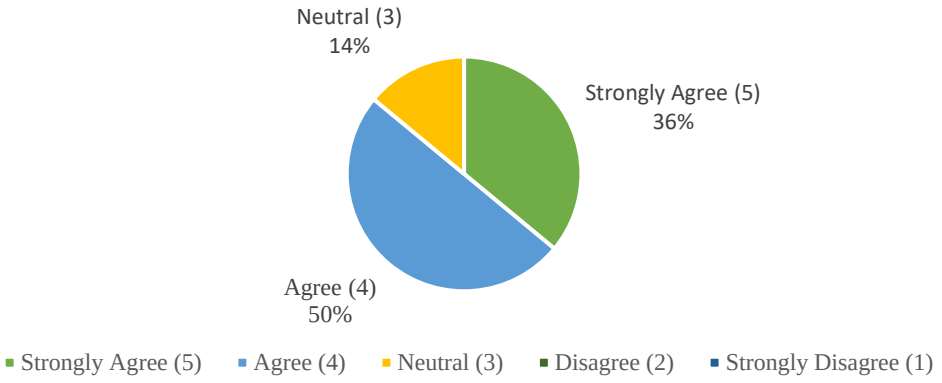
4. The E-bag's location address can be displayed via SMS if it is lost, stolen, or misplaced



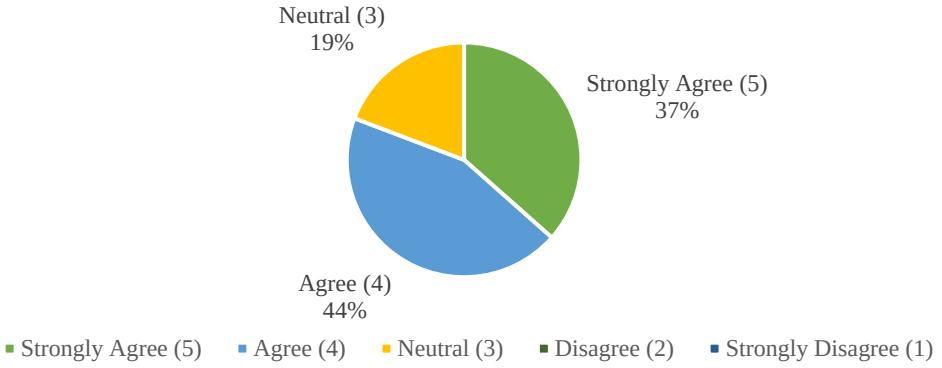
5. E-bags can capture an image of anyone who has handled the backpack using a safety camera.



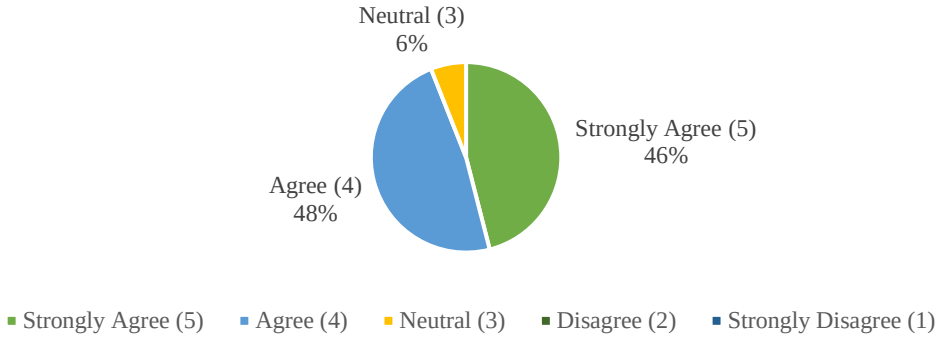
6. E-bags can alert the user through security features if someone else handles the bag.



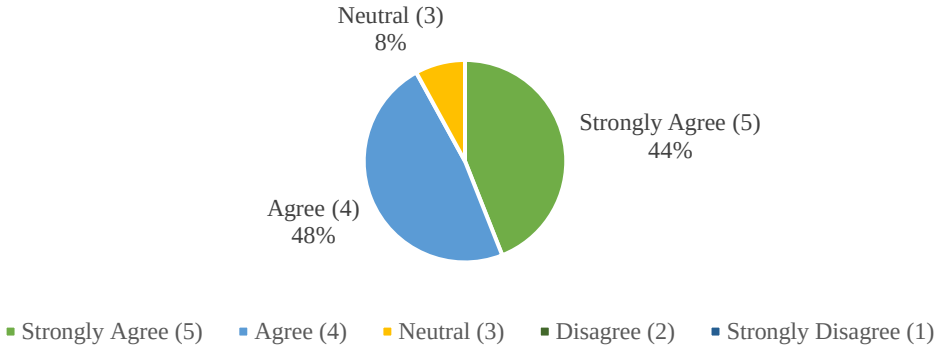
7. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app is easy to understand and use.



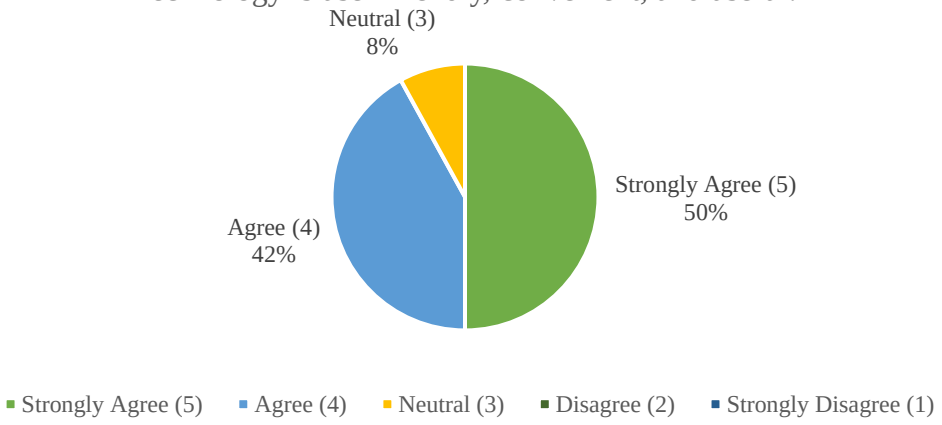
8. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can manage the backpack security system.



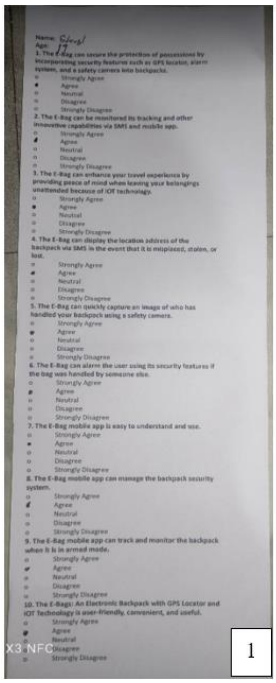
9. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology mobile app can track and monitor the backpack when it is in armed mode.

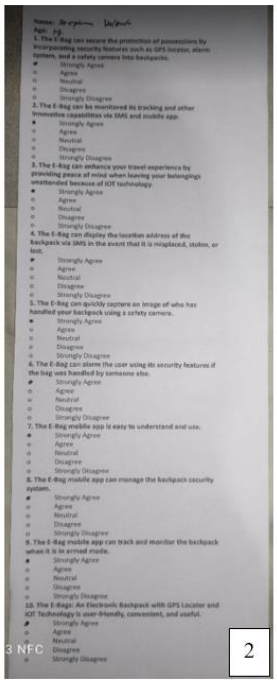


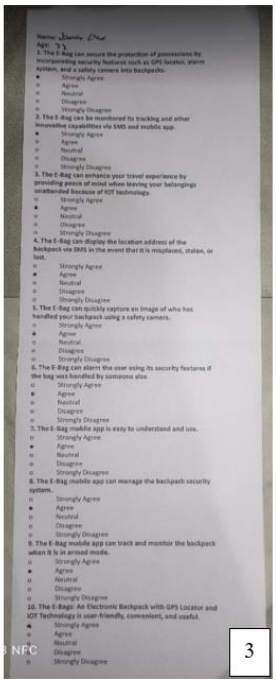
10. The E-Bags: An Electronic Backpack with GPS Locator and IOT Technology is user-friendly, convenient, and useful.

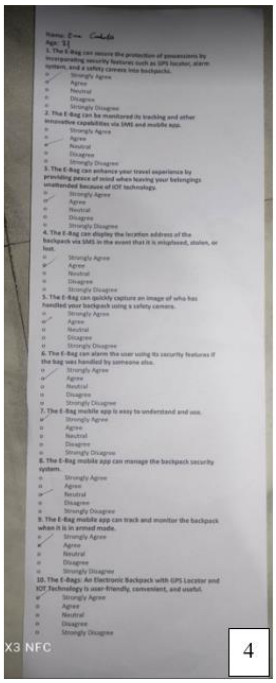


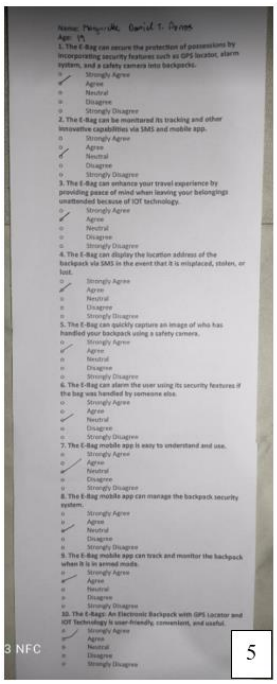
I.2 Transcript of Interview


1


2


3


4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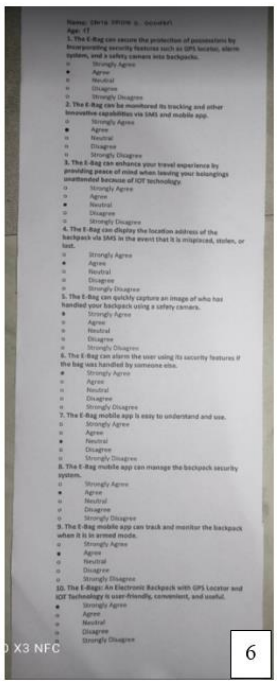
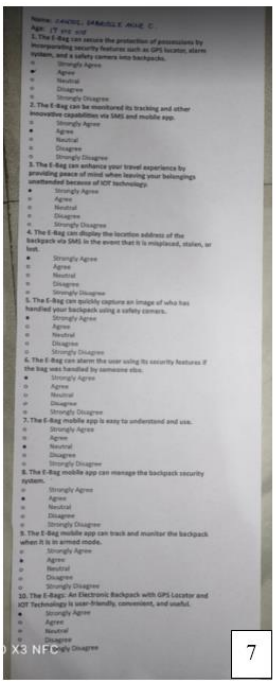
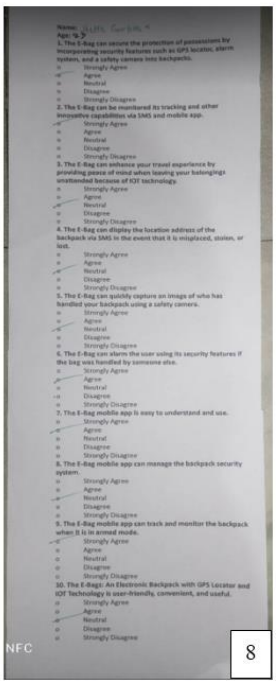

6


7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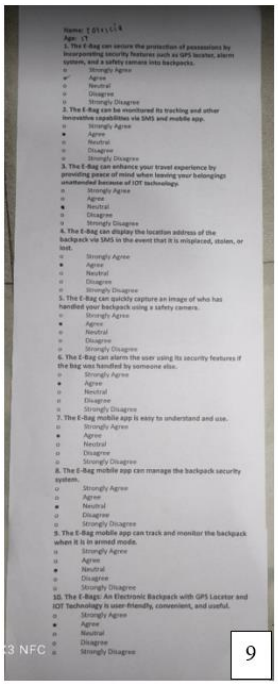
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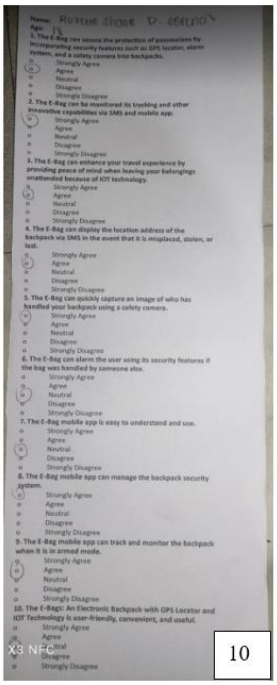
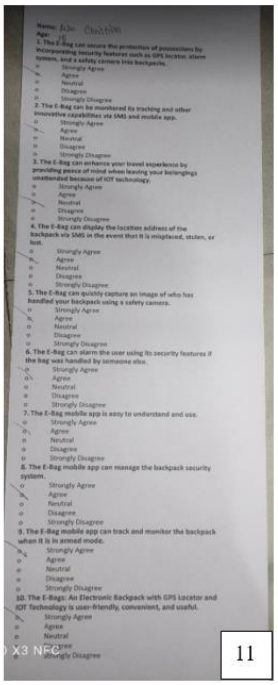

10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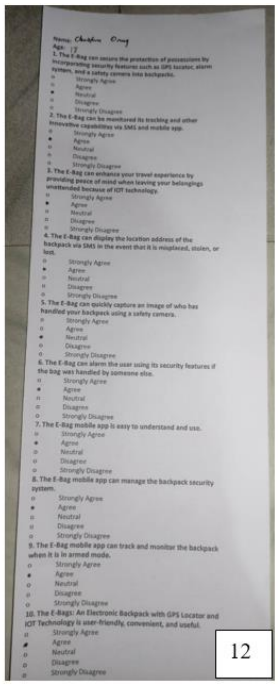
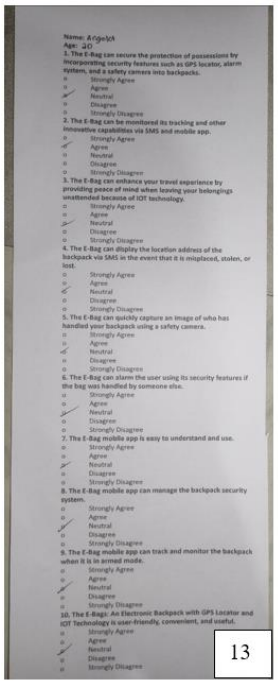
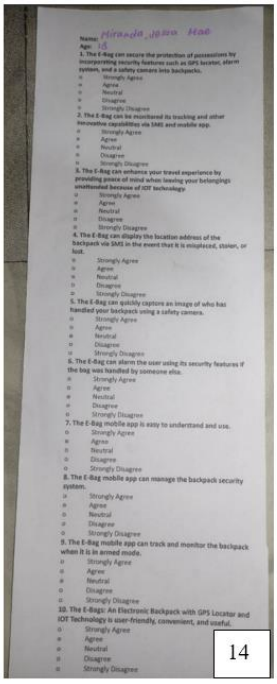
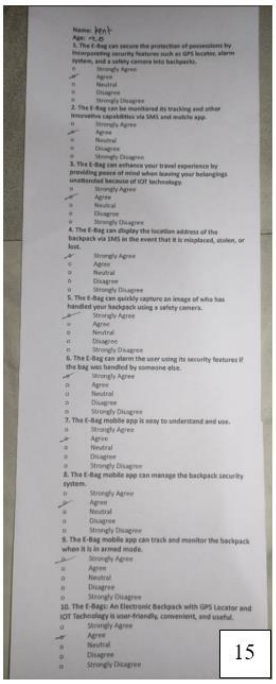

12


13


14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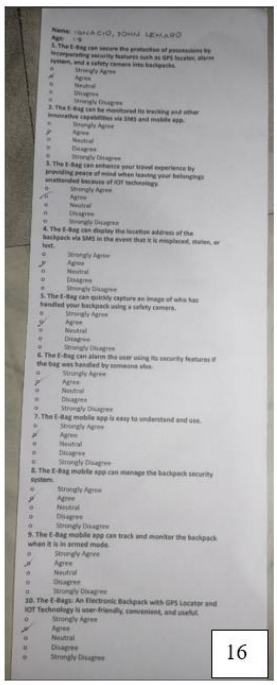
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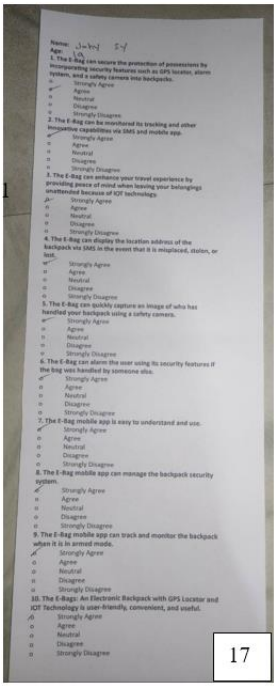
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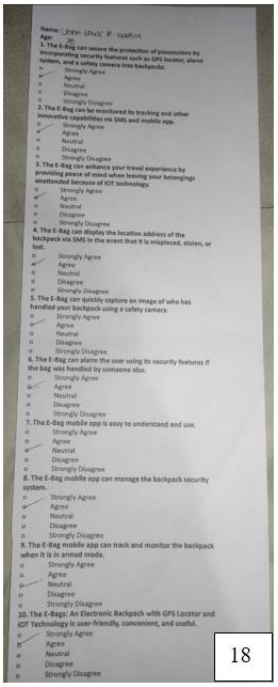
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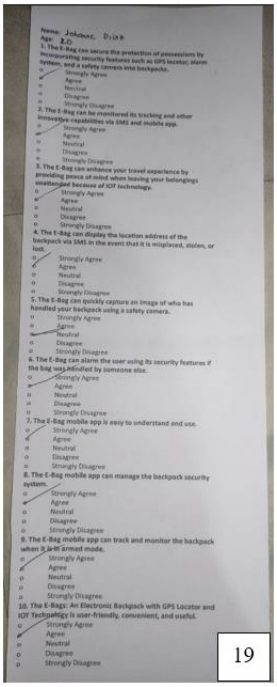
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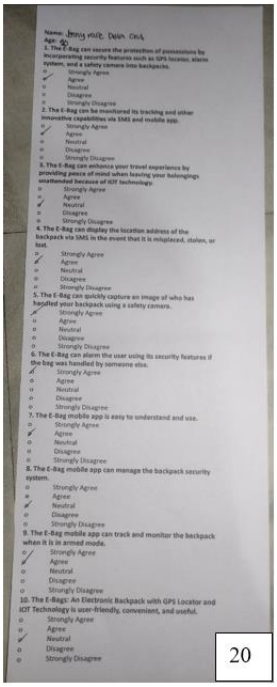
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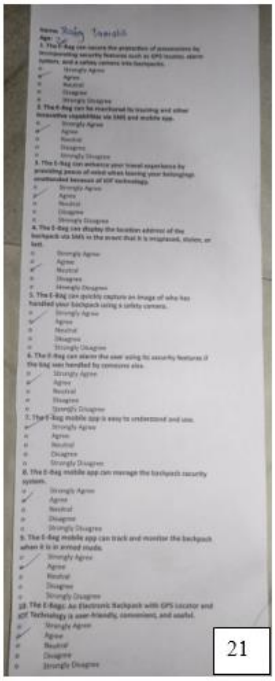
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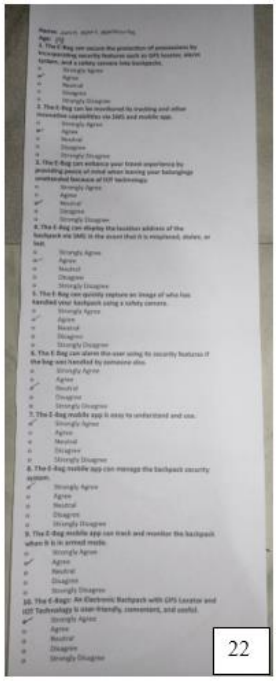
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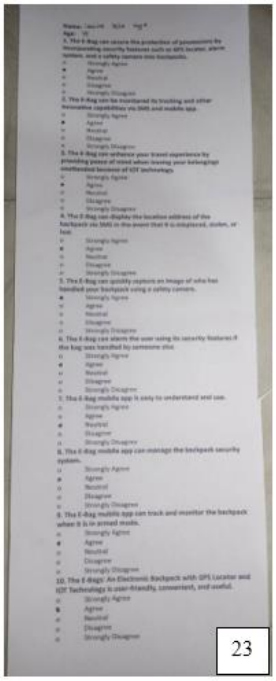
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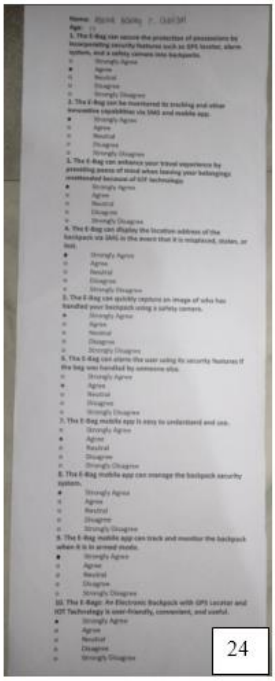
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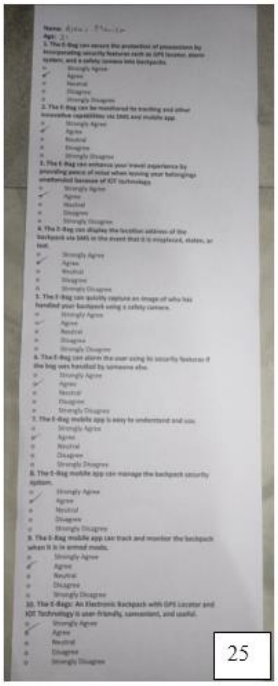
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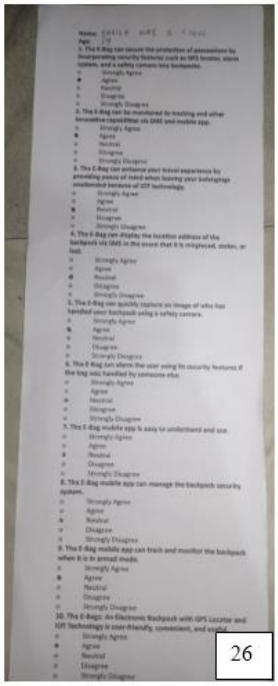
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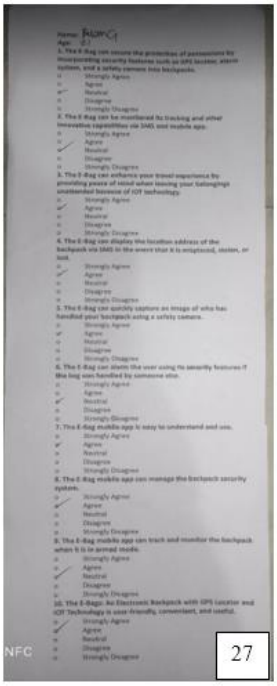
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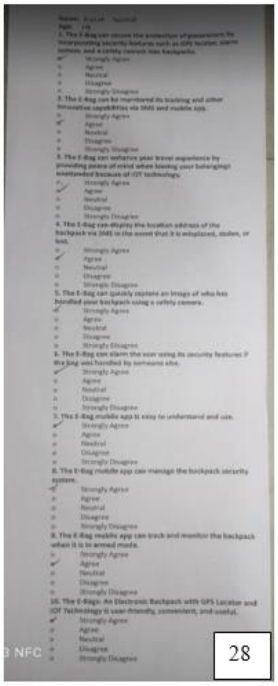
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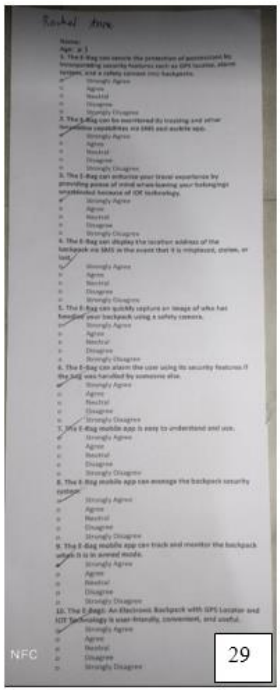
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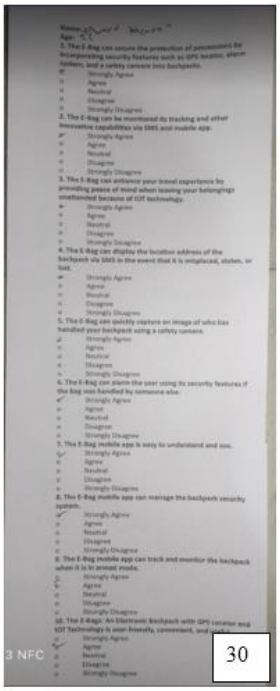
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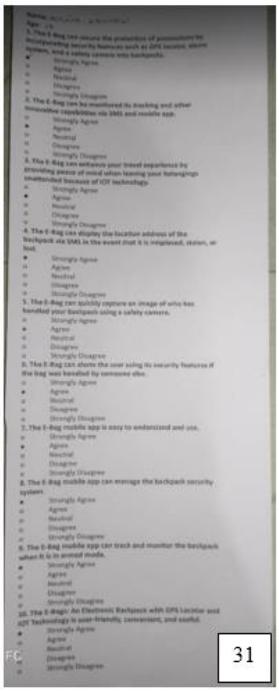
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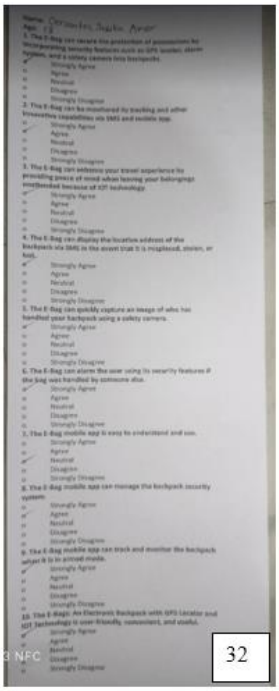
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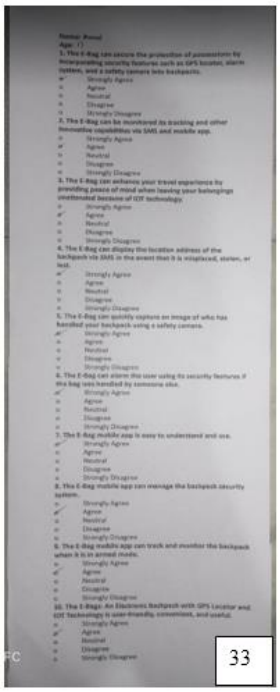
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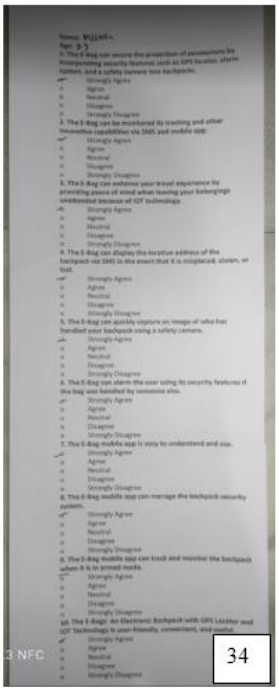
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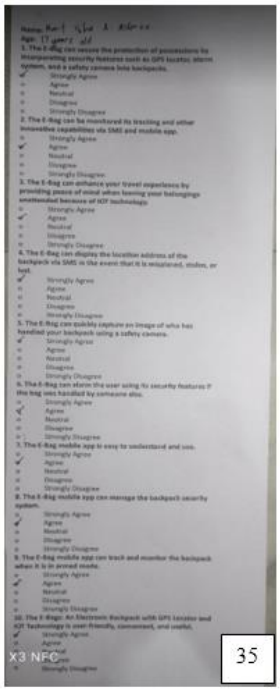
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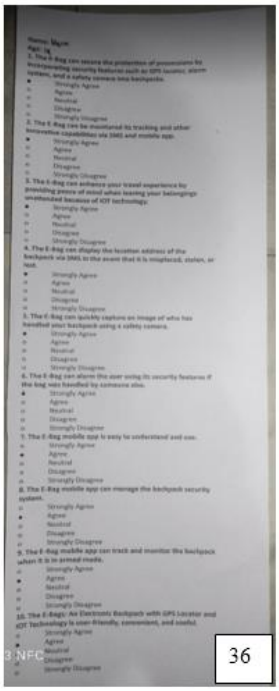
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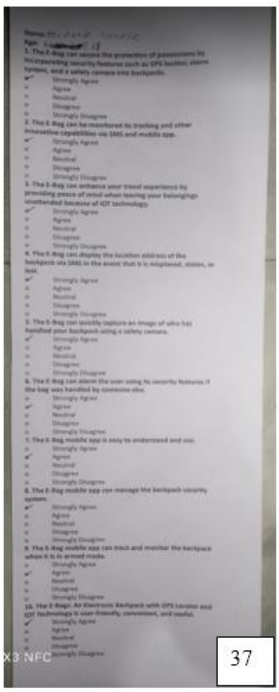
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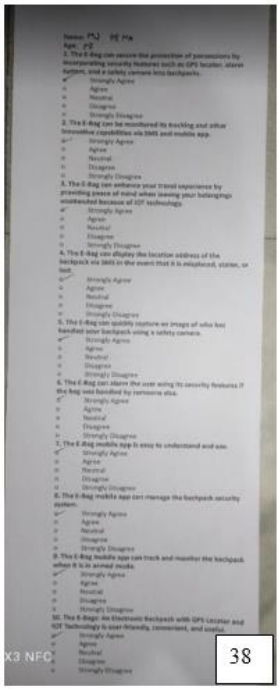
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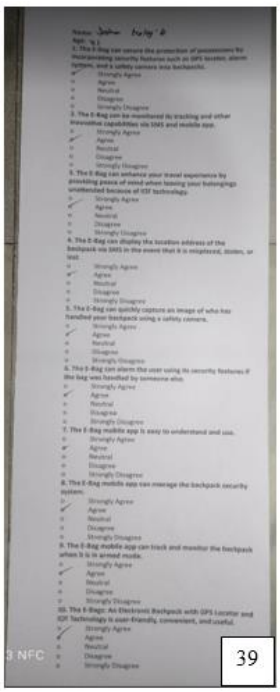
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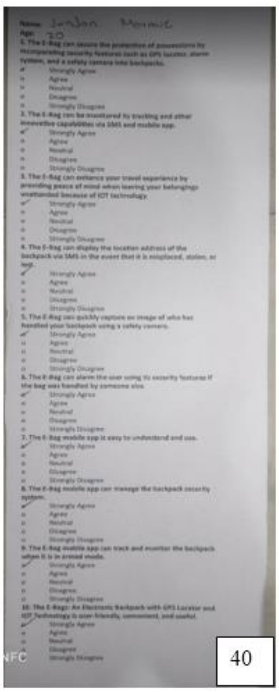
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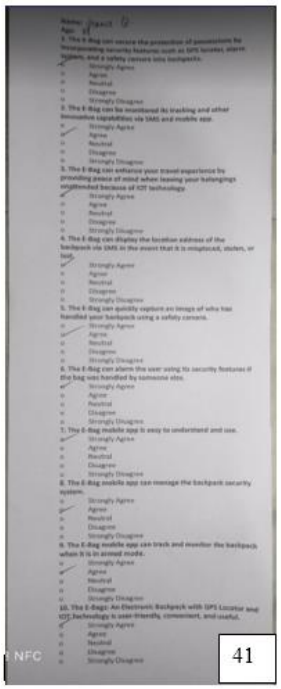
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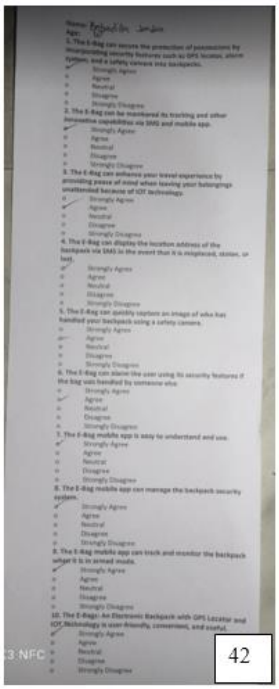
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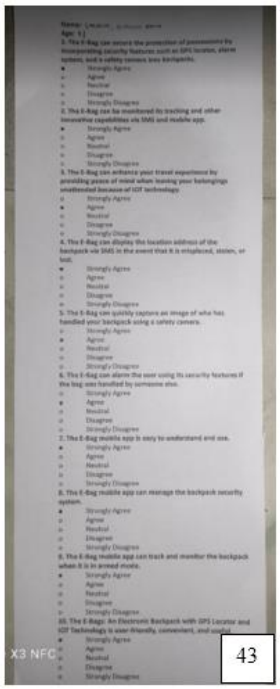
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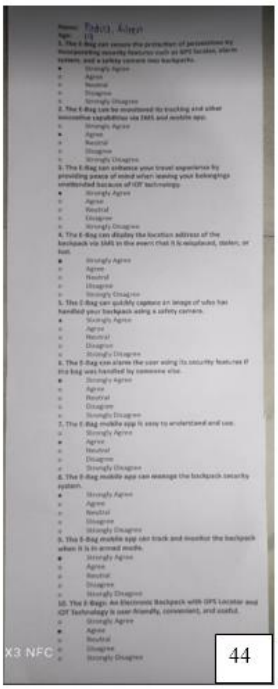
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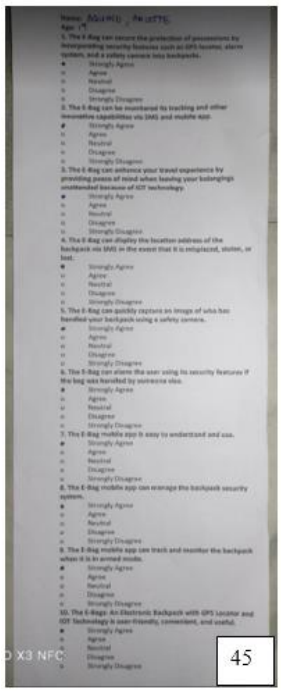
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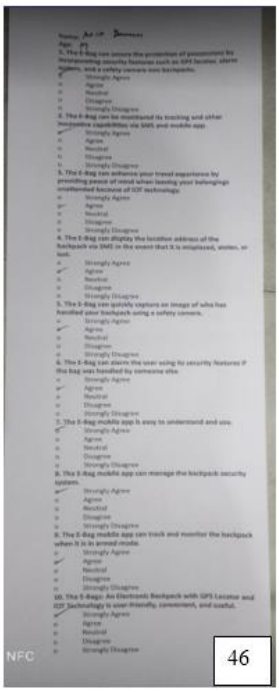
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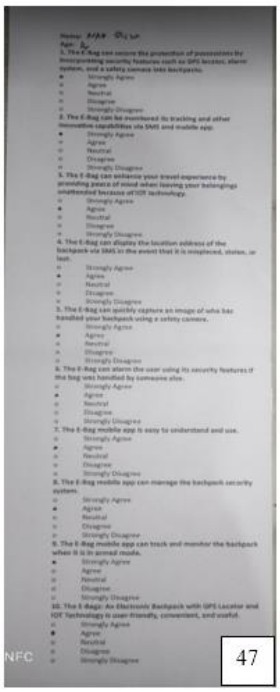
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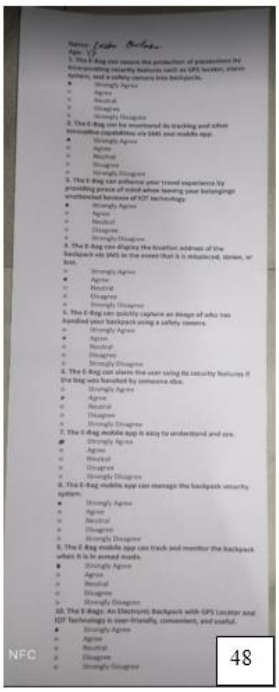
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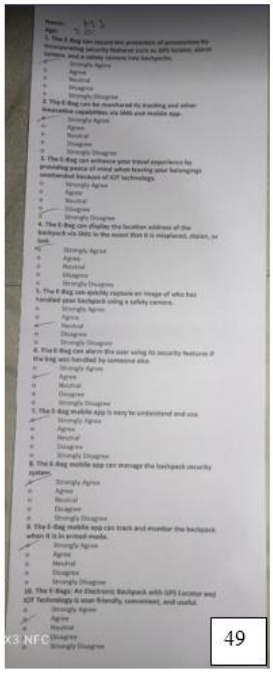

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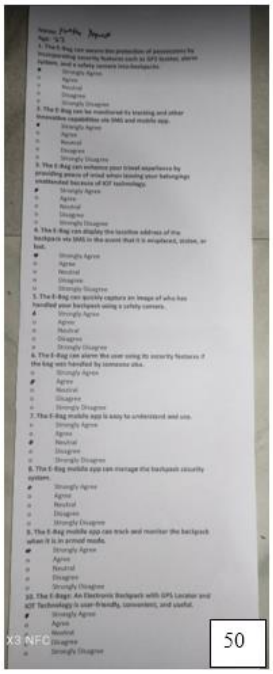

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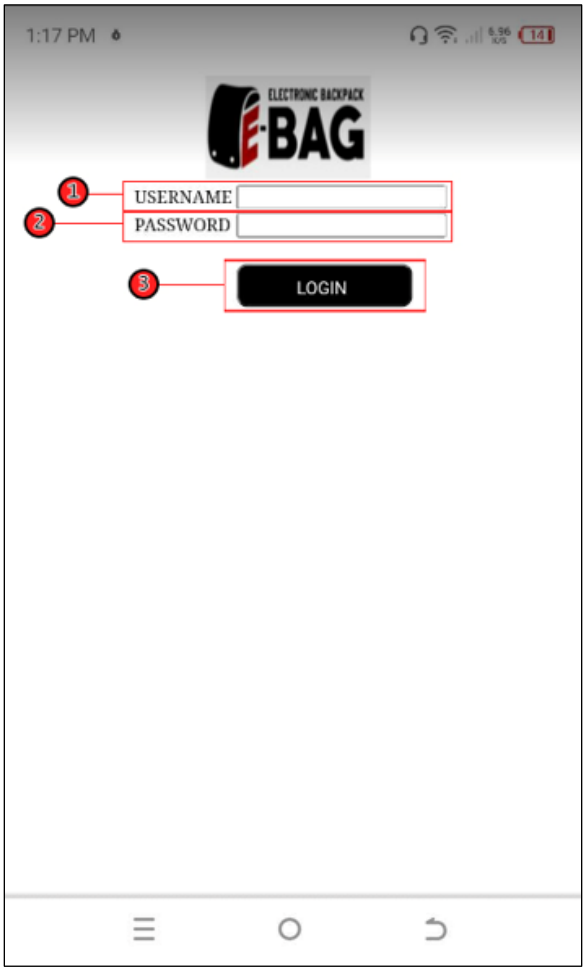
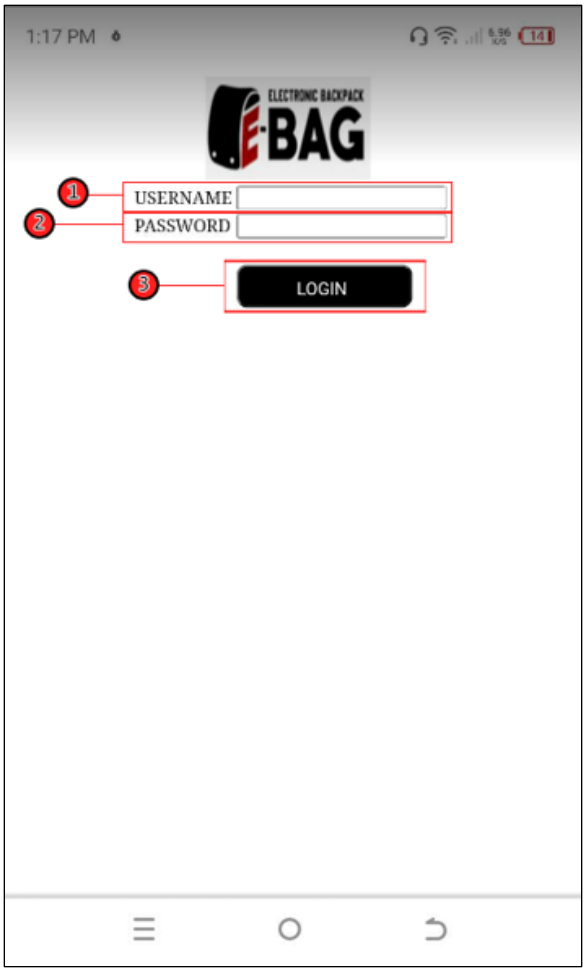
Appendix J: User's Guide

E-BAG: AN ELECTRONIC BAGPACK WITH GPS LOCATOR AND IOT TECHNOLOGY



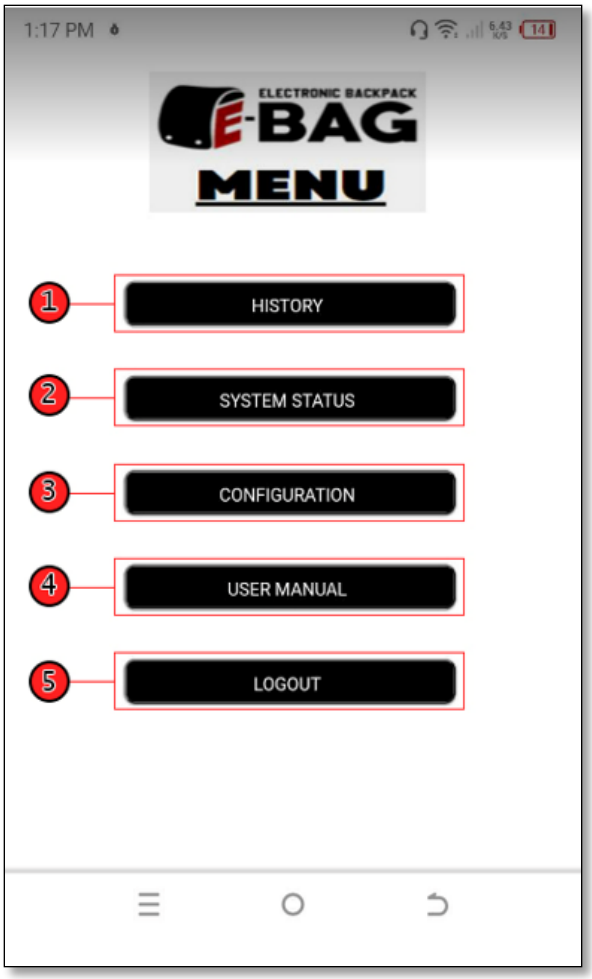
1.) Enter default username.
2.) Enter default password.
3.) Click "LOGIN" button to proceed to the web page where the user can download the mobile application.

1.) Click “BACK” button to go back to the previous page.
2.) Click “EDIT” button to change the manual on the website.
3.) Click “DOWNLOAD APP HERE” to be redirected to the downloading off the mobile application.



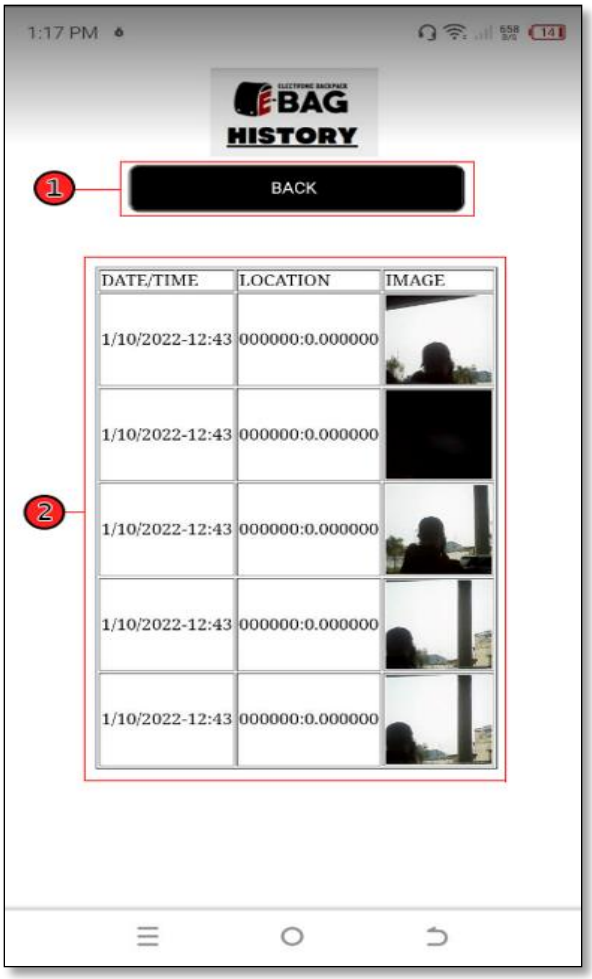
After the user download the app the login page will show again using the mobile application

1.) Enter default username.
2.) Enter default password.
3.) Click “LOGIN” button to proceed to the web page where you can download the app.



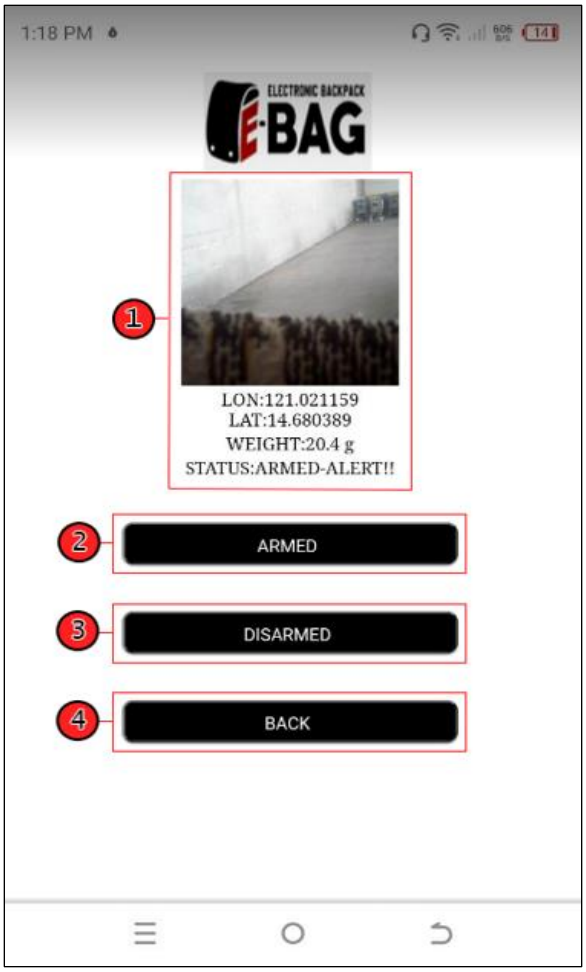
After the user login in the login page. It will proceed to the menu of the mobile application of the E-Bag

1.) Click “HISTORY” button to access the history logs.
2.) Click “SYSTEM STATUS” button to view the status of the bag and to set the device in armed or disarmed mode.
3.) Click “CONFIGURATION” button to access device settings.
4.) Click “USER MANUAL” button to see the instructions on how to operate the device.
5.) Click “LOGOUT” button to log off.



1.) Click “BACK” button to go back to the previous page.

2.) The table shows the history logs of the device.

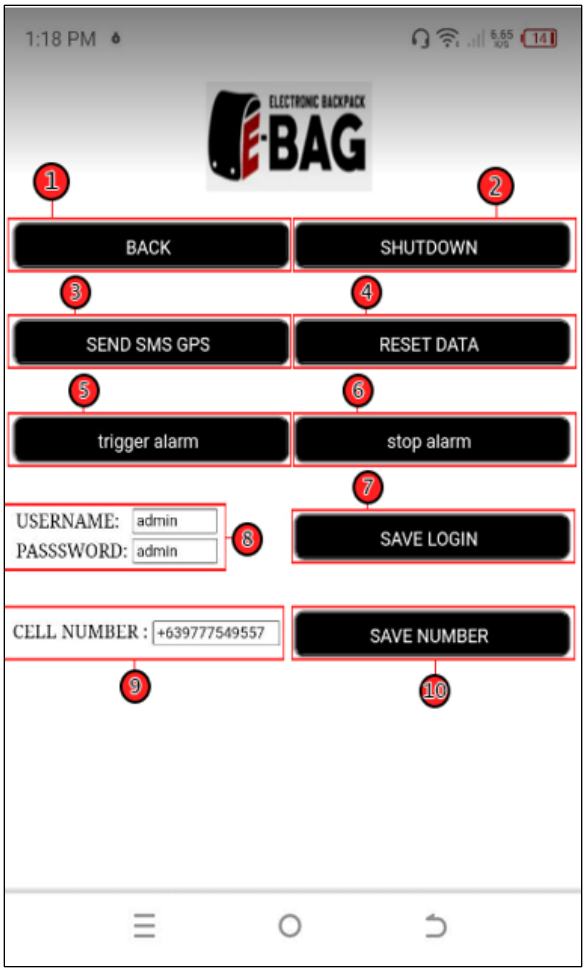


1.) User can see here the current location, weight, and status of the device.

2.) Click “ARMED” button to activate the device alarms.

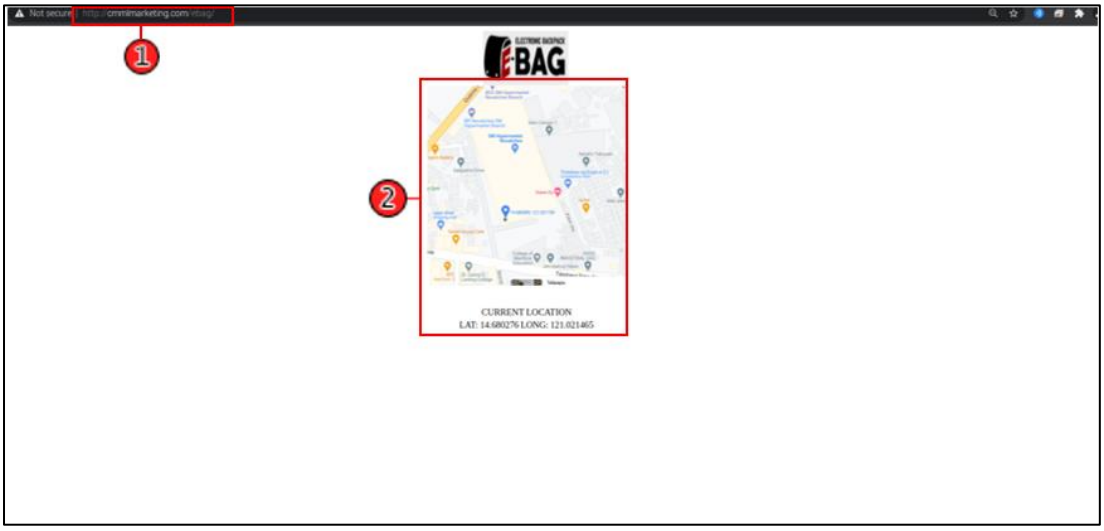
3.) Click “DISARMED” button to deactivate the device alarms.

4.) Click “BACK” button to go back to the previous page.



1.) Click “BACK” button to go back to the previous page.
2.) Click “SHUTDOWN” button to make the device go offline.
3.) Click “SEND SMS GPS” button to send the current location of the device to a mobile phone.
4.) Click “RESET DATA” button to clear the history logs.
5.) Click “trigger alarm” button to activate the device alarms.
6.) Click “stop alarm” button to deactivate the device alarms.
7.) Click “SAVE LOGIN” button to save any changes in username or password.
8.) User can change here their username or password.
9.) User can change the mobile phone number that is currently being used.
10.) Click “SAVE NUMBER” button to save any changes in the number.

1.) Click “BACK” button to go back to the previous page.



To view the web page

1.) Search the link cmmlmarketing.com/ebag/

2.) The map shows the current location of the E-Bag with its longitude and latitude.

Appendix K: Personal Technical Vitae

Curriculum Vitae of

NERIZZA M. BAYAS

#35 Int. Rivera St., Brgy. Tañong, Malabon City
bayas.186441@caloocan.sti.edu.ph
+639565092089

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2019-Present	STI College of Caloocan
Senior High School	2017-2019	Arellano University-Malabon
High School	2013-2017	Malabon National High School
Elementary	2007-2013	Tanza Elementary School

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2019	Institute of Computer Engineering of the Philippines	Member

SKILLS

SKILLS	Date Acquired
Computer Hardware and Software	2019
Microsoft Office (MS Word, Excel, PowerPoint, etc.)	2016

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
May 2023	JavaScript-Algorithms-and-Data-Structures- FreeCodeCamp
May 2023	Responsive Web Design - FreeCodeCamp
March 2023	Internet of Things using LoraWan technology - PacketHACKS
August 2022	HUAWEI: HCIA - CpE Practicum
August 2022	HUAWEI: HCIA - Cloud Service V3.0 Course
July 2022	HUAWEI: HCIA - IOT Course
July 2022	HUAWEI: Compute Cloud Services
September 2021	HUAWEI Training Webinar
September 2021	The Future of Mobile App Industry
November 2020	Cyber Security: Are you at Risk.
November 2019	"CpE Upgrade Towards to Technology"

Curriculum Vitae of

JOMARI C. JAMERLAN

340 Maginoo st brgy 104 Tondo,Manila
jomarijamerlan123@gamil.com
09892586916

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2019-Present	STI College Caloocan
Senior High School	2017-2019	STI College Recto
Junior High School	2013-2017	Antonio J. Villegas High School
Elementary	2011-2013	Gen V. Lim Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2019	Office Assistant	QMMC Compound, Lat: 14°37'20.28" Long: 121°4'22.08", Project 4, Quezon city

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2021	ICpEP – Student Edition	Member

SKILLS

SKILLS	Date Acquired
Computer Hardware	2019
Microsoft Office (MS Word, Excel, PowerPoint, etc.)	2016

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
October 2021	Cybersecurity
August 2021	Huawei Mobile App Development Webinar
March 2021	KUMUSTA KA? A webinar series about Stress Management & Essential Self-care Tips for students
November 2020	Cybersecurity Webinar

Curriculum Vitae of

MARK JOSEPH L. LUMIBAO

4595 Chico St. Gen. T. de Leon Valenzuela City
josephlived@gmail.com
09777549557

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2019-Present	STI College Caloocan
Senior High School	2017-2019	Divine Word Learning Center
Junior High School	2013-2017	Divine Word Learning Center
Elementary	2007-2013	Gen. T. de Leon Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2019	Service Crew	17/F, Citibank Bldg., 0702 Makati City (Golden Arches Development Corp.)

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2020-2021	ICpEP – Student Edition	Member

SKILLS

SKILLS	Date Acquired
Java	2016
Computer Hardware	2018
Microsoft Office (MS Word, Excel, PowerPoint, etc.)	2013

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
October 2021	Cybersecurity
August 2021	Huawei Mobile App Development Webinar
March 2021	KUMUSTA KA? A webinar series about Stress Management & Essential Self-care Tips for students
November 2020	Cybersecurity Webinar